



2019 K-Innovation ODA Program with Cambodia

Policy Consultation on National STI Policy Development and STI System Management for Cambodia's Policymakers

2019 K-Innovation ODA Program with Cambodia

Title	Policy Consultation on National STI Policy Development and STI System Management for Cambodia's Policymakers
Institute	Science and Technology Policy Institute (STEPI)
Partner Country	The Kingdom of Cambodia
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Program Investigator	Wangdong Kim, Chief Director, Division of Global Innovation Strategy, STEPI
Project Manager	Jung Won Lee, Senior Research Fellow, Office of Development Cooperation Research(IICC), STEPI
Researchers	Jung Won Lee, Senior Research Fellow, Office of Development Cooperation Research(IICC), STEPI Jieun Kim, Researcher, Division of Global Innovation Strategy, STEPI
Project Coordinator	Jieun Kim, Researcher, Division of Global Innovation Strategy, STEPI

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Policy Consultation on National STI Policy Development and STI System Management for Cambodia's Policymakers

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5th~7th Flrs., Building B, Sejong National Research Complex, 370 Sicheong-daero, Sejong-si, 30147. Republic of Korea

Tel: +82-44-287-2000 Fax: +82-44-287-2068

www.stepi.re.kr

For more information, visit

iicc.stepi.re.kr

Printed by Mirae Media

Tel: +82-2-815-0407 Fax: +82-2-822-1269

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Executive Summary

The Royal Government of Cambodia is striving to achieve its Rectangular Strategy IV through the 2019-2023 National Strategic Development Plan (NSDP) with the mid- and long-term vision of becoming an upper middle-income country by 2030 and a high-income country by 2050. Recognizing the important role of STI in national development, the General Secretariat of the National Science and Technology Council (GS-NSTC) established the “Cambodian STI Policy 2019-2025.” The Science and Technology Policy Institute (STEPI) supported and provided policy consultation to GS-NSTC for STI policy development as a project of the K-Innovation ODA Program. GS-NSTC and STEPI agreed that close and strategic linkage between the national economic development plan and science and technology development is necessary for the effective implementation of STI policy, for which Cambodia needs a national technology strategy as a next step based on scientific methods such as technology roadmap.

The aim of the project is to provide consultation on capacity building for national STI policy development and STI system management for Cambodian policymakers. The objective of the 2019 K-Innovation ODA Program with Cambodia is to develop and finalize the STI Policy 2019-2025 of Cambodia, share Korean experiences in the implementation of the STI policy including TRM, and provide technical advice for the implementation of the national STI Policy and development of pilot TRM.

The consultation process for the project had the following activities:

The STEPI team visited Cambodia from March 25 to 29, 2019 for a kick-off meeting and a field study. The workshop was held to share the experiences of Korea in STI policy and TRM. Key STI stakeholders were interviewed to provide consultation for the implementation of STI policy and analyze related issues.

The STEPI team held a capacity-building workshop to share the best practices of Korea and to practice the technology roadmap as an implementation strategy of STI policy. A total of

10 delegations from GS-NSTC and relevant institutions were invited to Korea from May 20 to 25, 2019.

The draft of the Cambodian implementation plan and pilot TRM of STI policy were reviewed by the research team for progress assessment.

A final workshop was held in October 2019 in Phnom Penh to present the developed pilot TRM for comments from the STEPI team as well as Cambodian stakeholders. Additionally, Korean experts held a consultation meeting with the pilot TRM team to assess the relevance and sustainability of recommendations and to build on possible follow-up programs.

The research team suggested implications and recommendations on how to develop Cambodia's capacity for national STI strategy and technology roadmap. Regarding Cambodia's STI system, it is necessary to strengthen the governance of STI policy. It is recommended that the General Secretariat of NSTC be expanded and reorganized to create a dedicated department dealing with science and technology innovation. It is also important to have an implementation system and a higher policy coordination system that enables practical policy discussion and coordination.

Regarding the technology roadmap, the fact that the pilot project was made to recognize the importance of the technology roadmap and to understand the basic concepts and procedures by GS-NSTC was a significant achievement. It is also important to be able to cultivate experts who can play the role of leader or champion in each sector by carrying out the national technology roadmap project in the second phase. Governments should not directly interfere with the roadmap contents but facilitate and coordinate instead the roadmap process itself. Governments should always manage the process so that it is fair and transparent during the roadmap process. Another role of the government is to support sufficient resources, such as supporting staff and budget, for the technology roadmap. Finally, it is always necessary to encourage experts from the private sector to participate in the roadmap actively.

The STI plan and policy, such as last year's "STI Policy 2019-2025" and this year's technology roadmap, should be implemented in line with the national development goals, especially when an independent STI ministry is established.

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CHAPTER 1

Project Overview

Policy Consultation on National STI Policy
Development and STI System Management for
Cambodia's Policymakers





Chapter 1. Project Overview

1. Introduction

The Royal Government of Cambodia is striving to achieve its Rectangular Strategy Phase IV through the 2019-2023 National Strategic Development Plan (NSDP) with the mid- and long-term vision of becoming an Upper Middle-Income Country by 2030 and a High-Income Country by 2050.

Driven by its long-term vision toward 2030, particularly as expressed in the Rectangular Strategy III and IV, Cambodia has prepared and adopted the Cambodian Industrial Development Policy 2015-2025 (IDP 2015-2025) as a guide to promoting the country's industrial development that will help maintain sustainable and inclusive high economic growth and reach Middle-Income economy status.

In this regard, Cambodia has promoted science, technology, and innovation (STI) development in order to achieve sustainable socioeconomic development goals. As a first step, Cambodia has established the National Science and Technology Council (NSTC), which is assisted by a secretariat called the General Secretariat of the National Science and Technology Council (GS-NSTC).

GS-NSTC aims, in broad terms, to contribute to the sustainable socioeconomic development of Cambodia through the formulation of the National STI plan, which will guide domestic research and development (R&D) infrastructure building, capacity expansion, and government research projects. It also has the mandate of improving the

quality of human resources for the development of national STI programs and building an STI information system for the effective dissemination of knowledge among policymakers and related entities.

In line with this mandate, GS-NSTC addressed organizational challenges and indicated the preferred topics of consultation in a project concept paper. Based on the confirmation and acceptance of the project proposal submitted by GS-NSTC, the K-Innovation ODA Program has affirmed the commitment to supporting Cambodia in making faster progress toward achieving Cambodia's development goals.

At the same time, the Korean government set up the official development assistance priority areas for Cambodia based on its development strategies and development needs, Korea's comparative advantage, and the bilateral policy discussions.

Accordingly, the scope of the K-Innovation Program is also based on the consideration of the Royal Government's development priorities and the lessons learned from the previous Korean development experience to avoid trial and error and introduce the most appropriate policies for Cambodia's specific needs.

The aim of the project is to provide consultation on capacity building for national STI policy development and STI system management for Cambodian policymakers, which includes the following aspects:

Capacity building on STI policy development and management of STI ecosystem for officials at GS-NSTC and/or relevant ministries in Cambodia

- 1) Advice on the draft national STI policy
- 2) Progress assessment
- 3) Sharing the Korean experience relevant to Cambodia

The Science and Technology Policy Institute (STEPI) is a Korean government-sponsored research institute and a leading global think-tank that conducts research and analysis on

issues pertaining to science, technology, and innovation. STEPI works closely with various ministries, regional governments, and relevant institutes to support policy design. It also has a long experience in providing STI policy consultancy for developing countries in the role of conducting STI development cooperation activities with partner countries.

As an output of the 2018 K-Innovation Program in Cambodia, GS-NSTC established a final draft of the “Cambodian STI Policy 2019-2025” with support and policy consultation from STEPI. In addition, GS-NSTC and STEPI agreed that close and strategic linkage between the national economic development plan and science and technology development is necessary for the effective implementation of the STI policy. Strategic priority should also be set among the STI areas in STI policy for efficient R&D investment. In other words, for the implementation of the STI policy, Cambodia needs a national technology strategy as a next step.

2. Objectives

The title of the project is “Capacity Building Program on National STI Policy Development and STI System Management for Cambodia’s Policymakers.” The program runs for about 12 months, from January 1, 2019 to December 31, 2019.

The objective of the 2019 K-Innovation ODA Program with Cambodia is to:

- Develop and finalize the STI Policy 2019-2025 of Cambodia
- Share Korean experiences in the implementation of the STI policy, including TRM
- Conduct a local capacity-building workshop for the pilot TRM
- Provide technical advice for the implementation of the national STI policy and development of pilot TRM

3. Project Framework

■ Consultation on the establishment of science and technology roadmap (TRM)

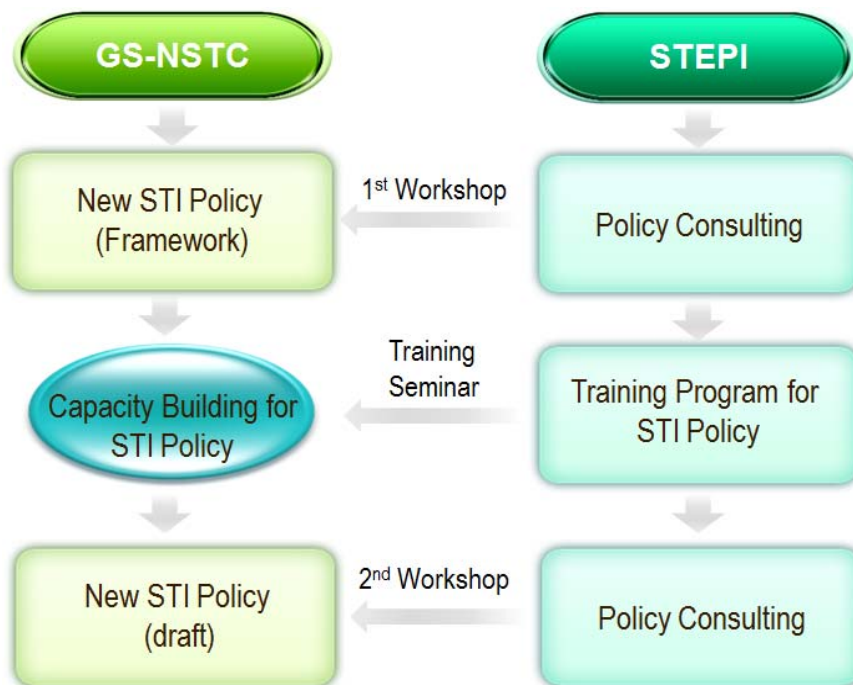
The consultation process for TRM will have the following activities:

- Establishment of key areas according to the priorities of the Cambodian government in the field of science and technology innovation ecosystem, government research and development program management, and science and technology manpower planning
- Research and diagnosis by Cambodia's domestic experts and STEPI researchers
- Local workshops on the formation of consensus on the direction of Cambodia’s science and technology innovation policy and the contents of policy consultation for major policymakers, government officials, and local public officials in Cambodia

■ **Strengthening the policymaking capacity of civil servants in charge of innovation policy in Cambodia**

A training workshop will be held in Korea for high-ranking and middle-level civil servants in GS-NSTC and related ministries and agencies in Cambodia.

[Figure 1-1] Conceptual Framework of the Project



4. Project Team

4.1 Research Team

Based on the research areas above, a research team was formed, and a high-level adviser was appointed in accordance with the relevant expertise in the fields of science and technology policy, related knowledge, and know-how for sharing the Korean development experience.

The research team is composed of three full-time experts including a Project Manager (PM), Dr. Jungwon Lee, and a policy planning expert in STEPI. External experts have been identified for the delivery of the workshop and lecture based on the topics to be covered, with one STEPI researcher serving as the team coordinator.

The research team is responsible for technical advice on Cambodia's national STI plan and management of the STI ecosystem in partnership with GS-NSTC. For sharing the relevant Korean experience during the training and consultation workshop and providing constructive comments for the progress evaluation of the program, the research team identified and selected additional experts deemed most appropriate based on education, experience, and expertise in the topic co-selected in the closing meeting of the first mission.

[Table 1-1] Research Team

Name	Organization	Position & Area of Expertise	Contact Information
Jungwon Lee	STEPI, Senior Research Fellow Project Manager (PM)	Innovation Ecosystem	leejw@stepi.re.kr
Jei-Young Lee	STEPI, Associate Research Fellow	ICT TRM Consultation	jeilee@stepi.re.kr
Dong Hoon Oh	Wisen Global, Partner	Technology Strategy	oh.donghoon@gmail.com

Name	Organization	Position & Area of Expertise	Contact Information
Sun Hark Bong	TBNA, CEO	Korean Experience	sunhark@tbna.co.kr
Hyun Lim	KISTEP, Senior Research Fellow	TRM Practice	hyim@kistep.re.kr
Jong Min Han	KISTEP, Research Fellow	TRM Practice	allbible@kistep.re.kr
Sungsoo Kim	KFRI, Senior Research Fellow	Agro-processing TRM Consultation	sung@kfri.re.kr
Sung Uk Chae	KIOM, Senior Research Fellow	Bio-health TRM Consultation	kendall@kiom.re.kr
Jieun Kim	STEPI, Researcher	Project Coordinator	kjieun@stepi.re.kr

4.2 Local Research Team

GS-NSTC is responsible for reinforcing the foundation for STI development, creating conditions, and enabling an environment for innovations to occur. The GS-NSTC team works in cooperation with the STEPI research team.

[Table 1-2] Local Research Team

Name	Office	Position	Contact Information
Has Bunton	GS-NSTC	Secretary General	hasbunton@hotmail.com
Ung Viseth	GS-NSTC	Deputy Secretary General	viseth_ung@yahoo.com
Nhem Solyvann	GS-NSTC	Department Director	solyvann.gsntc.kh@gmail.com
Neth Vansitha	GS-NSTC	Bureau Chief	nethvansitha2014@gmail.com
Pann Visal	GS-NSTC	Department Deputy Director	pannvial@gmail.com

Name	Office	Position	Contact Information
Tan Reasmey	ICT	Head	rtan@itc.edu.kh
Yin Chansothy	MAFF	Deputy Director	chansothy_yin@yahoo.com
Phon Reno	MAFF	Office Chief	ai_reno@yahoo.com
Sok Phal	GS-NSTC	Department Director	sokphal7778@gmail.com
Fidero Kouk	ITC	Dean	fidero@itc.edu.kh
Neth Raksmei	GS-NSTC	Deputy Director	smeyneth@gmail.com
Sek Sokna	GS-NSTC	Bureau Deputy Chief	sokna168@gmail.com
Sann Sary	MPTC	Undersecretary of State	-
Buoy Somethea	GS-NSTC	Department Director	bsomethea@gmail.com
Chan Pheakdey	MPTC	Department Deputy Director	-

5. Project's Main Activities

5.1 Activity 1: Preparation of the Project

From January to March 2019, the STEPI team conducts literature review related to the implementation of Cambodia's STI policy. Any relevant documents on the national STI of Cambodia to enhance understanding should be provided prior to Activity 2. A project program for the kick-off meeting and first workshop will be developed based on the project concept paper and shared with GS-NSTC for coordination.

5.2 Activity 2: Kick-off Meeting and Field Study

- The STEPI team visited Cambodia from March 25 to 29, 2019 for a kick-off meeting and a field study.
- The workshop was held to share the experiences of Korea in the identified topics. The GS-NSTC counterpart gave a presentation on the STI Policy of Cambodia in order to promote in-depth understanding on the situation of Cambodia. Afterward, the expectation of the project was discussed by identifying areas of interest by GS-NSTC in order to build consensus between GS-NSTC and the STEPI team on the expected results and outcome.
- Key STI stakeholders were interviewed to provide consultation on the implementation of the STI policy and analyze related issues. The relevant institutions and key actors including the ministries, public organizations, academe, private sector entities, and research institutes were identified by GS-NSTC. Meeting arrangements were also made by GS-NSTC. Each conducted interview was approximately 1.5 hours long, beginning with an introduction of the organization including its role in the STI ecosystem. Interview questions were provided one week prior to the interview to allow time to prepare the answers in advance, preferably in the form of a presentation.

5.3 Activity 3: Capacity-building Workshop in Korea.

- Based on the previous findings on Cambodia, the STEPI team held a capacity-building workshop to share the best practices of Korea and to practice the technology roadmap as an implementation strategy of the STI policy. STEPI provided guidelines for the preparation of the capacity-building and training workshop two weeks before and identify the relevant experts in Korea.
- 10 delegations more or less from GS-NSTC and relevant institutions were invited to Korea from May 20 to 25, 2019. Two- or three-day workshop was organized to share the best practices of Korea and to practice the technology roadmap. They visited the relevant Korean organizations and sites, including research institutes, for the on-site observation of their field of interest.

5.4 Activity 4: Policy Recommendation on Drafting the Implementation Plan and pilot TRM of Cambodia

A draft of the Cambodian implementation plan and the pilot TRM of STI policy were reviewed by the research team for progress assessment.

5.5 Activity 5: Final Workshop

A final workshop was held in October 2019 for GS-NSTC to present the developed draft implementation plan of STI Policy and pilot TRM for comments from the STEPI team as well as Cambodian stakeholders. Stakeholders from the public and private sectors, academe, and research institutes involved in STI policies were invited to the event, which will disseminate the details of the program. The STEPI research team delivered comments on draft policies, and the final report on the consultation will be distributed. Additionally, evaluation interviews are to be conducted to assess the relevance and sustainability of recommendations and to build on possible follow-up programs.

6. Project Schedule

The entire schedule of the project is as follows:

[Table 1-3] Project Schedule

Activities	Jan. '19	Feb. '19	Mar. '19	Apr. '19	May '19	Jun. '19	Jul. '19	Aug. '19	Sep. '19	Oct. '19	Nov. '19	Dec. '19
Project Identification												
Literature Review												
Kick-off Meeting and Workshop												
Capacity-building Workshop												
Interim Report												
Comments on the Progress												
Final Dissemination Workshop												
Final Report												

7. Project Outputs

STEPI provided the following deliverables:

- **Deliverable 1:** Project Proposal (March 25, 2019)
- **Deliverable 2:** Program Agenda for Kick-off Meeting and Workshop (March 25, 2019)
- **Deliverable 3:** Guidelines for Capacity-building Workshop in Korea (two weeks before the Workshop)
- **Deliverable 4:** Presentations and Training Materials for the Capacity-building Workshop (end of June or early July 2019)
- **Deliverable 6:** Interim Report (end of July 2019)
- **Deliverable 7:** Presentation Materials (October 2019) and Final Report (December 2019)



CHAPTER 2

Sharing Korea's Experiences in Technology Roadmap

Policy Consultation on National STI Policy
Development and STI System Management for
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Chapter 2. Sharing Korea's Experiences in Technology Roadmap

1. Technology Roadmaps of Korea

1.1 Introduction

Technology roadmap is a method of setting technology development goals as necessary to meet the demand of future markets and customers, including planning the R&D path in time series by providing answers to why, what, how, and when (Yim, 2012). It helps in decision making on the identification, selection, and development of an optimal technology alternative. Technology roadmap has been established extensively at the corporate level; given the growing influence of a country's scientific technology capabilities on national competitiveness, however, it is increasingly important to predict future technological changes and to select and nurture promising technologies at the national level.

The National Science and Technology Council of Korea describes the national technology roadmap as a methodology that provides a guideline for national S&T policies by sharing the goals and strategies of the industry, academe, and government agencies and forming consensus on key technologies that meet national demands. Established in 2002, the "National Technology Roadmap (NTRM)" can be considered a first attempt at a national science and technology roadmap for Korea. It presents the 5 major visions of S&T development and 99 key technologies to achieve these visions and describes a strategy for pushing forward the R&D for 10 years (2003-2012). It had been widely used as a basic

reference on the various science and technology R&D strategies and technology roadmaps produced thereafter. NTRM's 99 key technologies were reflected to the "First Basic Science and Technology Plan," 80 technologies in the "10 Next-Generation Growth Engine Projects (2004)," and characterization technologies in "Total Roadmap (2006)" (National Science and Technology Council, 2002).

2. Major Technology Roadmaps of Korea

2.1 National Critical Science and Technology Strategic Roadmap 2014-2023

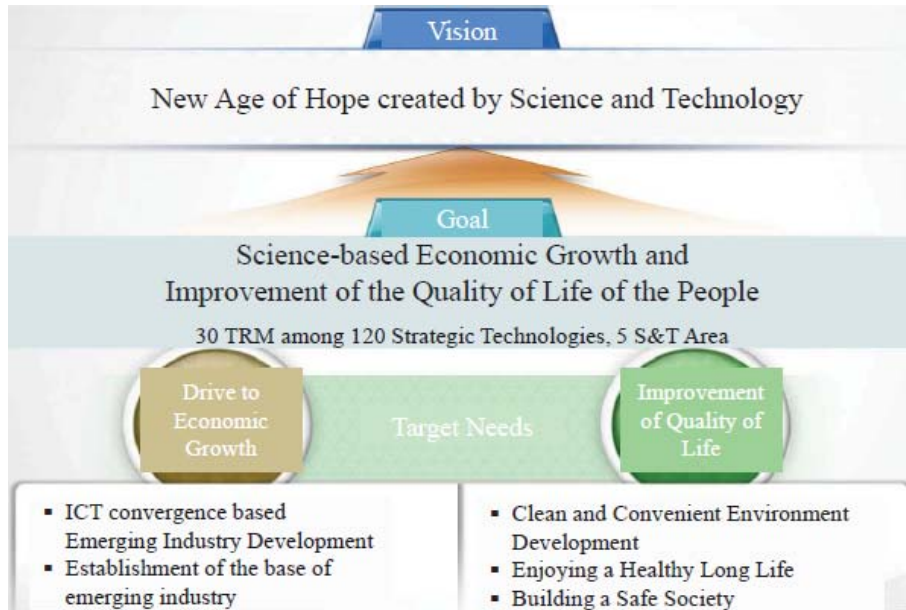
The National Critical Science and Technology Strategic Roadmap was established as a pan-ministerial strategic roadmap in 2014 in accordance with the “Third Framework for the Basic Plan on Science and Technology.” With Korea's entry into an ultra-aging society becoming faster and the potential economic growth rate possibly dropping below 1 percent after 2030, it was necessary to pre-respond to acceleration in global S&T development and rapid changes in the environment at home and abroad and to enhance investment effectiveness by strategically planning R&D. Recognizing the importance of the technology roadmap, the Korean government already established the NTRM in 2002 and the Total Roadmap in 2007, but they have not been effectively utilized due to the lack of regular modification and supplementation. The “Third Basic Plan of Science and Technology” was established in 2013, and a new national technology roadmap was drawn up accordingly for the technology areas selected as critical technologies among 120 national strategic technologies. Starting with the launching of the steering committee in July 2013, the draft document was prepared in December, and the government announced it in April 2014 following consultation with related ministries and public hearings in January and March 2014.

[Figure 2-1] Planning History of the National Critical Science and Technology Strategic Roadmap

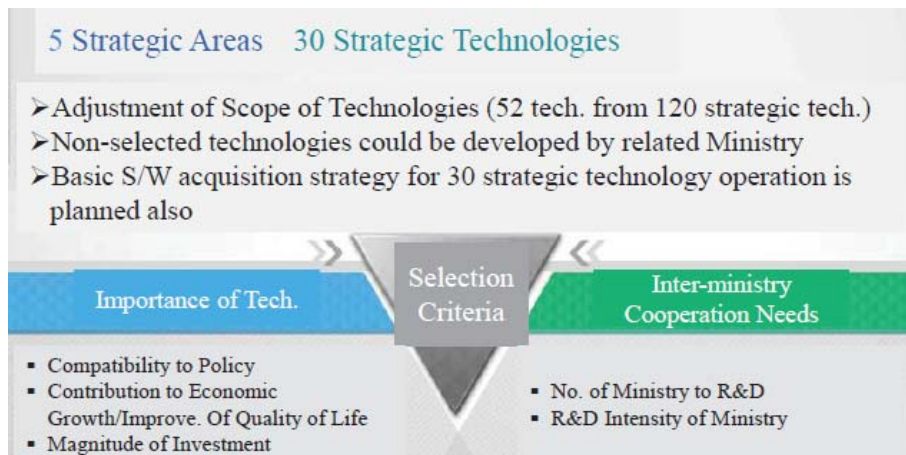


As shown as [Figure 2-2] and [Figure 2-3], under the vision of “a new era of hope opened with creative science and technology,” the roadmap aims to revive the economy based on science and technology and improve the quality of people’s lives. To realize this, the roadmap select two major fields -- driving economic sustainable growth and contributing to the improvement of quality of life -- and 5 areas of emerging industry development based on ICT convergence, expanding the base of the future new industry, developing a clean and convenient environment, realizing the age of health and longevity, and building a safe society. In addition, 30 specific key technologies were chosen considering the importance of the strategic technology (degree of correspondence with the national agenda, level of contribution to economic revival, and enhancement of people’s quality of life and ministerial investment in national strategic technology) and necessity of pan-ministerial cooperation (number of R&D participating ministries by technology and R&D intensity of ministries by technology).

[Figure 2-2] Vision, Goal, and Investment Area of the National Critical Science and Technology Strategic Roadmap



[Figure 2-3] Selection of Strategic Technologies of the National Critical Science and Technology Strategic Roadmap

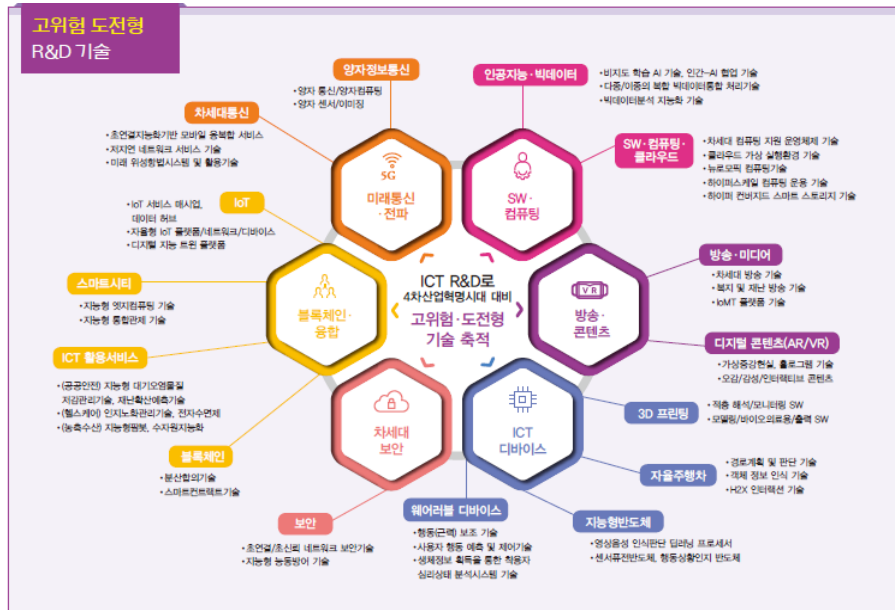


2.2 ICT R&D TRM 2023

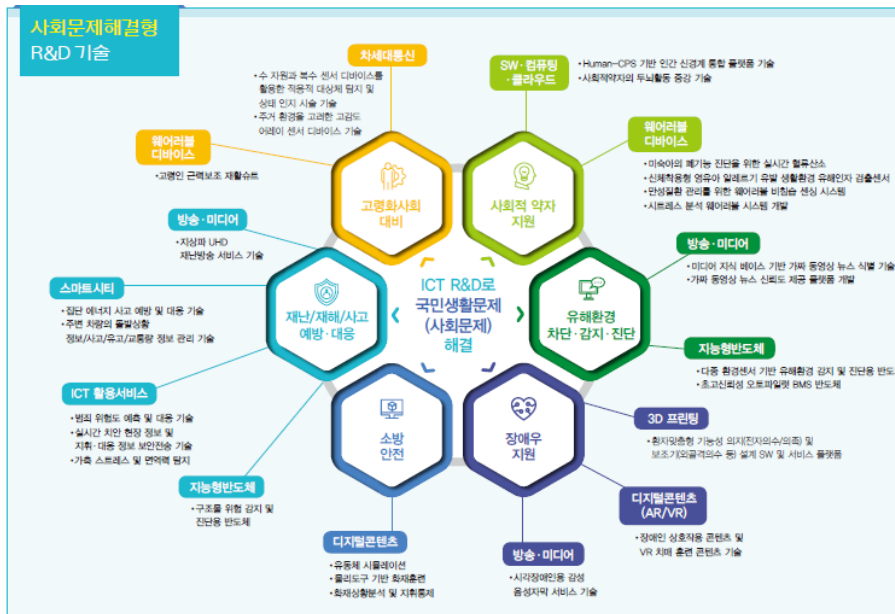
ICT R&D TRM 2023 was established as a mid- and long-term roadmap to establish strategies for technology development and set the key technologies for realizing an “ICT R&D Innovation Strategy” and discovering new growth engines by 2023. Specifically, this roadmap gave three directions of implementation: technologies: high risk and challenging R&D for accumulating technologies from a long-term perspective; social problem-solving R&D for contributing to the improvement of quality of life; and collaboration between science and ICT sectors for basic resource research. Four criteria are considered in selecting the key technologies: first, technologies requiring government support; second, high-risk and challenging field; third, technologies that must be developed by the government; lastly, technologies that can contribute to solving the social problems related to the six major public demand areas of city, transportation, welfare, environment, safety, and national defense.

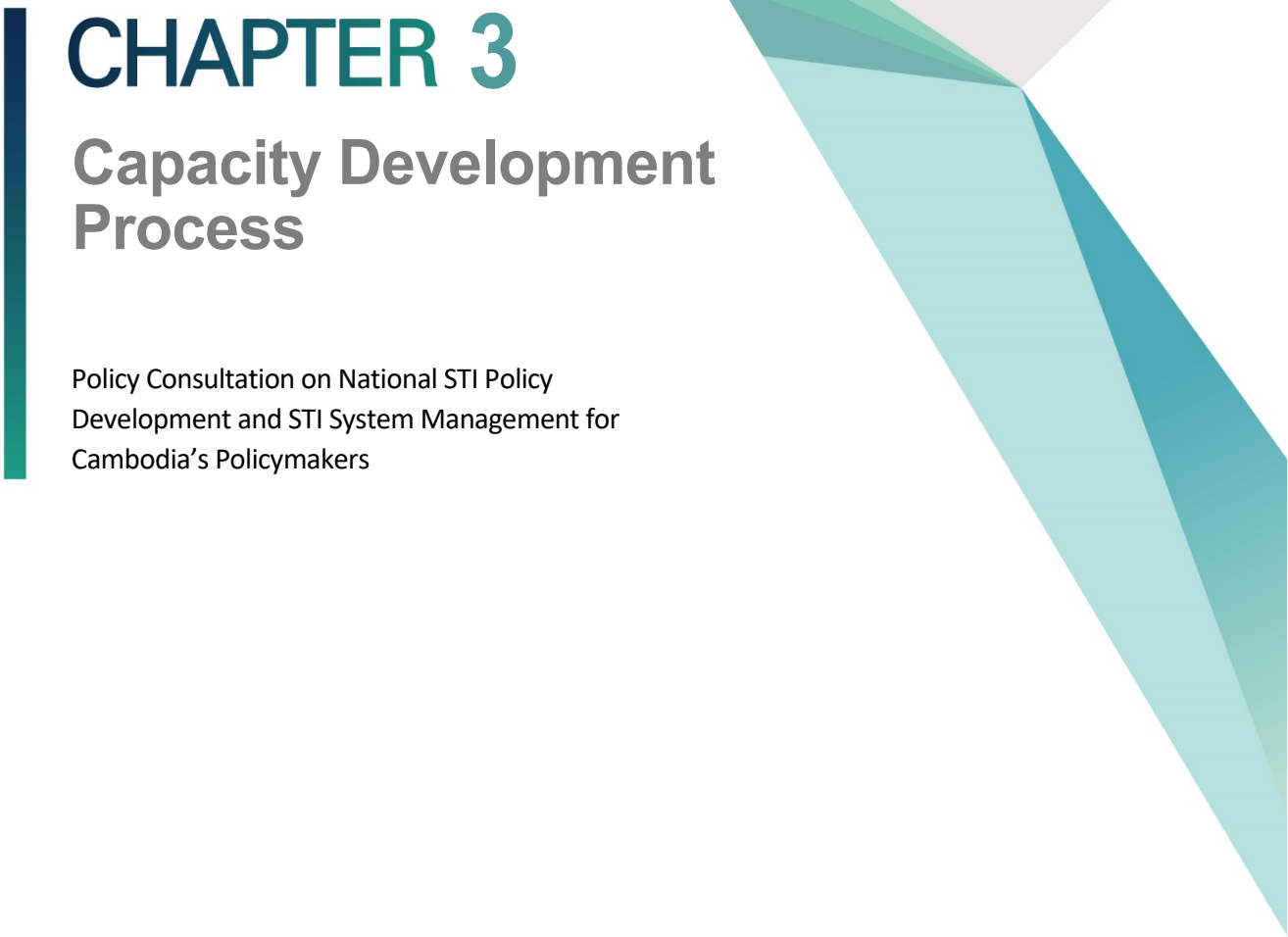
The roadmap chose 15 technologies that belong to the major areas of government policy - such as innovative growth engines and major tasks for responding to the fourth industrial revolution -- based on six major areas of ICT R&D: future communication, transmission, SW and AI, broadcasting and contents, devices, block chain and convergence, and next-generation security. Figures 2-4 and 2-5 summarize the results of the ICT R&D TRM 2023, which provides the details of high-risk and challenging R&D and social problem-solving R&D in each of the six major technology areas.

[Figure 2-4] Summary of ICT R&D TRM 2023: High Risk & Challenging R&D Technology



[Figure 2-5] Summary of ICT R&D TRM 2023: Social Problem Solving R&D Technology



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CHAPTER 3

Capacity Development Process

Policy Consultation on National STI Policy
Development and STI System Management for
Cambodia's Policymakers



Chapter 3. Capacity Development Process

1. Kick-off Workshop and Field Visit

1.1 Introduction (objectives, team, schedule, etc.)

1.1.1 Objectives

In response to the demand for international cooperation on Korea's science and technology innovation development model, Korea's economic growth with a science and technology innovation growth base.

Development of national science and technology innovation policy for Cambodian policymakers and implementation of policy consulting related to system construction.

1.1.2 Team

The research team will be responsible for the kick-off meeting and field visit to Cambodia. They will conduct a literature review related to the STI system and policy of Cambodia. Any related documents on the national STI of Cambodia to enhance understanding should be provided prior to the future Capacity Development Workshop.

Name	Organization	Position & Area of Expertise	Contact Information
Jungwon Lee	STEPI, Senior Research Fellow Project Manager (PM)	Innovation Ecosystem	leejw@stepi.re.kr
Sun-Hark Bong	TBNA, CEO	Technology Management	sunhark@tbna.co.kr
Dong Hoon Oh	Wisen Global, Partner	Technology Roadmap	oh.donghoon@gmail.com
Jieun Kim	STEPI, Researcher	Project Coordination	kjieun@stepi.re.kr

1.1.3 Schedule

[Table 3-1] Kick-off Workshop and Field Visit Schedule

Date	Time	Description	Remarks
March 25 (Mon.)	18:40	Departure from Incheon International Airport Incheon, Korea → Phnom Penh International Airport (Korean Air, KE 0689/Dr. Jungwon Lee, Ms. Jieun Kim)	Incheon Airport
	22:10	Arrival in Phnom Penh International Airport Move to Hotel and Check-in (TBD)	Phnom Penh Airport
March 26 (Tue.)	10:00 – 12:00	Preview of the 2019 Program and Future Plan	
	12:00 – 14:00	Lunch	-
	14:00 – 16:00	Embassy of the Republic of Korea in Cambodia	

Date	Time	Description	Remarks
	18:40	Departure from Incheon International Airport Incheon, Korea → Phnom Penh International Airport (Korean Air, KE 0689/External Experts)	-
	22:10	Arrival in Phnom Penh International Airport Move to Hotel and Check-in (TBD)	-
March 27 (Wed.)	10:00 – 12:00	Korea International Cooperation Agency (KOICA) Cambodia Office	KOICA Cambodia Office
	12:00 – 13:00	Lunch	
	14:00 – 17:00	Preliminary Meeting with GS-NSTC Introduction of the program & detailed project descriptions Brief identification of people at the meeting and their area of expertise and responsibility Identification of policy demands and expected outcomes	GS-NSTC
March 28 (Thu.)	10:00 – 12:00	Workshop (Open Session) Open session by Korean and Cambodian representatives Presentation on the following topics: “Introduction to national strategic planning for STI development”; “Introduction to TRM”; “Korean experiences in TRM”; and “STI policy in Cambodia”	CKCC
	12:00 – 13:00	Lunch	TBD
	14:00 – 17:00	Stakeholder meeting concerning capacity building for the Technology Roadmap of Cambodia	TBD

Date	Time	Description	Remarks
March 29 (Fri.)	10:00 – 12:00	Wrap-up Meeting with GS-NSTC Preliminary observation – Result of the field study Finalization of project scope, direction, and expected activities Discussion on the follow-up activities and schedule Expected Attendees: STEPI Team and GS-NSTC staff who are involved in this program	GS-NSTC
	12:00 – 13:00	Lunch	TBD
	14:00 – 16:00	STEPI team meeting concerning the report on the workshop and field study	TBD
March 30 (Sat.)	18:00	Move to Phnom Penh International Airport Phnom Penh International Airport → Incheon, Korea (Korean Air, KE 690)	Phnom Penh Airport
	23:20	Departure	
March 31 (Sun.)	06:35	Arrival in Incheon International Airport, Korea	Incheon Airport

1.2 Field Visit (information by organization to visit)

1.2.1 Meeting with the Embassy of the Republic of Korea in Cambodia

A meeting with the Korean Embassy in Cambodia was held on March 26, 2019 to provide the overall plans for this year's Cambodian STI program. The Korean Embassy in Cambodia shared information about ODA between Korea and Cambodia. While China has expanded its official and informal support for Cambodia recently, the demand of a new ODA project for Cambodia has increased rapidly since President Moon Jae-in's visit to Southeast Asia and the announcement of a "New Southern Policy." Note, however, that there was no specific discussion on science and technology-related cooperation between the leaders of the two countries.

The Korean Embassy in Cambodia and STEPI discussed ways of enhancing the effectiveness of Korea's credit assistance and free handout projects in Cambodia. The participants of the two countries should make efforts to maintain the network even after the project is over. The actual demand of each project should be identified to avoid redundancy between domestic and overseas projects. In addition, in terms of price competitiveness of Korea's science and technology ODA compared to China, it would be much more advantageous to focus on advanced science and technology rather than construction and infrastructure.

1.2.2 Meeting with KOICA's Cambodia Office

On March 26, 2019, the research team had a meeting with Myung Jin Kim, the assistant country director of KOICA's Cambodia office. The STEPI team explained the Cambodian STI policy as the main performance of last year's project and gave an introduction of this year's project plan. The assistant director introduced the current and future cooperation plans between KOICA and Cambodia.

According to her, the Cambodian government prioritizes fostering industry over science

and technology, which require long-term investment since the top priority of the Cambodian government is economic development. Moreover, it seems difficult to open a new policy window given the current political situation of Cambodia. This point is parallel with the STEPI research team's opinion about a gap between policy establishment and actual enforcement. It is important to draw strategies that can move the hearts of the budget decision makers from policy consultation because their will is a critical factor for having the policy consultation result realized. The implementation of Rectangular Strategy must be supported by science and technology; therefore, policymakers should recognize its importance and have the willingness to secure budgets for it.

In addition, as an assistant director of the project implementation organization, she asked the STEPI research team to draw specific policy implementation ideas along with policy suggestions so that the policy consultation report would not remain a mere book.

1.2.3 Meeting with the General Secretariat of the National Science and Technology Council (GS-NSTC)

The research team explained the importance of policy-related officials' capacity development and project plan for this year. GS-NSTC shared the current status of the approval process of national STI policy established by policy consultation last year.

The research team and GS-NSTC discussed the project implementation plan related to TRM establishment. The aim of this year's project is to establish a pilot TRM for practice as a preliminary step to national TRM GS-NSTC establishment. GS-NSTC chose ICT, agro-processing, and bio-health as the theme of the pilot TRM. Agreeing on the importance of those three fields, the STEPI research team suggested narrowing down the technology area to a very specific technology or item. Since TRM is aimed at developing specific technology development strategies, it is very important to select a specific target technology, which is also important to invite experts suitable for future policy advice.

Next, the STEPI research team and GS-NSTC discussed organizing each TRM team member and the final output of the project. The STEPI research team emphasized that making TRM demands a high degree of expertise and asked that not only policymakers but also science and technology experts in the private sector, such as universities and corporations, be invited as members of each TRM team. In addition, both sides agreed that the pilot TRM report would be written in English, and that Korean experts' consultation would be provided in English as well.

Regarding the major timeline for this year, both sides agreed on holding the invitation workshop for capacity development in Seoul in May and the final dissemination workshop in Phnom Penh in October. Especially, since the establishment of TRM is a very practical task, the invitation workshop participants were limited to the actual participants in the pilot TRM establishment.

[Figure 3-1] Kick-off meeting, Seminar, and Field Study



2. Capacity Development Workshop

2.1 Introduction

2.1.1 Objective

The workshop was designed to support Technology Roadmap (TRM) capacity building for officials of the relevant ministries including GS-NSTC and experts of each technology field by sharing the method of building a technology roadmap.

This workshop provided Cambodian policymakers and experts with a good opportunity to use Korean knowledge and know-how as basis for reflection for formulating Cambodia's pilot TRM. Furthermore, the Cambodian research team had a chance to visit key STI-related organizations in Korea in order to learn best practices on the history, roles, functions, and operations.

2.1.2 Research Team

[Table 3-2] Capacity Development Workshop Research Team

Name	Organization	Position & Area of Expertise	Contact Information
Jungwon Lee	STEPI, Senior Research Fellow Project Manager (PM)	Innovation Ecosystem	leejw@stepi.re.kr
Hyun Lim	KISTEP, Senior Research Fellow	TRM Practice	hyim@kistep.re.kr
Jong Min Han	KISTEP, Research Fellow	TRM Practice	allbible@kistep.re.kr
Jieun Kim	STEPI, Researcher	Project Coordination	kjieun@stepi.re.kr

2.1.3 Participants in the Capacity Development Workshop

[Table 3-3] List of Capacity Development Workshop Participants

No.	Name	Organization
1	Ung Viseth	General Secretariat of the National Science and Technology Council
2	Buoy Somethea	GS-NSTC
3	Sok Phal	GS-NSTC
4	Pann Visal	GS-NSTC
5	Yin Chansothy	Ministry of Agriculture, Forestry, and Fisheries
6	Phon Reno	MAFF
7	Neth Raksmeay	GS-NSTC
8	Sek Sokna	Ministry of Health
9	San Sary	Ministry of Posts and Telecommunications
10	Chan Pheakdey	MPTC

2.1.4 Brief Introduction of the Visiting Institution

■ Korea Institute of Science and Technology, KISTORIUM

The Korea Institute of Science and Technology (KIST) is a government-funded research institute established in February 1996 for the purpose of developing basic scientific research (original technology) of Korea and dissemination of its performance. Following the establishment of its Seoul headquarters, two branches were founded in Gangneung in May 2003 and in North Jeolla Province in January 2008. In the 1960s and 1970s, this institute had served as a think tank for industrialization; in the 1980s and 1990s, research was conducted to catch up with the advanced technologies of developed countries. Since the 2000s, it has been conducting research and development related to leading original technology.

As the first comprehensive science and technology institute, KIST opened KISTorium, a kind of museum showing the history of Korea's science and technology development in February 2006 to mark the 40th anniversary of KIST's foundation and which was reopened in 2017 after renovation. KISTORium is known to have about 25,000 visitors a year who can take a look at the history of Korea's science and technology development. This history museum features different themes by floor. Visitors can see the history of KIST, photo zone reproducing KIST's laboratory in the '70s, and scientific and technological achievements of KIST, through exhibits showing the course of science and technology development of Korea. Aside from showcasing the past of Korea and KIST, the museum also provides a special experience using future technology such as 3D display and virtual reality.

■ National Science Museum

The National Science Museum is a scientific culture facility that studies and exhibits various kinds of scientific equipment. At its center is the Astronomical Hall, which is the largest dome in Korea. One can vividly see the phenomenon of the celestial sphere and the human development of space. It also boasts of a world-class space theater. The 242 seats are available on a first-come-first-served basis. It can recreate the appearance of a night sky, the highlight being the view of the starlight at the planetarium. The Astro-vision is also a special visual device giving a magnificent, realistic view of space development. There are exhibits featuring more than 4000 items based on nature and natural science at the Permanent Exhibit Hall. There are also exhibit halls featuring Korean nature, weapons, and folk musical instruments. One can see the stages of collecting energy from nature and its use in transportation. Outside the Hall is Cheomseongdae, a weather-evaluating machine from the Goryeo period, including models of plant-eating dinosaurs as well as a steam engine. The exhibits aid in understanding the theories of science and the wonders of nature.

2.1.5 Capacity Development Workshop Schedule

[Table 3-4] Capacity Development Workshop Schedule

Date	Time	Description	Remarks
May 20 (Mon.)	23:55	Departure from Phnom Penh International Airport	
May 21 (Tue.)	07:15	Arrival in Incheon International Airport	
	10:00-10:30	Hotel Check-in	
	10:30-13:00	Rest	
	13:00-14:00	Lunch	
	14:00-15:00	Program Orientation	
	15:00-18:00	Rest	
	18:00-19:30	Dinner	
May 22 (Wed.)	09:00 – 10:50	[Lecture 1] Introduction to Technology Roadmap Korea's Experience: National Technology Roadmap	
	10:50 – 11:00	Coffee Break	-
	11:00 – 13:00	[Lecture 2] Introduction to the Roadmap Making Pilot Project (hands-on project) Step 1: Vision Building	
	13:00-14:30	Lunch & Coffee Break	
	14:30-18:00	[Lecture 3] Step 2: Environment Analysis Step 3: Strategic Products/Services	-
	18:00-19:00	Dinner	-
May 23 (Thu.)	19:00-10:50	[Lecture 4] Step 4: Key Technology	
	10:50 – 11:00	Coffee Break	
	11:00-13:00	[Lecture 5] Step 5: Charting (linkage between technologies and products/markets) and Identification of policy demands and expected outcomes	
May 23 (Thu.)	13:00 – 14:30	Lunch	
	14:30 – 16:50	Group Presentation	
	16:50 – 17:00	Coffee Break	
	17:00-18:00	Wrap-up	
	18:00	Dinner	
May 24 (Fri.)	09:30-11:00	Move from hotel to KIST	
	11:30 – 12:30	Site Visit: KIST	
	12:30 – 13:00	Move from KIST to Gwacheon	

Date	Time	Description	Remarks
	13:00-14:00	Lunch	
	14:30-16:00	Site Visit: National Science Museum	
	1700-18:00	Dinner	
	18:00-18:30	Completion Ceremony	
May 25 (Sat.)	11:00	Hotel Check-out	Phnom Penh Airport
	13:00-14:00	Lunch	
	14:00-15:00	Move to Incheon International Airport	
	19:30	Departure	

2.2 Wrap-up Discussion for the Capacity Development Workshop

2.2.1 Participants Comments on the Capacity Development Workshop

The STEPI research team collected comments from the Cambodian participants by conducting a survey on the capacity development workshop. The result showed high satisfaction with the workshop overall, but some participants gave opinions on enhancing the quality of the workshop. With regard to the whole schedule of the workshop, they thought that the workshop period is relatively short and hoped that future training programs would have longer period and more lectures.

The Cambodian participants also signified expectations for continuous cooperation with Korea, a country with long, innovative experience in science and technology development and leading technologies in ICT, agro-processing, and bio-health. Since there are several government-affiliated science and technology institutes in each field, the participants wished to cooperate with the relevant institutes in the form of MOU or joint research. They also mentioned that STEPI will have a critical role in this kind of cooperation.

[Figure 3-2] Capacity Development Workshop



Site Visit



Wrap-up and Completion Ceremony



3. Dissemination Workshop

3.1 Introduction

3.1.1 Objective

The workshop aimed to disseminate the performance of this year's project and provide advice on pilot TRM reports by experts to enhance the quality of reports and Cambodian researchers' experience in TRM formulation.

In the open session, Korean experts gave presentations about the importance of TRM establishment for national capacity building on science and technology policy implementation and shared Korean experiences in developing the fintech, nutricosmetic, and agro-processing fields with attendees. This workshop provided Cambodian policymakers with a good opportunity to use Korean knowledge and know-how as basis for reflection for formulating Cambodia's STI policy with site visits to key STI-related organizations and research institutes in Korea in order to learn their best practices on the history, roles, functions, and operations.

3.1.2 Research Team

[Table 3-5] Dissemination Workshop: Research Team

Name	Organization	Position & Area of Expertise	Contact Information
Jungwon Lee	STEPI, Senior Research Fellow Project Manager (PM)	Innovation Ecosystem	leejw@stepi.re.kr
Jei-Young Lee	STEPI, Associate Research Fellow	Fintech TRM Consultation	Jeilee@stepi.re.kr
Sungsoo Kim	KIOM, Senior Research Fellow	Agro-processing TRM Consultation	sung@kfri.re.kr
Sung Uk Chae	KFRI, Senior Research Fellow	Bio-health TRM Consultation	kendall@kiom.re.kr
Jieun Kim	STEPI, Researcher	Project Coordination	kjieun@stepi.re.kr

3.1.3 Schedule

[Table 3-6] Dissemination Workshop Schedule

Date	Time	Description	Remarks
October 1 (Tue.)	18:30	Departure from Incheon International Airport Incheon, Korea → Phnom Penh International Airport (Korean Air, KE 689/Dr. Jungwon Lee, Dr. Sung Soo Kim, Dr. Jei Young Lee, Ms. Jieun Kim)	
	22:25	Arrival in Phnom Penh International Airport Move to Hotel and Check-in	
October 2 (Wed.)	10:00 – 11:00	Korea International Cooperation Agency (KOICA) Cambodia Office	
	11:00 – 12:00	Embassy of the Republic of Korea in Cambodia	
	12:00 – 13:00	Lunch	
	14:00 – 17:00	Preliminary Meeting with GS-NSTC Brief identification of people at the meeting and their area of expertise and responsibility Confirmation on workshop preparation Discussion on future plan	
	18:30	Departure from Incheon International Airport Incheon, Korea → Phnom Penh International Airport (Korean Air, KE 689/Dr. Sung Uk Chae)	
	22:25	Arrival in Phnom Penh International Airport Move to Hotel and Check-in	
October 3 (Thu.)	08:00 – 12:00	Workshop (Open Session) Open session by Korean and Cambodian representatives Presentation on the following topics: “TRM on ICT	

Date	Time	Description	Remarks
		(Fintech)”; “TRM on Bio-Health (Vitamin)”; “TRM on Agro-Processing (Cassava)”	
	12:30 – 13:30	Lunch	
	15:00 – 17:00	Consultation meeting concerning the Technology Roadmap of Cambodia (closed) Expected participants: Members of three TRM teams, STEPI team, and Korean experts	
October 4 (Fri.)	10:00 – 12:00	Wrap-up meeting with GS-NSTC (if required)	
	12:00 – 13:00	Lunch	
	14:00 – 16:00	STEPI team meeting concerning the report on the workshop and field study	
	18:00	Move to Phnom Penh International Airport	
	23:45	Departure Phnom Penh International Airport → Incheon, Korea (Korean Air, KE 690)	
October 5 (Sat.)	06:45	Arrival in Incheon International Airport, Korea	

3.2 Field Visit (information by organization to visit)

3.2.1 Meeting with the Embassy of the Republic of Korea in Cambodia

A meeting with the Korean Embassy in Cambodia was held on October 2, 2019 to provide an introduction to this year's project progress. The Korean Embassy in Cambodia shared information about the current status related to the three fields of the pilot TRM.

First, regarding fintech, Cambodia's financial environment is rather poor; the country's monetary policy has failed to take root because people prefer US dollars more since they experienced the instability of the Cambodian currency (Riel). Due to the people's low will and awareness with regard to the use of the financial sector, only 20 percent of the people have bank accounts. Interest in utilizing ICT in the financial sector is increasing; in the current situation, however, applying ICT to the payment system is deemed most likely to be realized. In addition, it is necessary to cooperate with the national bank of Cambodia to apply fintech in Cambodia's finance sector.

Next, the Cambodian government has been trying to distribute and cultivate cassava, but its yield is not as abundant as expected. Searching the current status of Korean corporations regarding agro-processing and bio-health -- such as Hyundai and MH Bio in Cambodia -- is recommended.

The STEPI research team explained that the second phase of the project for establishing the national TRM would be launched next year. Emphasizing the importance of national TRM establishment was urged when a newly appointed ambassador visits the Cambodian government since science and technology development is important for Cambodia's socioeconomic growth and national TRM is critical as an implementation strategy of STI policy.

3.2.2 Meeting with KOICA's Cambodia Office

A meeting with the Korean Embassy in Cambodia was held on October 2, 2019 to discuss with newly appointed associate director Yeakyung Ji the project progress and a new project that will be started next year. The associate director shared information on the policy trend of Cambodia and the current project status of KOICA's Cambodia office.

She mentioned that the human resource training policy is a top priority in every field to supply well-educated, high-quality human resources. The number of projects of the Cambodia office is lowest among the new southern countries despite the increased demand for cooperation with Cambodia since the launching of the new southern policy. Therefore, the Cambodia office is trying to expand its project, including finding new project items and expanding its budget.

Regarding the 3 fields of pilot TRM, especially fintech, she noted that KOICA has worked with Cambodia's central bank to establish a payment system, and that it was supposed to open the service within the year, but it is uncertain whether this would be distributed smoothly due to low demand from the public. There is no specific plan for science and technology-related cooperation, but she expected to get some ideas for new projects from the policy consultation project of STEPI.

3.3 TRM Report Consultation by Korean Experts

3.3.1 Fintech

Dr. Jei Young Lee gave advice on the fintech TRM report. He asked Cambodian researchers to conduct an accurate investigation on the number of Internet and mobile users and the status of fintech corporations in Cambodia. In addition, the strategic priorities for fintech products and services should be derived from evaluations considering the economic effect, strategic importance, and potential by experts; based on this, researchers can choose more specific, realizable target products and services.

In terms of technology, the electronic tax settlement system that the Cambodian government wants to implement is equivalent to payment and settlement services using fintech, and it is important to have deep understanding of the key element technologies (transmission/payment devices, data collection and analysis technologies, information security technologies, etc.) to realize the system. The TRM should show a future plan of how to develop simple payment technology having reliability, stability, and ease with these element technologies.

3.3.2 Bio-health

Dr. Sung Uk Chae gave advice on the bio-health TRM report. He pointed out that vitamin tablet processing technology, which is the main topic of the submitted report is too narrow and specific to reflect Cambodia's bio-health problem and solve it. Thus, it was suggested to establish the system and database at the national level for managing Cambodia's resources before taking a focus on the vitamin. First, it is necessary to establish a national life-cycle health system to develop the technology for improving infant malnutrition. Medication development is also an industry in which the human resource and S&T development should be preceded by a long-term national strategy.

In addition, considering the current status of Cambodia's biotechnology, it was proposed to prioritize the establishment of a systemic database about traditional medicine and native botanical species. Despite sufficient and various native medicinal plants in Cambodia, it is not utilized strategically. Cambodia should move beyond simple cultivation to the production of special products through discovering effective materials, and introduce an open innovation system and promote international joint research to complement the infrastructure for basic research. In this way, Cambodia can promote national interests preventing infringement on ownership of natural and biogenetic resources from other countries under the Nagoya protocol.

3.3.3 Agro-processing

Dr. Sungsoo Kim had a positive opinion of the overall contents of the agro-processing TRM report. He emphasized the importance of gradual and innovative action plans and the application of appropriate technology on each stage of agro-processing to realized the technology roadmap as intended. Cambodia has abundant agricultural production while there is a lack of technology, infrastructure, and professionals. Thus, it has to be started from the small farm unit of the processing plant and enlarge to the small-medium size of the industrial processing plant and then the large scale of the total processing plant. Not only the size of plants but also the appliance of appropriate technology on agricultural products is important to develop agro-processing in Cambodia. Depending on the quality of the agricultural product, distribution technology is required to export the high-quality agricultural product at the national or global market, and food processing technology is required for the reproduction of relatively low-quality as high-value-added products.

4. Pilot TRM Reports of Cambodia

4.1 Introduction

As the 2019 project, the STEPI research team has conducted policy consultation for establishing the technology roadmap of Cambodia with GS-NSTC. After target technologies were chosen -- fintech, bio-health, and agro-processing -- ten Cambodian trainees visited Korea to learn how to establish TRM in May. After they got back to Cambodia, GS-NSTC arranged three TRM teams whose members are from diverse relevant ministries and academic organizations, including the trainees who participated in the workshop in May. The Cambodian researchers submitted the first draft of pilot TRM reports on 17 September, and the STEPI research team reviewed the reports and prepared comments to develop each report. After the presentations by Cambodian researchers at the dissemination workshop on 3 October, the STEPI research team gave feedback about the reports. As a result of the whole process, the revised reports were submitted on 19 October.

Below is the list of the Cambodian researchers who participated in the project.

[Table 3-7] List of Cambodian Researchers

Team	Name	Organization	Position
ICT	Sann Sary	MPTC	Undersecretary of State
	Buoy Somethea	GS-NSTC	Department Director
	Chan Pheakei	MPTC	Department Deputy Director
	Rith Sela	GS-NSTC	Department Deputy Director
	Mam Piseth	GS-NSTC	Vice Bureau Chief
	Sarot Srang	ITC	Department Chief
	Sok Kimheng	ITC	PhD Researcher
Bio-Health	Sok Phal	GS-NSTC	Department Director
	Kuok Fidero	ITC	Dean
	Tourt Sodanea	Private Sector	Nutritionist
	Ung Porsry	-	Researcher
	Sek Sokna	MoH	Deputy Chief of Bureau
Agro-Processing	Pann Visal	GS-NSTC	Department Deputy Director
	Phon Reno	MoAFF	Chief of Bureau
	Tan Raksmeay	ITC	Head
	Tuy Vitou	MoAFF	Officer

4.2 Pilot TRM Reports (draft)

4.2.1 FinTech

First Draft



KINGDOM OF CAMBODIA

Nation - Religion - King

Cambodia's Technology Roadmap for ICT

Financial Technology

October

2019

Executive Summary

Cambodia's financial sector continued to expand robustly last year, with outstanding loans and deposits as the end of the year maintaining double-digit growth and bad loans remaining low. The financial industry maintained solid growth, and it has been expanding its area of coverage nationwide, establishing itself as a major contributor to national economic growth. The financial industry consists of 43 commercial banks, 14 specialized banks, 7 microfinance deposit-taking institutions, 73 microfinance institutions, and 302 rural credit institutions.

Through the rapid expansion of the financial industry and the economic diversification of the Royal Government of Cambodia at the same time, including preparing for the digital economy and responding to the Fourth Industrial Revolution, the development of the financial and banking sectors is being spurred, including the lifestyle change of people. "The industry is being modernized parallel to the growth of financial technology, and it has seen upgrades in financial infrastructure and electronic payment services." Fintech is steadily becoming one of the most promising digital economy sectors, and it continues providing financial services to the Cambodian people.

Part 1: Introduction

Background:

For the past two decades, Cambodia has undergone a significant transition, reaching lower middle-income status in 2015. Authorities aspire to attain upper middle-income status by 2030. Driven by garment exports and tourism, Cambodia's economy has recorded an average growth rate of 7.7% between 1995 and 2018, making it one of the fastest-growing economies in the world. As global demand peaked in 2018, economic growth is estimated to be 7.5% compared to 7% in 2017 and is expected to remain robust over the medium term¹.

Science, Technology, and Innovation (STI) have emerged as the major drivers of national development globally. Nonetheless, the responsibilities of Science, Technology, and Innovation are crucially relevant interaction with cross-cutting issue from various organizations and institutions. In 2014, the Royal Government of Cambodia (RGC) established through Royal Decree NS/RKT/0916/1136 dated 08 September 2014 the National Science and Technology Council (NSTC), chaired by **Prime Minister Samdech Akka Moha Sena Padei Techo Hun Sen**. In 2015, RGC established the General Secretariat of the National Science and Technology Council (GS-NSTC) as a secretary of NSTC to coordinate and promote the development of S&T in Cambodia.

The Royal Government of Cambodia is moving in the right direction. As announced by the Ministry of Posts and Telecommunications in 2018, the government aims to transform the country into a predominantly digital economy by 2023 with the existing Cambodia ICT Master Plan 2014-2020, Law on Telecommunications, and Policy on Telecom/ICT Development 2020. The country is taking steps to develop its S&T sector including ICT infrastructure. The Ministry of Education, Youth, and Sports is promoting STEM education from primary school to post-secondary levels. This is in keeping with IDP 2015-2025, which emphasizes the need to embrace modern S&T and invest in R&D and highlights the

¹ <https://www.worldbank.org/en/country/cambodia/overview>, 2019.

linkages between S&T and other priority sectors. Cambodia's TVET Policy 2017-2025 sets four goals: to improve TVET quality to meet national and international market demand; to increase equitable access to TVET for employment generation; to promote public-private partnerships and aggregate resources from stakeholders to support the sustainable development of the TVET system; and to improve the governance of the TVET system. The Cambodia National Science and Technology Master Plan 2014-2020 aims to establish the S&T foundation, to secure R&D capability, to create a sound S&T environment, and to improve the core industry capacity. All in all, extra effort and policy coordination and visionary, inspiring, and determined leadership at all levels are still needed for industrialization.²

Furthermore, innovative technologies have been developed and applied to support the provision of financial services. Over recent years, however, the range of financial innovations, the prevalence of their use, and their pace of evolution have increased substantially. In light of these developments, there is a challenge for regulators and supervisors to allow opportunities presented by FinTech to be realized fully and properly without undermining consumer protection as well as the level playing field, integrity of the financial markets, and stability of the financial system taken as a whole.³

Cambodia's financial sector continued to expand robustly last year, with outstanding loans and deposits maintaining double digit growth as of the end of the year and bad loans remaining low. The financial industry maintained solid growth, and it has been expanding its area of coverage nationwide, establishing itself as a major contributor to national economic growth. The financial industry consists of 43 commercial banks, 14 specialized banks, 7 microfinance deposit-taking institutions, and 73 microfinance institutions. The combination of banks and microfinance institutions grew by 19 per cent to \$24.5 billion by the end of last year, with deposits increasing by 15.3 per cent to \$22.1 billion. The industry's total assets grew to \$40 billion at the end of last year, increasing 19.4 per cent from the end of 2017. "The industry is being modernized parallel to the growth of

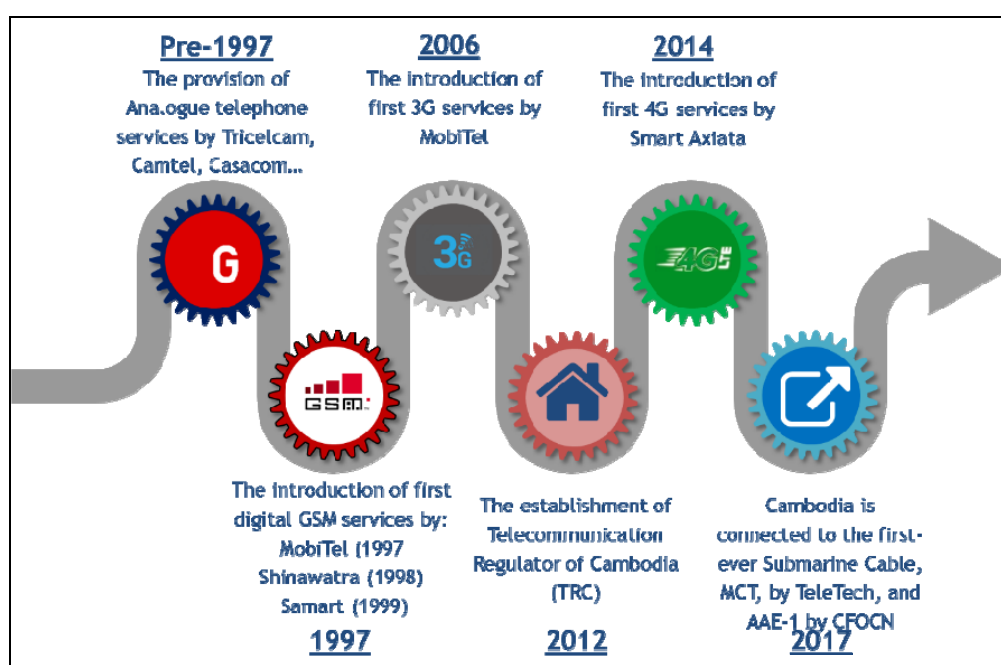
² Cambodia's Outlook Brief, CDRI, 2018.

³ European Banking Authority (EBA) Roadmap on FinTech, 2018.

financial technology, and it has seen upgrades in financial infrastructure and electronic payment services,” “These factors have reduced operating costs and increased efficiency, supporting the increase of business and investment activities.”⁴

Current industry: its products, customers, suppliers, and manufacturing processes

Figure 1. Cambodia’s Telecom Evolution



Source: MPTC

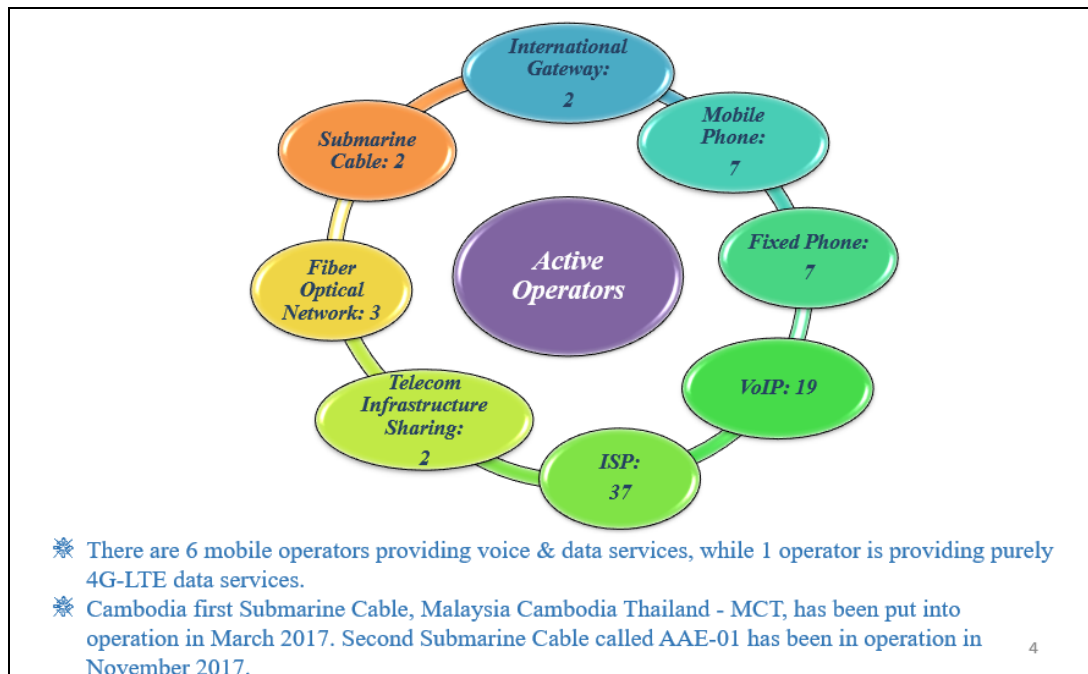
As indicated in Figure 1, **2G** was first introduced in 1992 but was first introduced in Cambodia in 1997. It is the second-generation cellular telephone technology and the first to use the digital encryption of conversations. 2G networks were the first to offer data services and SMS text messaging, but their data transfer rates are lower than those of their successors.

⁴ National Bank of Cambodia (NBC) report, 2018.

3G networks succeed 2G ones, offering faster data transfer rates; they are also the first to enable video calls. This makes them especially suitable for use in modern smartphones, which require constant high-speed Internet connection for many of their applications. 3G was first introduced in Cambodia in 2006.

4G is the fourth generation of mobile phone communications standards. It is the successor of 3G, providing ultra-broadband Internet access for mobile devices. The high data transfer rates make 4G networks suitable for use in USB wireless modems for laptops and even home Internet access. 4G was first introduced in Cambodia in 2014, and Cambodia will launch the first 5G in late 2019.

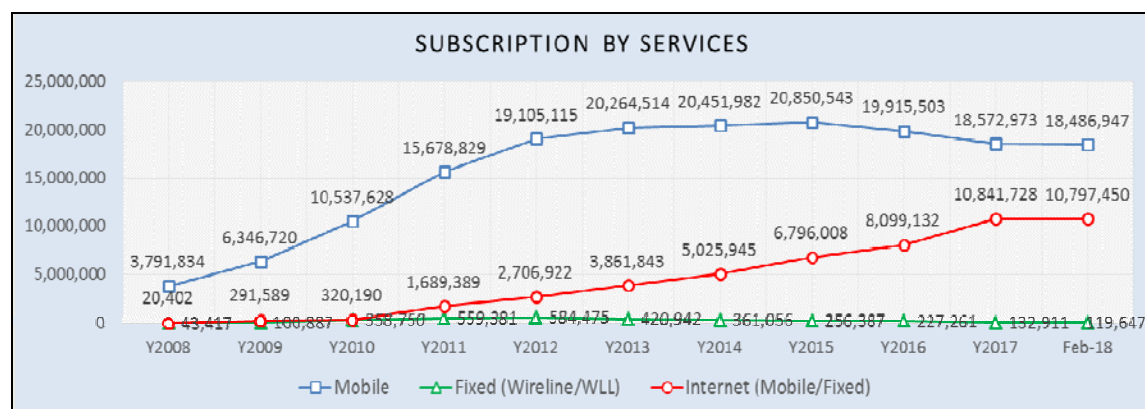
Figure 2. Telecom Operator Licenses, Active Telecommunication Operators, 2018



Source: MPTC

Figure 2 shows the telecom operator licenses and active telecommunication operators in 2018. There are 7 mobile operators providing voice and data services. There are 37 Internet Service Providers available anywhere in the urban area, and they continuously expand their service rapidly. There are 3 fiber optical networks that provide fast Internet/Intranet connection. There is also submarine 2, which was operated in March and November 2017.

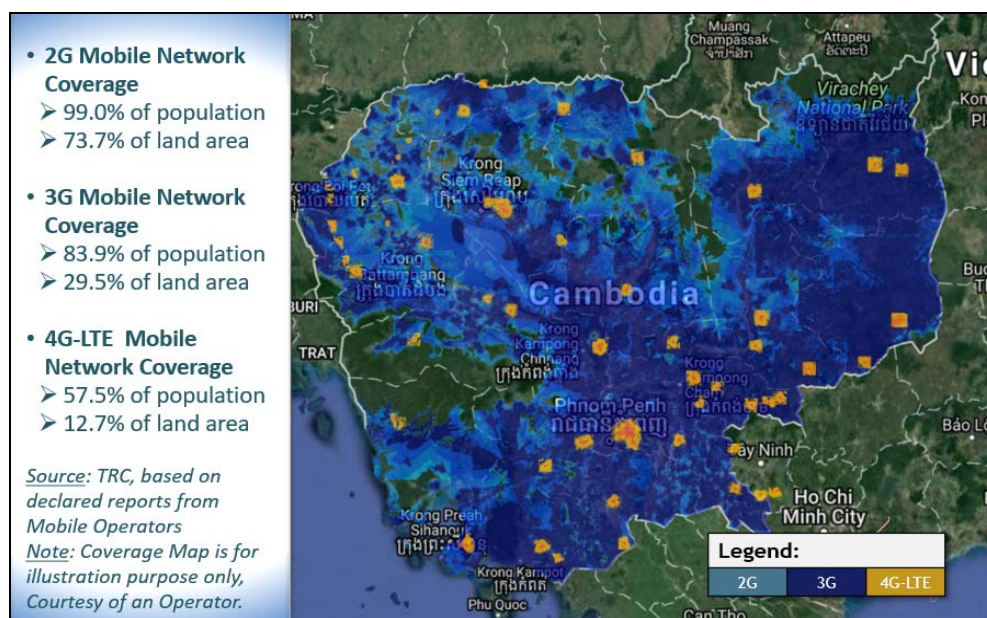
Figure 3. Telecom Subscription 2008-2017



Source: MPTC

Figure 3 shows the Telecom Subscription trend from 2008 to 2017. There was a rapid increase in mobile subscription from 3 million in 2008 to almost 20 million in 2018, with the total population in 2018 accounting for 15,288,489.⁵

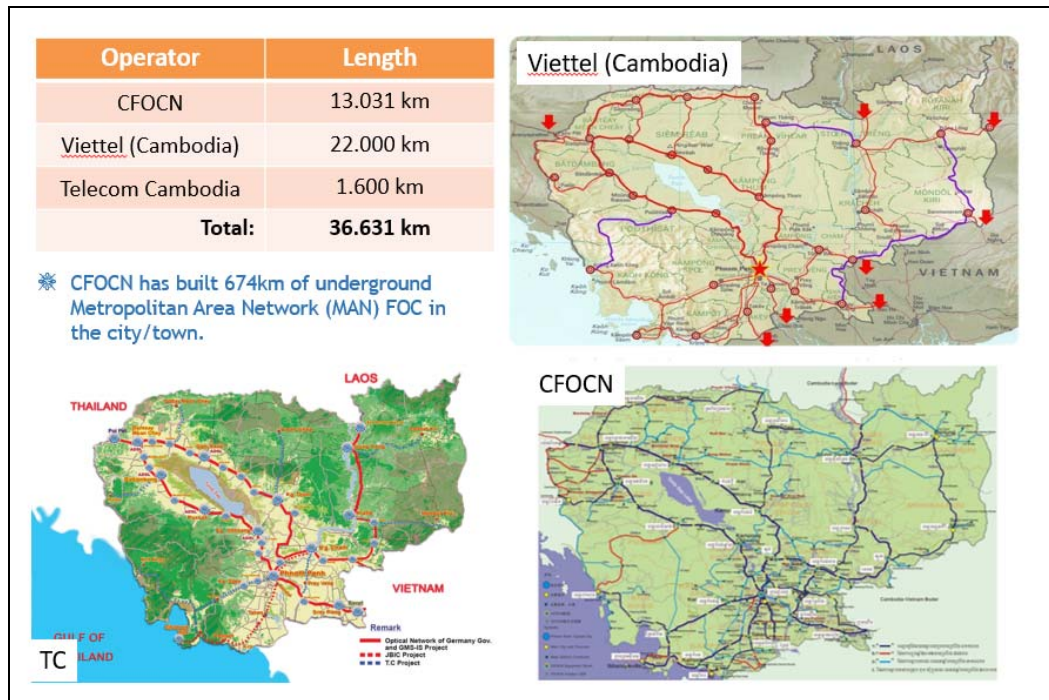
Figure 4. Mobile Service Coverage 2018



Source: MPTC

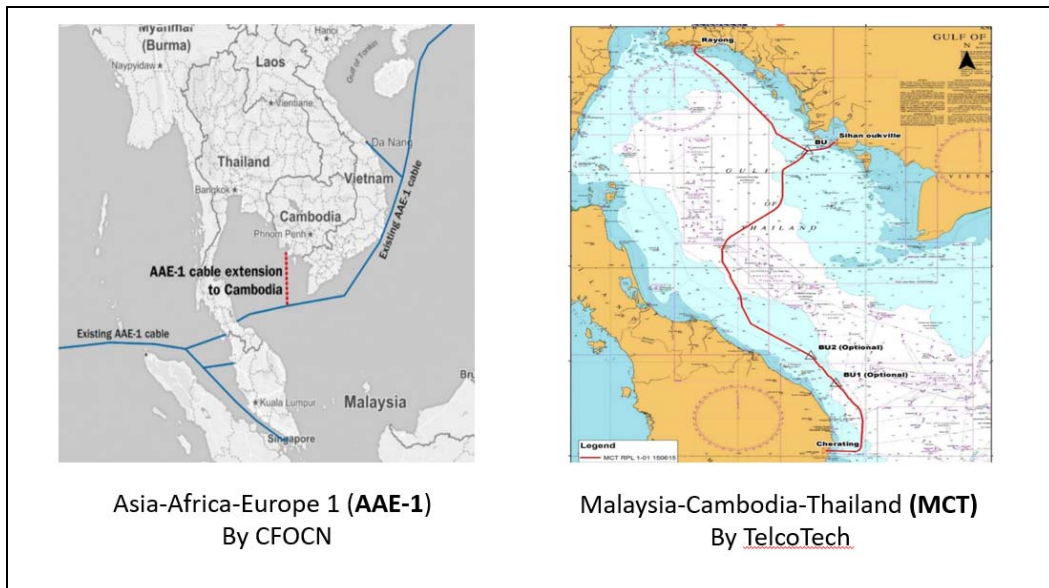
⁵ National Institute of Statistics, Ministry of Planning, Provision Population Total, June 2019.

Figure 5. F.O. Network Infrastructure 2018



Source: MPTC

Figure 6. Submarine Cables



Source: MPTC

Figure 7. Cambodia's Fintech Industry



Source: MPTC

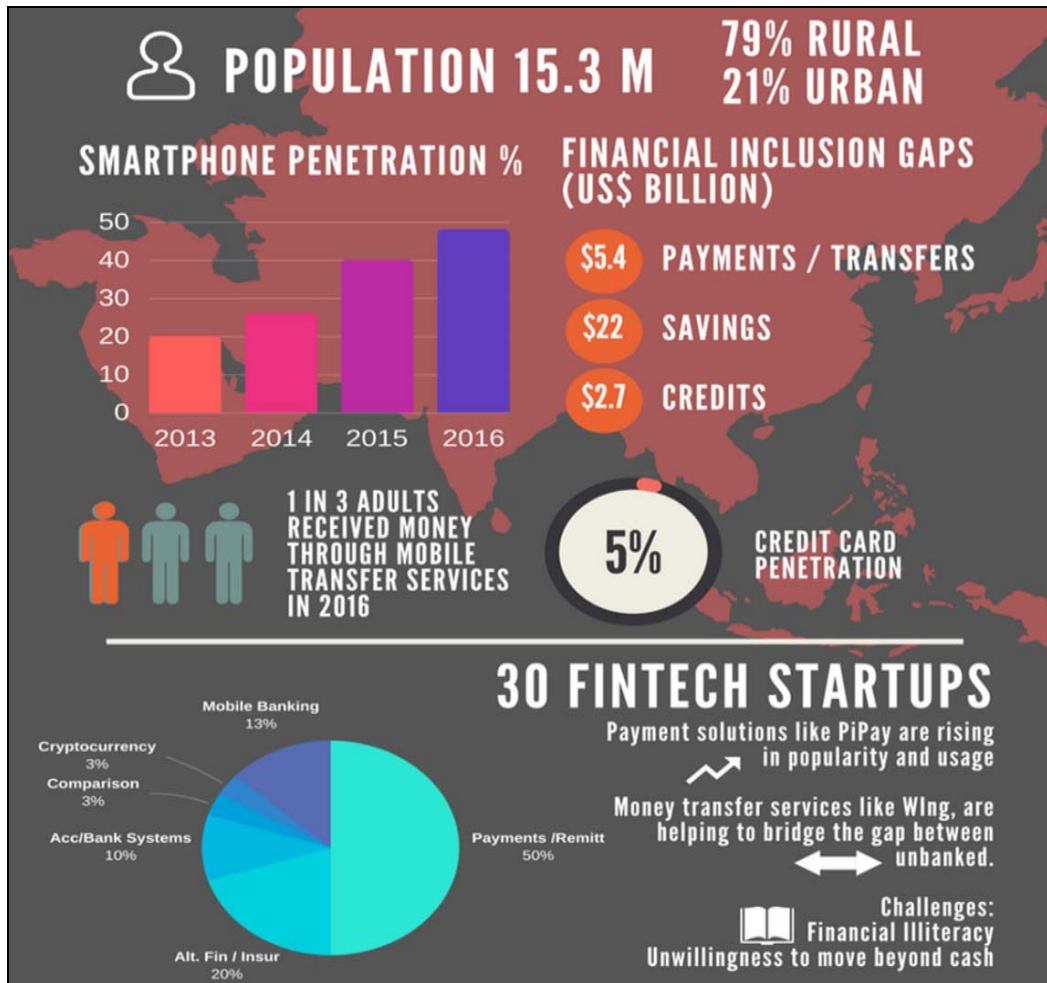
The National Bank of Cambodia has already launched in July 2019 "Bakong," the mobile payment that uses one type of Block chain technology called "Hyperledger Iroha." It is safe, easy, fast, and free of charge. *"The Bakong system is a peer-to-peer fund transfer service available to retail customers of participating banks in Cambodia."* This project aims to offer banking services to the unbanked, especially in the rural areas.

a. Market trends and projections (ASEAN and global)

ASEAN as an engine of economic growth and prosperity has caught the eye of global investors, and there is an abundant supply of early-stage funding in the region. Investment in Southeast Asia FinTechs jumped 33 per cent year-on-year to US\$252 million in 2016. Total investment in the region this year is set to exceed US\$338 million. Beyond the traditional forms of funding from angel investors and venture capitalists (VCs), crowdfunding, venture debt, and bank venture funds have also contributed to the rise in dry powder available for investing in ASEAN FinTechs. While FinTech are still in their early stage in ASEAN, digital platforms such as e-commerce have proliferated, backed by

Internet giants having the financial muscle to make large, billion-dollar investments in the next unicorns of ASEAN. The battle for the consumer wallet and mindshare will continue to drive investment as Internet giants seek to establish a foothold in ASEAN, starting in Singapore, Indonesia, and Thailand.⁶

Figure 8. Cambodia's FinTech industry



Source: Fin Tech Cambodia

Figure 8 shows that 1 in 3 adults received money through mobile transfer services in 2016.

⁶ State of FinTech in ASEAN, United Overseas Bank Limited, 2018

Among the financial technology services, payment/remittance accounted for 50 percent, followed by alternative finance and insurance with 20 percent, mobile banking with 13 percent, Account/Bank systems with 10 percent, and crypto-currency with only 3 percent.

Figure 9. Number of Payment FinTechs in ASEAN by Country, 2017

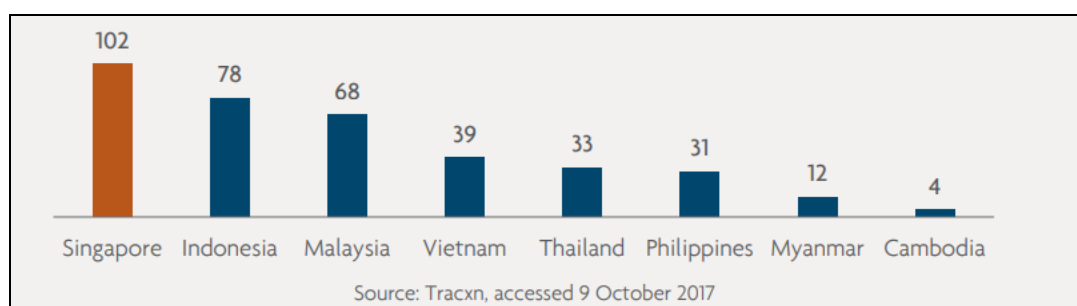


Figure 9 shows the number of payment FinTechs by ASEAN country in 2017. Among the ASEAN countries, Singapore is receiving the highest payment by using FinTechs followed by Indonesia. In contrast, Cambodia has just started on FinTechs.

Figure 10. Number of Saving and Investment FinTechs by ASEAN Country, 2017

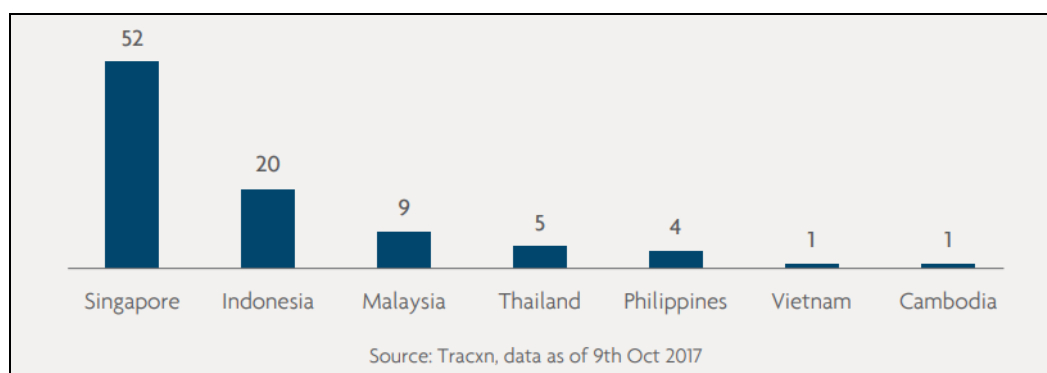
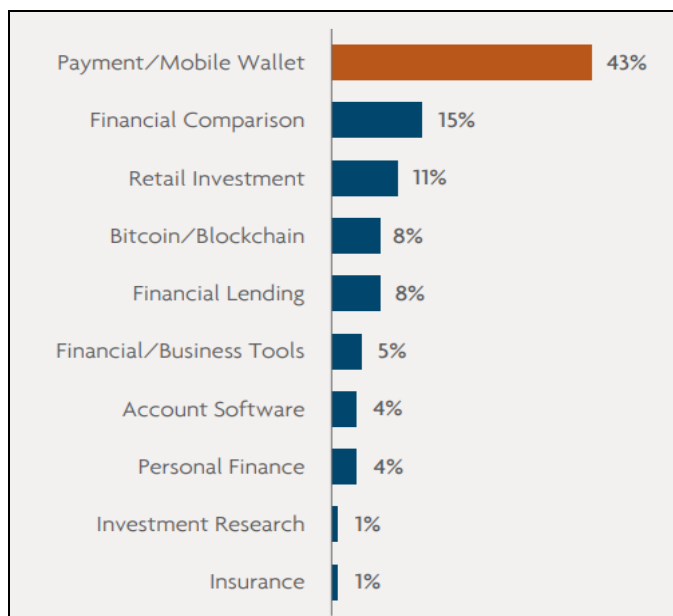


Figure 10 shows the number of saving and investment FinTechs by ASEAN country in 2017. Among the ASEAN countries, Singapore still has the highest number of savings and investments using FinTechs followed by Indonesia, whereas Cambodia has just started on FinTechs although it is continuously increasing use rapidly.

Figure 11. Density of ASEAN FinTech Industry by Category



Source: Tracx, data as of 9 Oct. 2017

Figure 11. shows the density of ASEAN FinTech industry by category. By FinTech industry category, payment/mobile wallet constituted the highest proportion with 43 percent. Insurance and Investment Research accounted for the lowest percentage of FinTech industry with 1 percent.

Figure 12. Adults with an Account (2014) % by ASEAN Country

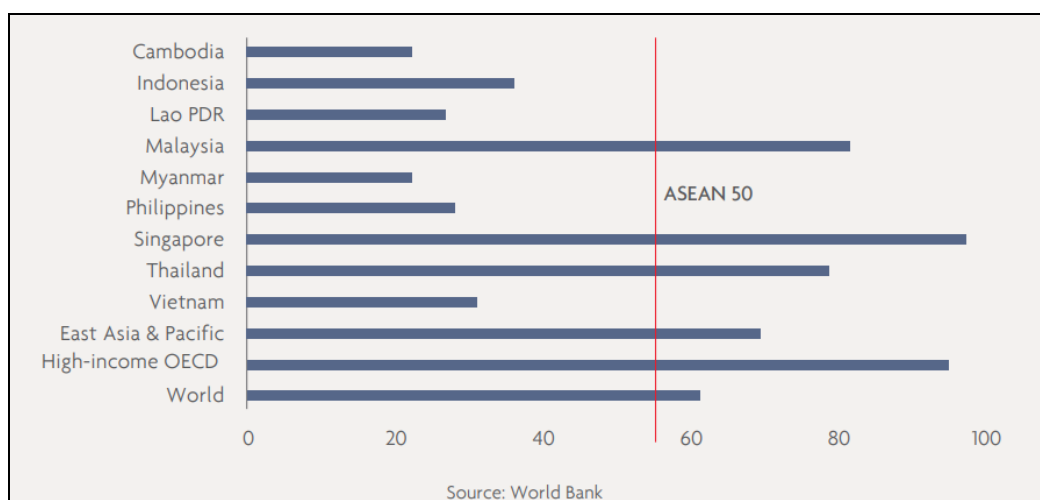
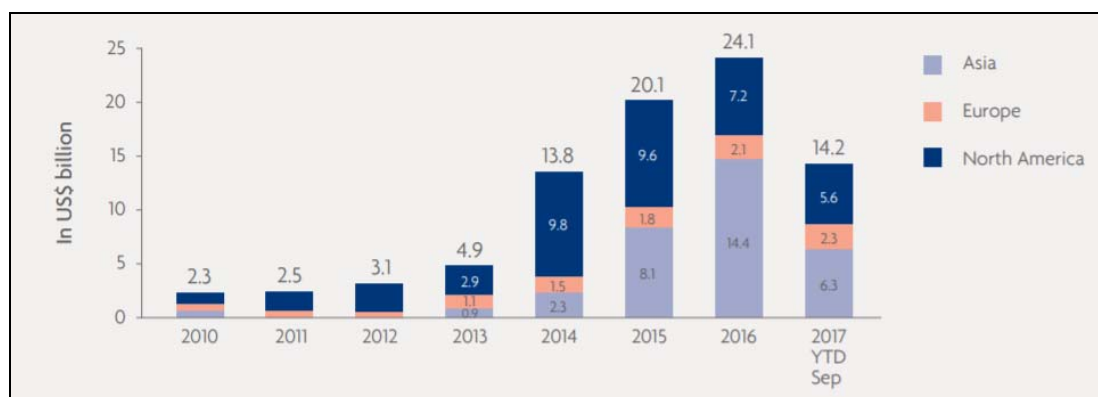


Figure 12 shows the proportion of adults with an account by ASEAN country in 2014. The World Bank estimated that only 22% of Cambodian adults had a bank account compared to 78% in Thailand or 98% in Singapore. In Myanmar, 23% of the citizens have a bank account.

Figure 13. Rise of FinTech in ASEAN, Global FinTech Investment Activity



Source: Tracx, data as 9th Oct 2017

Figure 13 shows the Rise of FinTech in ASEAN, particularly the Global FinTech investment activity in 2016. The global FinTech industry attracted more than US\$24 billion in investment, which is ten times the level received in 2010. FinTech investment in Asia exceeded that in North America for the first time in 2016, led by blockbuster deals in China with Alipay and Lu.com raising US\$4.5 billion and US\$1.2 billion, respectively.

b. Vision

Make Financial Technology Services Easy, Secure, and Fast.

c. Goals and objectives

- Anytime, anywhere, any device communication (Ubiquitous Computing) of Financial Technology Services
- Innovation in content and services of Financial Technology Services

Part 2: TRM Development

2.1 Future prospects

Cambodia might not come to mind when drawing up a list of fintech hubs; across Southeast Asia, however, the benefits of addressing the digital divide and harnessing the power of fintech should be clear-cut. Taken together, both these steps can increase the level of access to capital and meet financial inclusion-critical needs in developing Asia.

Broadly defined, “fintech” – which is short for the technologies delivering innovations as well as new challenges and opportunities to the once staid banking and finance sectors -- is also enabling the rise of new companies and transformative businesses. The emergence of FinTech has enabled the growth in other sectors, particularly e-commerce and other tech startups.

With little or no access to formal banking services, too many people in Cambodia go without the basic protections of a savings account, possibly facing relatively higher costs for sending or receiving money -- this in a region where remittances were valued at \$236 billion in 2016⁷.

Still, the sustained benefits of fintech will only be realized if a proper ecosystem is created and maintained -- one that addresses the concerns of regulators while benefiting innovators and most importantly consumers.

Narrowing the digital divide will also continue to be a fundamental need, with increased mobile phone ownership and Internet penetration as key factors in spurring the consumer adoption of mobile financial services.

Indeed, the true measure of success for fintech in Cambodia and elsewhere should not be the deal size or quantity but the expanded horizons. True success is when fintech helps once-poor farming communities access funds to bring their crops to market, helps small

⁷ World Bank, International debt statistics, 2019

shopkeepers grow bigger, or provides seed money for a young entrepreneur ready to turn a great idea into a concrete reality.

Beyond the fintech hype and jargon, the human element of financial technology should not be forgotten in Cambodia as this nation, too, moves forward. Future assessments of fintech must go beyond counting fortunes made and businesses disrupted or created, but must also include a measure of everyday Cambodians being helped.

With the emergence of FinTech in the Kingdom, we are witnessing the start of a significant transformation. The financial sector is being re-imagined through the lens of financial technology. This is the “perfect storm” -- FinTech provides an unprecedented opportunity to close the USD 24+ billion gap between the demand for financial services and the formal supply⁸. The accelerated adoption of new financial technologies, new business models, and lower barriers to entry on the supply side can enable a massive increase in financial inclusion.

The advancement of next-generation technology provides a more level playing field for FinTech firms, allowing them to work around inflexible legacy systems and “silo-ed” channels. This has lowered barriers to entry. The cost advantage is on the FinTech side.

Although only a small amount of FinTech development and investment has occurred in Cambodia to date, the situation is likely to change soon. In Cambodia are many of the necessary elements to develop a strong FinTech ecosystem. There is every reason to believe that the Kingdom will see the development of a dynamic ecosystem that can provide broader economic benefits.

Despite its nascent stage of development, FinTech in Cambodia has drawn significant attention at the regional level. Two Cambodian startups made it to the top ten FinTechs in Asia by Startup bootcamp FinTech in Singapore for two consecutive years.

Financial inclusion, particularly in the payment vertical, is the main focus of FinTechs in

⁸ Asian Development Bank, Accelerating Financial Inclusion in Southeast Asia with Digital Finance, 2017.

Cambodia. This will remain the case for the foreseeable future, though alternative Financing is gaining pace and SME solutions will also be important.

2.2 Environmental analysis

With the increasing number of tech startups, Cambodia's fintech startup landscape remains modest but is growing steadily nonetheless.

Unsurprisingly, the payment and remittance categories are the most crowded with at least 16 startups. These include: Wing, a mobile banking services provider targeting the unbanked; PayGo, a digital wallet; TrueMoney, a mobile payment provider; and Prasac, one of the largest microfinance institutions (MFIs) in Cambodia.

Cambodia also has startups tackling other spaces including the following: alternative finance such as TosFUND -- a non-profit form of crowd funding -- and Karprak.com, an online peer-to-peer lending platform; insurtech like Bima; credit rating like Agribuddy; accounting services such as Banhji; and block chain technology such as CryptoAsia.

Cambodia has a population of 15.3 million people with some 5 million Internet users. Most importantly, Cambodia has a mobile phone penetration rate of 166% and a smartphone penetration rate of 39.5%.

With almost 80% of the population having no formal bank account and credit card penetration rate of 5%, experts and industry participants have raved about the potential of fintech to improve financial inclusion.

2.2.1. Trends/Drivers

■ Social

- **Demographic dividend:** Entering a period when the working-age population has good health, quality education, decent employment, and lower proportion of young dependents is the greatest demographic opportunity for a country's development.
- **Lifestyle changes:** Rapid development and social stability are factors for the transformation of social life into a digital lifestyle. At the same time, there will be a need for a faster, more secure payment method.

■ Technology

- **Block chain:** It is the distributed ledger technology (DLT) underlying bitcoin cryptocurrency discovered in 2008 due to the distributed nature; the immutable property of using cryptographic functions allows this technology to be used in different domains such as fintech, logistics, health, and many more to solve the problem of trust and provide auditable proof of origin through its data structure. It has proven to be a potential underlying technology for building future decentralized applications through the use of smart contract suitable for the current collaboration world. By allowing digital information to be distributed but not copied, block chain technology created the backbone of a new type of Internet. Originally devised for the digital currency Bitcoin, the tech community has now found other potential uses for the technology, which will open a new portal for the country to make use of this technology.
- **AI:** As exciting as it should be, not only does this technology currently dominate the tech world but also enjoys overwhelming support from the Cambodian government. Young students enter the robotics competition every year and sometimes best the representatives of neighboring countries.
- **Edge computing:** With the increasing uses of IoTs such as smartphone, smart TV, smartwatch, etc., wherein most would collect data -- sometimes massive amounts of it (Big Data) -- and some of it is processed locally to reduce the backhaul traffic and then send it to a data center or the cloud for processing, that is when edge computing is absolutely necessary for critical processing such as financial operation.
- **Increasing the R&D proportion compared to GDP:** Striving to be an upper middle-income country by 2030, the government's commitment is to increase the R&D expenditure up to 1% of GDP.
- **Patent protection:** On paper, Cambodia's patent protection looks strong. The kingdom:
 - Passed the Law on Patents, Utility Model Certificates, and Industrial Designs (Law on Patents) in 2003

- Is a signatory to the Paris Convention for the Protection of Intellectual Property
- In 2016, became the 151st contracting state of the Patent Cooperation Treaty (PCT)
- Signed an Agreement on the Validation of European Patents with EPO in 2017
- By June 2017, however, the country had granted only 17 patents, according to the Patent Office.¹⁶ Cambodia's patent applications are accepted through the Ministry of Industry and Handicraft.

There has been significant progress in terms of recognition of international patents, however. In 2017, Cambodia became the first Asian country to grant validation to European patents. EU patent holders who follow a validation process can gain protection of their patent in Cambodia without having to apply for a Cambodian patent.¹⁷ (pharmaceutical patents excluded). A similar rule applies to Chinese invention patents.

Table 1. STEEP Analysis of Trends and Drivers

STEEP	Drivers
Social	Demographic dividend (young population) Lifestyle changes (urbanization of population, increase in mobile penetration)
Technology	Block chain AI Edge computing (Big Data, IoTs, cloud...) Government spending for R&D by increasing the proportion as compared to the GDP New development in technology transfer, Increasing tech startup Patent protection
Economic	Gross Domestic Product trends Energy availability and cost
Environment	Global climate change (green technology; by using FinTech, does not spend time for travel) Achieving Cambodia's Sustainable Development Goal
Political	Stability of government Political commitment especially for S&T development Tax laws (tax holiday for tech startup) Special incentives for R&D investment Sharing information

Note: Group Discussion

2.2.2 Opportunities/Threats

Going forward, Cambodia faces both opportunities and challenges. Here, we highlight the major issues including, but not limited to, FinTech, Block chain, Patent Protection, Demographic Dividend, and Lifestyle Changes. Having the strong foundation of tax laws, labor force availability, political stability, increasing R&D investment and expenditure, the rise of tech startups, ever growing interest in e-commerce, and preference for cashless transaction can be seen. Driving these trends are deeper changes in the economy toward high value-added as well as the emergence of a young, dynamic population. Nonetheless, maintaining this path also presents challenges including how to manage the transfer of money, technology & security, standardize IT devices, apply for and protect patents online as well as the lack of tech researchers and AI-based tech services and need for tech Foreign Direct Investment (FDI).

Table 2. Driver Analysis

Drivers	Opportunity/Threat
Technological evolution	Increase of mobile online (O) Technology cost has been decreasing very fast (O) Collection tax (O, T) Transaction tax (T) Fast and secure transfer (O) Managing money transfer technology (T) Lack of Researchers (T) Lack of patent applications online (T) Lack of Cybersecurity specialist (T) Efficient IT device (T) Patent protection (T) Standardized IT devices (T) Lack of AI-based tech services (T)
Changing customer expectations	Lifestyle changes Increase of people moving from rural to urban area (O) Younger population-labor force availability (O)
Availability of funding	Need for tech Foreign Direct Investment (FDI) (T) More investment of micro-macro finance (O)

Drivers	Opportunity/Threat
Regulatory support by the government	Commitment of government to FinTech investment(O) Promote of FinTech startup (O) Government set priority to R&D investment especially for FinTech (O) Tax laws (tax holiday for FinTech start-up) (O) Special incentives for R&D investment (O)

Noted: Group Discussion

2.3 Strategic products and services

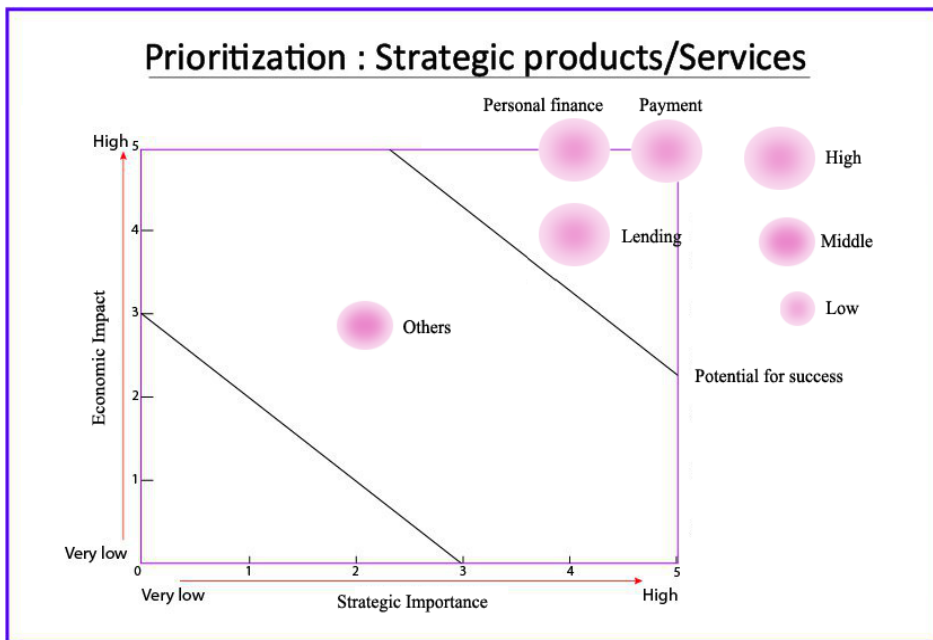
2.3.1 Product feature concepts

Table 3. Linkage of Key Technologies

Product/Service	Code	Evaluation			Sum
		Economic Impact	Strategic Importance	Potential for Success	
Payments (billing, automation, and streamlining)	S1	5	5	2	11
Personal finance (online/mobile banking)	S2	5	4	2	10
Lending (direct lending and underwriting, peer-to-peer lending)	S3	4	4	1	8
Others (money transfer and FX services, cryptocurrency payments)	S4	3	2	1	6

Note: Group Discussion

Figure 14. Prioritization of Strategic Products and Services



2.3.2 Grouping

Key Product Development

Payments (billing, automation, and streamlining)

Personal finance (online/mobile banking)

Lending (direct lending and underwriting, peer-to-peer lending)

2.3.3 Prioritization

Table 4. Identification and Prioritization of Key Products

	Opportunity	Threats
Opportunities and Threats	Labor force availability Political stability Promotion of tech startup Priority to R&D investment Tax laws Incentives for R&D investment	Management of money transfer technology Lack of Tech Researchers Lack of patent applications online Need for tech Foreign Direct Investment (FDI) Standardized IT devices Efficient IT device Lack of AI-based tech services
Products/Services	Payments (billing, automation, and streamlining) Personal finance (online/mobile banking) Lending (direct lending and underwriting, peer-to-peer lending) Blockchain-based market such as cryptocurrency	

2.3.4 Performance dimensions

Table 5. Identification of Performance Dimension

Performance Dimension	Performance Target
Functionality	SMEs registration: 85% by 2025, 99% by 2030
Cost	Cheaper services
Security	SMEs' data protection, Customers' data records
Accuracy	Smart and user-friendly
Scalability	Professional support, Low maintenance
Stability	Real-time transaction (24/7)

Table 6. Technology Alternatives

Technology Alternatives
App, Software as a Service, AI
Cloud computing
Cryptographic Encryption
Block chain
Cyber-physical system, scaled IT approach, edge computing
5G technology

2.4 Key technologies

2.4.1 Technology solutions

Table 7. Technology Solutions

Technology
Cryptographic Encryption, block chain, AI, Big Data
5G technology, edge computing, cyber-physical system, Open source, Forecasting software

2.4.2 Grouping

Table 8. Technology Tree

Tech Area	Tech Level 1	Tech Level 2
Service	Cryptographic Encryption, Block chain, AI	Big Data
Infrastructure	5G technology, edge computing, open source	Cyber-physical system, Forecasting software

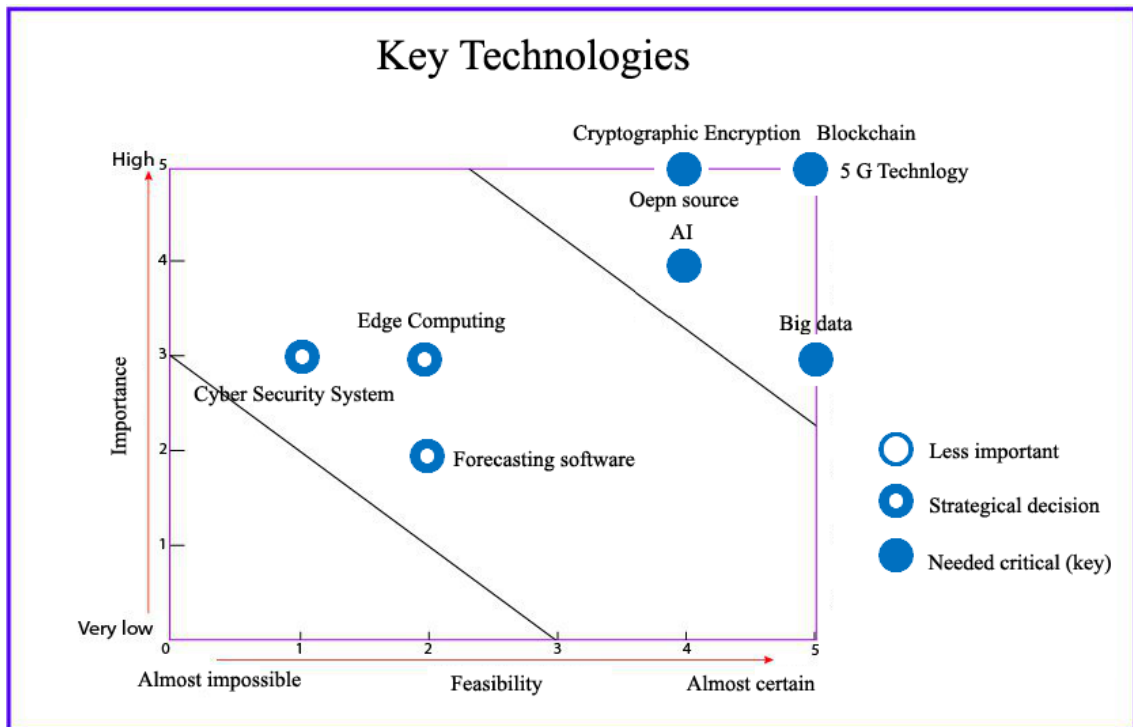
2.4.3 Prioritization

Table 9. Prioritization of Key Technologies

Technology	Code	Evaluation Criteria		Sum
		Importance	Feasibility	
Cryptographic Encryption	T1	5	4	9
Block chain	T2	5	5	10
AI	T3	4	4	8
Big Data	T4	3	5	6
5G technology	T5	5	5	10
Edge computing	T6	3	2	5
Cyber-physical system	T7	3	1	4
Open source	T8	5	4	9
Forecasting software	T9	2	2	4

Note: Group Discussion

Figure 15. Key Technologies



2.5 Macro TRMs for strategic products

Vision	Easy, secure, and fast financial technology services									
Goals	Anytime, anywhere, any device communication Innovation in content and services									
	21	22	23	24	25	26	27	28	29	30
Future Prospect	5G technology Open source			Cryptographic Encryption Block chain Big Data AI Edge computing				Big Data AI Edge computing		
Strategic Services or Products	Functionality: SMEs registration: 85% by 2025, 99% by 2030									
	Cost: Cheaper services									
	Security: SMEs’ data protection, Customers’ data records									
	Accuracy: Professional support, Low maintenance									
	Scalability: Smart and user-friendly									
	Scalability: Real-time transaction (24/7)									
Block chain										
Cryptographic encryption										
Big Data										
Artificial intelligence										
5G technology										
Open source										
Edge computing										
R&D Strategy	Cooperation with researchers, research institute, and university			Cooperation with researchers, research institute, and university				Cooperation with researchers, research institute, and university		
Key Infrastructure	R&D investment			R&D investment				R&D investment		

Part 3: Conclusion

3.1 Recommendations

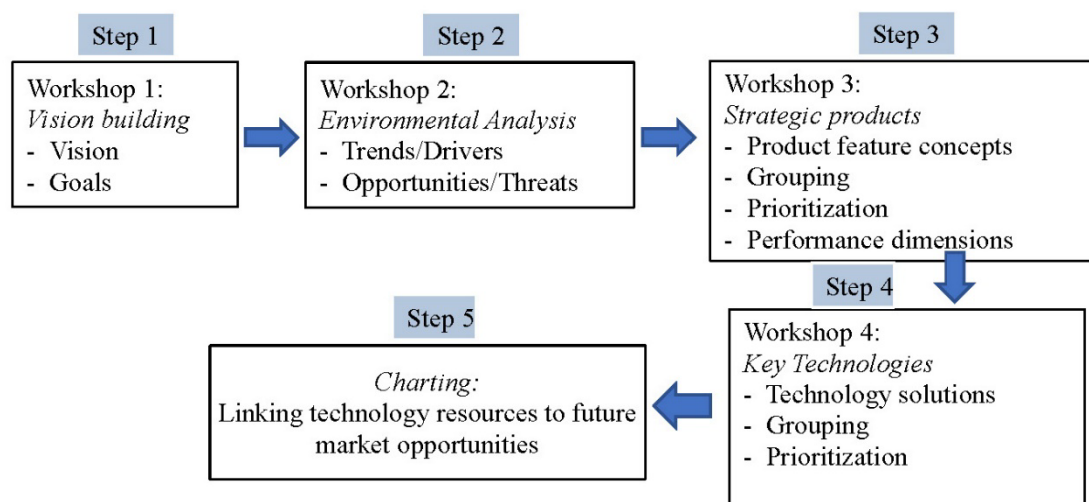
1. Expand broadband access to more people and businesses to help bridge the digital divide between rural and urban areas.
2. Promote and encourage financial service technology startup by providing tax exemption.
3. Create a unified payment infrastructure to support mobile financial services, reducing the costs and complexities of cross-border payments and promoting financial inclusion.
4. Expand and encourage people to create and use bank accounts.
5. Strengthen and promote the use of financial technology service.
6. Create the FinTech startup ecosystem for Cambodian entrepreneurs.
7. Promote local FinTech startup to the regional and global levels.
8. Promote and strengthen human resources such as graduates with basic ICT skills, ICT researchers, and ICT R&D experts.
9. Harmonize government policies and coordinate legal and regulatory frameworks in the areas of digital identity, privacy, trade, and e-commerce.

3.2 Implementation

Technology Areas	Key Technologies	Final Target	Element Technologies		
			Short-term (1-3 y.)	Mid-term (4-7 y.)	Long-term (8-10 y.)
Services	Block chain	No. of applications used		X	X
	Cryptographic encryption (stra.)	No. of encryptions applied	X	X	X
	Big Data (stra.)	No. of data collected		X	X
	Artificial intelligence (key)	No. of data collected		X	X
Infrastructure	5G technology (key)	No. of sim cards or e-sim cards connected	X	X	X
	Open source (key)	No. of open sources applied	X	X	X
	Edge computing (stra.)	No. of people/devices used	X	X	X

Part 4: Appendices

4.1 Roadmapping process



4.2 Participants

No.	Name	Position	Organization
1	HE Mr. Sann Sary	Undersecretary of State	MPTC
2	Mr. Buoy Somethea	Director of Policy and Planning, Administration and Finance and ICT Department	GS-NSTC
3	Mr. Chan Pheakei	Deputy Director of the ICT Industry Department	MPTC
4	Mr. Rith Sela	Deputy Director of Policy and Planning, Administration and Finance and ICT Department	GS-NSTC
5	Mr. Mam Piseth	Vice Bureau Chief	GS-NSTC
6	Mr. Sarot Srang, PhD	Chief of ICT and Mechanics	ITC
7	Mr. Sok Kimheng	PhD researcher on Block chain, Smart Contract, IoT, and Privacy, Institute of Technology of Cambodia (ITC), University of Namur in Belgium (UNAMUR)	ITC

4.2.2 BioHealth_TRM (draft)



ព្រះរាជាណាចក្រកម្ពុជា

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ផែនទីបង្ហាញផ្លូវវិទ្យាសាស្ត្រ បច្ចេកវិទ្យា និងនវានុវត្តន៍ ឆ្នាំ២០២០-២០៣០

វិស័យអាហារបំប៉នសុខភាព

» គម្រោងទិសវិទ្យាសាស្ត្រ និងបច្ចេកវិទ្យា ដើម្បីជំរុញកំណើននិងអភិវឌ្ឍន៍ទីផ្សារផលិតផល«

Science and Technology Orientation for Growth and
Product Development Marketing

Outline of Technology Roadmap for Bio-Health

Executive Summary

In recent years, advance technology such as encapsulation technology has emerged as a promising method of delivering bioactive materials to humans through multivitamins, skincare products, pharmaceuticals, packaging, foods, and dietary supplements. Some players in the dietary supplements and personal care products industries view nanotechnology as a major breakthrough, and they would like to use it as a way of increasing the benefits of nutraceuticals and cosmetic products. Apart from nanotechnology, there are encapsulation or emulsion, advanced manufacturing, advanced materials, and alternative source of energy with a view to encouraging innovation among local companies and developing new products. An emulsion is a mixture of two or more liquids that are normally completely or partially immiscible, where fine droplets in one liquid are dispersed in the other by using various emulsifiers and different emulsification techniques. Many techniques were discovered and involved to prepare emulsions such as high-pressure homogenization, membrane emulsification, microfluidics, sonication, or phase inversion. Recently, microchannel emulsification was developed to produce monodispersed emulsions, resulting in narrow droplet size distribution compared to other emulsification methods. Moreover, the economy of Cambodia is highly dependent on the agricultural sector, but the valorization of the local product still has many limitations. The efficient extraction method of nutrients from the local agricultural product is another important step, and it could potentially support the important micro-nutrients for the citizens. The extraction method could be done in different ways including solid and liquid extraction; solvent extraction is usually applied, however. As such, it is almost certain to have a remarkable impact on the food product and healthcare capabilities in the next 10-20 years' timeframe. Therefore, considering deliberation on the impact of extraction and encapsulation technology is essential.

Part 1. Introduction

1.1 Background

Home food or commercial food product supply is crucial for human health. Nonetheless, malnutrition is still a main concern associated with food consumption, particularly in a developing country such as Cambodia. People are probably not getting enough nutrients due to their food consumption habits or family income. On the other hand, people pay more attention to healthcare for the purpose of beauty and health. Many food products and healthcare supplements were introduced to the market, becoming an alternative solution for promoting the status of human health. Note, however, that the combination of different micronutrients in the supplement is still under discussion and investigation by technology, and there are still many aspects and concerns with regard to those supplements. Scientists work on various researches to produce the supplement with high bio-availability and efficacy. Supplements are developed and produced in response to consumer trends and wishes and wants for quality and safety by technology. The availability of local agricultural products assures the source of nutrient supply for the people, but the lack of extraction methods is another challenge for the country.

The Royal Government of Cambodia places population health at the center of its socioeconomic development. Significant progress in terms of expanded health infrastructure and increased health service coverage during the last decade has resulted in an unprecedented level of health service utilization. Without good health, an individual cannot acquire knowledge through education, and he/she cannot be productive in an occupation. Technology Vision 2030 places great emphasis on Bio-health and healthcare with a view to transforming Cambodia into a healthy, developed nation.

As a result of positive developments in the technology-related multivitamins, supplements, nutraceuticals, cosmetic products, environmental protection, and education, the commercialization and adoption of new technologies resulting from nanoscale R&D are expected to yield a positive economic return in the form of benefits such as creation of businesses, jobs, and trade.

Thus, unsurprisingly, there are various opinions as to what constitutes technology. Global investments in extraction and encapsulation technology have begun to yield economic benefits as products incorporating nanotechnology enter the marketplace. Nano-enabled products are estimated to have recorded \$290 billion in revenues in 2015, according to Lux Research.

The societal impact of nanotechnology is the potential benefits and challenges that the introduction of novel nanotechnology devices and materials may have for society and human interaction. Major benefits of nanotechnology include improved manufacturing methods, multiple micro-vitamins, better food production methods, nutrition, water purification systems, energy systems, physical enhancement, nanomedicine, and large-scale infrastructure auto-fabrication. If nanotechnology is going to revolutionize manufacturing, healthcare, energy supply, communications, and probably defense, then it will transform labor as well as the workplace, medical system, transportation, and power infrastructures. None of these will be changed without significant social disruption.

TRM from market-driven planning

Over the last decade, the Cambodian population has become more knowledgeable on the importance of healthcare and health supplements, and nutrition products are becoming more popular.

The demand for supplement products has increased in the last several years, particularly in Phnom Penh and larger provincial towns in Cambodia. Two major US supplement brands have gained popularity in Cambodia recently: Unicity and Herbalife.

In addition to the increased demand arising from population growth, increased demand for supplement products globally is driven by socioeconomic changes such as rising incomes, increased urbanization, and aging populations wherein the contribution of dietary supplements to healthy aging is increasingly recognized.

Based on Table 1, 32 percent of Cambodian children under age 5 are stunted, and 9 percent are severely stunted (Table 16.1 and Figure 16.1). The analysis by age group indicates that stunting is apparent even among children less than 6 months old (16 percent). In general, stunting increases with the age of the child, rising from 13 percent among children 6-8 months old to 40 percent among children aged 36-47 months before declining to 36 percent among children aged 48-59 months. There is very little difference in the level of stunting by gender. Stunting is highest when the birth interval is less than 24 months (37 percent). Size at birth is an important indicator of children's nutritional status. Nearly 2 out of 3 children (63 percent) reported to have been very small at birth are stunted. Children whose mothers are underweight are more likely to be stunted (44 percent) than the children of normal weight mothers (32 percent). The disparity in stunting prevalence between rural and urban children is substantial -- 34 percent of rural children are stunted compared to 24 percent of urban children.

At least 10 percent of children under age 5 are wasted, and 2 percent are severely wasted. There is substantial correlation between wasting and size at birth. Babies who are very small and small at birth are more likely to be wasted (24 percent and 17 percent, respectively) than those of average or larger size at birth (9 percent). The prevalence of wasting among children of thin mothers (BMI below 18.5) is more than twice that of children whose mothers are either normal weight or overweight/obese. Wasting is higher among rural children than urban children (10 percent versus 8 percent) and is highest in Takeo and Otdar Meanchey (15 percent each). Wasting prevalence varies inconsistently according to the age of the child and does not differ substantially by sex. It is highest among children whose mothers have no education and those in the lowest two wealth quintiles.

Table 1: Nutritional Status of Children

Percentage of children under age 5 classified as malnourished according to three anthropometric indices of nutritional status: height-for-age, weight-for-height, and weight-for-age, by background characteristics, Cambodia 2014

Background characteristic	Height-for-age ¹			Weight-for-height				Weight-for-age				Number of children
	Percentage below -3 SD	Percentage below -2 SD ²	Mean Z-score (SD)	Percentage below -3 SD	Percentage below -2 SD ²	Percentage above +2 SD	Mean Z-score (SD)	Percentage below -3 SD	Percentage below -2 SD ²	Percentage above +2 SD	Mean Z-score (SD)	
Age in months												
<6	6.1	16.1	(0.4)	5.4	12.8	6.0	(0.3)	3.2	11.5	1.3	(0.6)	426
6-8	1.2	13.1	(0.6)	2.3	6.5	4.8	(0.3)	1.9	8.5	2.2	(0.6)	252
9-11	3.9	16.6	(0.9)	2.3	14.2	3.1	(0.7)	3.0	15.4	0.5	(1.0)	225
12-17	6.4	28.1	(1.3)	3.1	10.6	3.5	(0.7)	2.6	21.2	1.0	(1.1)	515
18-23	11.3	33.8	(1.5)	2.8	10.9	0.7	(0.7)	5.4	19.6	0.3	(1.2)	545
24-35	10.8	38.5	(1.6)	1.8	8.0	1.8	(0.6)	4.3	24.9	0.9	(1.3)	1,013
36-47	10.9	39.8	(1.8)	1.9	9.3	0.7	(0.8)	7.1	31.4	0.1	(1.5)	978
48-59	9.4	36.0	(1.7)	1.2	8.7	0.5	(0.8)	5.1	30.9	0.2	(1.6)	939
Sex												
Male	9.4	32.9	(1.4)	2.7	9.9	2.3	(0.6)	4.5	23.2	0.8	(1.2)	2,497
Female	8.4	31.9	(1.4)	1.9	9.3	1.6	(0.7)	4.8	24.6	0.5	(1.3)	2,395
Birth interval in months³												
First birth ⁴	8.3	30.2	(1.4)	2.3	9.6	2.1	(0.7)	3.3	24.3	0.7	(1.3)	1,683
<24	11.7	37.4	(1.5)	2.3	10.5	2.3	(0.8)	9.3	31.1	0.9	(1.4)	336
24-47	9.5	34.1	(1.5)	2.4	9.9	2.0	(0.7)	6.3	25.0	0.6	(1.3)	1,102
48+	7.8	31.0	(1.3)	2.7	9.5	1.8	(0.7)	3.6	21.0	0.5	(1.2)	1,139
Size at birth³												
Very small	25.6	63.1	(2.3)	3.6	23.6	4.8	(1.0)	16.5	58.9	2.6	(2.0)	96
Small	13.9	40.5	(1.7)	2.3	16.7	1.0	(0.9)	7.1	38.5	0.0	(1.6)	368
Average or larger	7.8	30.2	(1.4)	2.4	8.6	2.0	(0.7)	4.1	21.8	0.7	(1.2)	3,772
Mother's interview status												
Interviewed	8.7	32.0	(1.4)	2.4	9.7	2.0	(0.7)	4.6	24.1	0.6	(1.3)	4,261
Not interviewed but in household	2.5	36.0	(1.2)	1.4	11.4	5.1	(0.5)	2.7	23.1	2.1	(1.0)	107
Not interviewed and not in the household ⁵	11.9	35.2	(1.5)	1.6	8.4	0.8	(0.5)	5.3	22.3	0.2	(1.2)	525
Mother's nutritional status⁶												
Thin (BMI <18.5)	13.7	44.0	(1.8)	2.5	18.5	0.7	(1.0)	11.0	39.7	0.4	(1.7)	461
Normal (BMI 18.5-24.9)	8.5	31.8	(1.4)	2.3	8.6	2.4	(0.6)	4.0	23.9	0.7	(1.3)	2,726
Overweight/obese (BMI ≥25)	6.2	26.2	(1.3)	2.0	6.7	1.7	(0.5)	2.5	15.7	0.9	(1.1)	671
Residence												
Urban	5.9	23.7	(1.1)	2.0	7.5	3.1	(0.4)	2.6	14.8	1.9	(0.9)	674
Rural	9.4	33.8	(1.5)	2.4	9.9	1.8	(0.7)	5.0	25.4	0.4	(1.3)	4,219
Province												
Banteay Meanchey	6.9	28.6	(1.3)	0.7	7.8	0.7	(0.5)	5.3	17.0	0.0	(1.1)	241
Kampong Cham	8.6	33.5	(1.4)	1.5	8.1	2.2	(0.7)	4.2	25.7	0.8	(1.3)	692
Kampong Chhnang	13.5	42.8	(1.7)	3.1	11.2	2.2	(0.9)	5.1	35.6	0.0	(1.5)	173
Kampong Speu	10.0	40.5	(1.7)	2.5	11.5	1.3	(0.8)	6.9	29.4	0.0	(1.5)	318
Kampong Thom	10.7	36.4	(1.3)	3.1	13.0	3.4	(0.8)	7.1	27.7	1.9	(1.3)	217
Kandal	3.5	28.1	(1.3)	3.2	9.2	0.2	(0.8)	4.7	26.2	0.0	(1.3)	298
Kratie	10.5	38.4	(1.6)	2.7	6.5	0.5	(0.7)	4.4	25.1	0.0	(1.4)	180
Phnom Penh	4.9	17.9	(0.9)	1.0	8.4	3.7	(0.4)	2.2	12.9	3.4	(0.8)	391
Prey Veng	8.7	32.7	(1.5)	2.9	8.6	1.8	(0.6)	3.4	22.2	0.3	(1.3)	379
Pursat	18.4	38.8	(1.8)	5.7	12.3	4.7	(0.6)	7.9	31.6	0.4	(1.4)	200
Siem Reap	11.3	35.9	(1.5)	2.3	9.5	1.0	(0.7)	6.9	26.2	0.4	(1.3)	323
Svay Rieng	8.2	32.8	(1.4)	2.7	7.6	3.6	(0.6)	3.6	20.8	0.5	(1.2)	190
Takeo	6.4	30.7	(1.3)	5.0	14.6	1.5	(0.8)	4.4	22.7	0.0	(1.3)	258
Oddar Meanchey	14.0	36.3	(1.3)	7.2	15.1	5.3	(0.7)	5.2	26.4	0.0	(1.3)	78
Battambang/Pailin	5.0	24.9	(1.2)	0.3	7.9	0.7	(0.6)	1.8	18.2	0.5	(1.1)	388
Kampot/Kep	8.3	25.2	(1.4)	0.9	8.2	1.9	(0.7)	3.5	21.1	0.7	(1.3)	195
Preah Sihanouk/ Koh Kong	10.4	33.4	(1.4)	3.1	10.5	1.8	(0.6)	6.5	22.0	0.0	(1.2)	105
Preah Vihear/ Stung Treng	14.0	44.3	(1.8)	1.3	13.8	2.3	(0.7)	5.9	30.7	0.1	(1.5)	142
Mondul Kiri/ Ratanak Kiri	14.6	39.8	(1.6)	1.4	8.2	1.2	(0.6)	6.0	26.2	0.3	(1.4)	125
Mother's education⁷												
No education	13.3	38.5	(1.6)	2.8	12.2	1.5	(0.7)	7.4	29.7	0.4	(1.4)	577
Primary	8.7	34.1	(1.5)	2.1	9.3	2.1	(0.7)	4.7	24.6	0.6	(1.3)	2,397
Secondary and higher	6.8	26.8	(1.3)	2.7	9.9	2.0	(0.7)	3.5	22.3	0.6	(1.2)	1,278
Wealth quintile												
Lowest	14.1	41.9	(1.7)	2.8	11.0	1.1	(0.8)	7.1	31.0	0.0	(1.5)	1,182
Second	9.8	37.1	(1.6)	2.3	11.4	2.1	(0.7)	5.7	27.5	0.5	(1.4)	998
Middle	7.8	31.7	(1.4)	2.1	8.4	1.3	(0.7)	4.3	23.3	0.2	(1.3)	978
Fourth	6.9	29.1	(1.3)	2.5	9.3	2.4	(0.7)	3.5	22.0	0.8	(1.2)	844
Highest	4.1	18.5	(0.9)	1.9	7.4	3.3	(0.4)	1.7	13.0	2.1	(0.8)	891
Total	8.9	32.4	(1.4)	2.3	9.6	2.0	(0.7)	4.7	23.9	0.6	(1.3)	4,893

Note: Table is based on children who stayed in the household on the night before the interview. Each of the indices is expressed in standard deviation units (SD) from the median of the WHO child growth standards adopted in 2006. The indices in this table are NOT comparable to those based on the previously used NCHS/CDC/WHO reference. Table is based on children with valid dates of birth (month and year) and valid measurement of both height and weight. Total includes 25 cases with missing information on size at birth.

¹ Recumbent length is measured for children under age 2 and in the few cases when the age of the child is unknown and the child is less than 85 cm; standing height is measured for all other children.

² Includes children who are below -3 standard deviations (SD) from the WHO child growth standards population median

³ Excludes children whose mothers were not interviewed

⁴ First-born twins (triplets, etc.) are counted as first births because they do not have a previous birth interval.

⁵ Includes children whose mothers are deceased

⁶ Excludes children whose mothers were not weighed and measured, children whose mothers were not interviewed, and children whose mothers are pregnant or gave birth within the preceding 2 months. Mother's nutritional status in terms of BMI (body mass index) is presented in Table 16.10.

⁷ For women who are not interviewed, information is taken from the Household Questionnaire. Excludes children whose mothers are not listed in the Household Questionnaire.

Vitamin and Calcium Deficiency among Children

Among children, the overall prevalence of vitamin A deficiency (RBP <0.70 µmol/L) is 9 percent, just below the threshold for a major public health problem (Table 2). Moreover, the high prevalence of children with marginal vitamin A status (29 percent) is a significant public health concern.

In contrast, 8 percent of children suffer from folic acid (vitamin B9) deficiency. The prevalence of folic acid deficiency among children is higher in urban areas (18 percent) than rural areas (7 percent). The prevalence of vitamin D deficiency is also high, with 15 percent of children having vitamin D levels below 50 nmol/L and 33 percent having levels below 70 nmol/L.

Table 2 Blood level of vitamins A, B12, B9, and D and calcium in children

Among children born since January 2009, percentage with deficiencies of vitamin A, vitamin B12, vitamin B9, vitamin D, and calcium, according to residence, Cambodia 2014

	Urban	Rural	Total
Vitamin A deficiency			
Deficient (RBP <0.70 µmol/L)	13.7	8.7	9.2
Marginal (RBP <1.05 µmol/L)	39.9	27.9	29.2
Vitamin B12 deficiency			
Deficient (<150 pmol/L)	0.8	1.8	1.7
Vitamin B9 deficiency			
Deficient (<10 nmol/L)	17.5	6.8	8.0
Vitamin D deficiency			
<50 nmol/L	17.9	15.0	15.3
<70 nmol/L	36.7	32.7	33.1
Calcium deficiency			
<1.15 mmol/L	0.4	0.6	0.6
<0.90 mmol/L	0.4	0.5	0.5
Number of children	87	688	775

Note: Blood level of vitamin A is measured in micromoles per liter (µmol/L), vitamin B12 in picomoles per liter (pmol/L), vitamin B9 and vitamin D in nanomoles per liter (nmol/L), and calcium in millimoles per liter (mmol/L).

Table 3 presents the levels of deficiency for different vitamins among children by age. The prevalence of vitamin B12 deficiency and calcium deficiency is substantially higher among children aged 6-11 months than among children 12 months and older. In contrast, the proportion of children with vitamin D deficiency increases with age (Table 3). The proportions of children who suffer from vitamin A and folic acid deficiency vary only slightly by age.

Table 3 Blood level of vitamins A, B12, B9, and D and calcium in children by age

Among children born since January 2009, percentage with deficiencies of vitamin A, vitamin B12, vitamin B9, vitamin D, and calcium, according to age group, Cambodia 2014

	6-11 months	12-23 months	24-59 months	60+ months	Total
Vitamin A deficiency					
Deficient (RBP <0.70 µmol/L)	11.1	8.8	9.3	8.7	9.2
Marginal (RBP <1.05 µmol/L)	33.8	26.1	27.7	35.5	29.2
Vitamin B12 deficiency					
Deficient (<150 pmol/L)	11.4	1.0	1.3	0.3	1.7
Vitamin B9 deficiency					
Deficient (<10 nmol/L)	8.4	4.3	7.8	11.3	8.0
Vitamin D deficiency					
<50 nmol/L	9.1	12.9	14.8	21.2	15.3
<70 nmol/L	9.1	27.4	34.9	39.5	33.1
Calcium deficiency					
<1.15 mmol/L	4.2	1.1	0.3	0.0	0.6
<0.90 mmol/L	4.2	0.4	0.3	0.0	0.5
Number of children	46	105	486	138	775

Note: Blood level of vitamin A is measured in micromoles per liter (µmol/L), vitamin B12 in picomoles per liter (pmol/L), vitamin B9 and vitamin D in nanomoles per liter (nmol/L), and calcium in millimoles per liter (mmol/L)

To alleviate these issues and death, considering taking multivitamins or supplement to treat stunting in height and weight and using some common natural remedies used to treat or reduce the symptoms of stunting and diabetes (fish oil, potassium, vitamin A, B9, B12, D, K, pyridoxine (vitamin B6), green tea, glucosamine sulfate, etc.) not only reduce the risk of premature death but may also have an important impact on childhood and adult health.

According to industry insiders, young Cambodians open to the Korean culture and lifestyle

are the main driver for the development of the industry. In the first three months of year 2013, the Kingdom imported cosmetic products worth \$9.9 million, representing an increase of 130 per cent compared to the same period last year as shown by import data from the Ministry of Commerce, instead of importing cosmetics from overseas.

By application, the dietary supplement market is divided into medical supplements, beauty supplements, and sports nutrition among others. By ingredient, the market has been segmented into vitamins and minerals, amino acids, botanical, etc. Dietary supplements come in both capsule and powder forms and provide the user with something that is missing from their usual diet. Fiber is a well-known dietary supplement, since it can be difficult for some people to get enough fiber from food alone. Green supplements are another popular type, enabling getting the nutrients from multiple servings of vegetables and other plants all at once.

<https://www.phnompenhpost.com/business/cambodia's-beauty-industry-blossoming>

<https://dhsprogram.com/pubs/pdf/FR312/FR312.pdf>, Cambodia Demographic and H

1.2 Current industry: products, customers, suppliers, and manufacturing processes

Supplementary product

Nutrition plays a vital role for normal growth and to maintain physical and mental fitness throughout one's life (Awasthi and Kumar, 1999). Nutrient deficiencies may contribute to growth retardation indirectly by reducing the intake of other growth-promoting factors, such as energy and protein. Moreover, several micronutrients, including zinc, iron, and vitamin A, are associated with immune function and risk of morbidity, which in turn affect growth (Rivera and Martorella, 1988). According to FIDR 2017, Cambodia is facing changes in people's dietary patterns with many options of nutrient-poor, unbalanced, and processed food and sugary beverages.

Childhood and adolescence diets not only influence the immediate health of children but may also have an important impact on adult health. The childhood diet must be adequate to support normal growth and development, and appropriate amounts of minerals are required since deficient intake of certain minerals can cause diseases and lead to abnormal development (ACC/SCN, 2000).

Customers

According to an economic estimation of the burden of malnutrition in Cambodia using the 2010 Cambodian Demographic Health Survey (CDHS) data led by the Council for Agricultural and Rural Development (CARD), the negative health outcomes from child malnutrition present a heavy economic burden of increased health expenditure coupled with decreased productivity and affect the rights of women and children to be healthy participants in society. The Royal Government of Cambodia, with the support of UNICEF/WFP, estimated that malnutrition has caused annual loss of over USD 250–400 million to Cambodia's gross domestic product (Bagriansky, et al, 2014).

The 2014 CDHS shows that, in nutrition, Cambodia did not meet the 2015 Cambodia Millennium Development Goal (CMDG) targets; 32.4% and 23.9% of children are still stunted and underweight, respectively, compared to the set target reductions to 24.5%

and 19.2%, respectively, by 2015. Therefore, the government is currently working on expanding the range of low-cost yet effective nutrition interventions to break the current cycle of increased mortality and poor health, impaired cognitive and physical growth, reduced learning capacity, and ultimately lower work performance, productivity, and earnings (RGC, 2015).

The assessment explored the economic losses associated with 14 nutrition indicators collected during the 2014 Cambodian Demographic Health Survey (a nationally representative survey) (RGC, 2015) via 4 discrete pathways: (i) mortality in children with consequent lost value of future workforce (Pathway 1); (ii) child cognition deficit resulting in inferior school performance and adult productivity (Pathway 2); (iii) current value of reduced productivity among working adults (Pathway 3); and (iv) current value of excessive and preventable healthcare and welfare utilization (Pathway 4). The same methodology was used in other countries, such as Laos (Bagriansky, et al, 2013), Albania, and Cambodia via the 2010 Cambodian Demographic Health Survey. The 14 nutrition indicators are described in Table 1 and are divided into three groups: (i) four indicators for pregnant women; (ii) eight indicators for children under five years; and (iii) two indicators for adults.

Table 4: Fourteen Nutrition Indicators: Cases and Risk Groups Derived from the Demographic Health Survey 2014, Cambodia

Risk Group	Nutrition Indicator	Prevalence	Cases or at Risk (000)	Risk or Deficit
393 thousand pregnant women	Low body mass index	14%	55.1	Infant and maternal mortality
	Short stature	5%	21.2	
	Anemia	53%	209.3	
	Annual projected neural tube defects (NTDs) due to folate deficiency ^a	-	0.722	
1.6 million children <5 years	Sub-optimal breastfeeding	35%	273.8	Mortality and Morbidity

Risk Group	Nutrition Indicator	Prevalence	Cases or at Risk (000)	Risk or Deficit
of age	Low weight for height	10%	137.1	Mortality
	Low weight for age	24%	341.3	
	Low height for age	32.4%	462.6	Growth, Development, and Productivity
	Vitamin A deficiency ^b	9%	124.2	Mortality
	Zinc deficiency ^c	67.5%	963.8	Mortality and Morbidity
	Iodine deficiency ^d	12.8%	182.8	Growth, Development, and Productivity
	Iron deficiency ^e : Childhood	7%	104.2	Growth, Development, and Productivity
10 million adults 15–64 years	Iron deficiency: Women	2.9%	159.7	Strength, Endurance, and Productivity
	Iron deficiency: Men	1.1% ^f	55.6	

Maternal malnutrition and sub-optimal breastfeeding along with child underweight, wasting, zinc, and vitamin A deficiencies bring about a higher risk of death among children less than 5 years of age, ranging from about one-third to more than 10 times the risk of well-nourished children (Table 5). These risks, taken from global literature -- together with national prevalence data from CDHS and after statistical adjustment -- are used to forecast 4454 thousand deaths annually, or nearly one-third of the total child mortality in Cambodia (estimated at 13,767 dead children under 5 years of age; Bagriansky, et al, 2013).

Table 5: Relative Risk of Mortality

	Relative Risk	
Child Mortality Associated with Maternal Nutrition Status		
Anemia	1.25	
BMI < 18.5	1.71	
Height < 145cm	2.2	
Neonatal Mortality Attributed to Low Birth Weight		
2000–2499 g	2.8	
<2000 g	8.1	
Post-Neonatal Mortality Attributed to Low Birth Weight		
Post-Neonatal Mortality	1.98	
Mortality Associated with Severe, Moderate, and Mild Wasting and Underweight		
	Wasting	Underweight
Pneumonia among children with anthropometric indicator z-score < –3SD	9.7	10.1
Pneumonia among children with anthropometric indicator –3SD < z-score < –2SD	4.7	3.1
Pneumonia among children with anthropometric indicator –2SD < z-score < –1SD	1.9	1.9
Diarrhea among children with anthropometric indicator z-score < –3SD	12.3	11.6
Diarrhea among children with anthropometric indicator –3SD < z-score < –2SD	3.4	2.9
Diarrhea among children with anthropometric indicator –2SD < z-score < –1SD	1.6	1.7
Measles among children with anthropometric indicator z-score < –3SD	9.6	7.7
Measles among children with anthropometric indicator –3SD < z-score < –2SD	2.6	3.1
Measles among children with anthropometric indicator –2SD < z-score < –1SD	1	1
Others among children with anthropometric indicator z-score < –3SD	11.2	8.3
Others among children with anthropometric indicator –3SD < z-score < –2SD	2.7	1.6
Others among children with anthropometric indicator –2SD < z-score < –1SD	1.7	1.5

Infant Mortality by Breastfeeding Behavior				
	0–6 Months			6–24 Months
	Predominant	Partial	None	None
Diarrhea	2.28	4.62	10.53	2.1
Pneumonia	1.75	2.49	15.13	1.92
All Mortality	1.48	2.85	14.4	3.68
Diarrhea- and Measles-related Mortality Associated with Vitamin A deficiency				
Diarrhea- and Measles-related Mortality	1.32			
Mortality Associated with Zinc Deficiency				
ARI	1.96			
Diarrhea	2.01			
Mortality Associated with Neural Tube Defects (NTD), Estimated by the United States Centers for Disease Control and Prevention				
NTDs	80% of 2 births out of 1000 births			

Note: SD (Standard deviation), BMI (Body Mass Index), ARI (Acute Respiratory Infection).

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1.3 Market trends and projections (ASEAN and global)

Market trends and projections

The Kingdom of Cambodia has a population of 15.3 million in 2019 based on the general population census (NIS, 2019). **Figure X** shows Cambodia's total population trend during the period 1962-2019. The primary industry in Cambodia is agriculture, with almost 80% of the population residing in rural areas and engaging in agriculture; moreover, the agricultural sector is composed of about 53.8% crop production, about 27.3% fisheries, about 12.8% livestock, and about 6.1% forestry (FAO, 2011). Despite the fact that thousands of small and medium enterprises (SMEs) have increased and expanded for the last few years, the production of functional food and/or health supplement is still very limited. This could be one of the potential sources of income for the country where the abundance of fish, seafood/weed, and diverse fruits is unexplored. In fact, the indigenous Cambodian fruits have been drawing great attention as a prospective source of functional food ingredients; these include Noni fruits (*Morinda citrifolia*), also known as Indian mulberry, Noni, Jamun (*Syzygium cuminii*), tamarind (*Tamarindus indica*), Aonla (*Phyllanthus emblica*), banana, and mango.

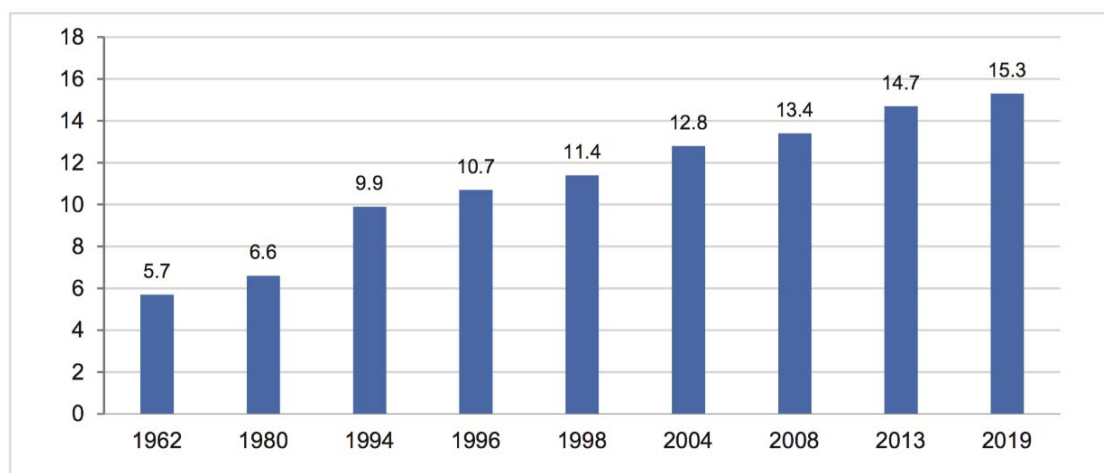


Figure. Cambodia's total population in millions, 1962-2019 trend (NIS, 2019).

With the continuous increase of global population, the market for functional foods and/or health supplements has tremendously increased in value terms of 26.7% from 2009 to 2013 (Leatherhead Food Research, 2014); moreover, it has been estimated that, with annual average growth rate of about 8.5%, the global market for these functional foods and health supplement product would exceed US\$ 305.4 billion by 2020. This growth in the market for functional food and/or health supplements could be attributed to the increasing concern over healthcare and its cost, steady increase in life expectancy, and desire for aging well (Bigliardi and Galati, 2013) for which the global markets for these products include the United States and Japan, followed by Asia-Pacific and European markets. In 2013, it was found that China had the highest expenditure on health and wellness retail products, followed by Brazil, the United States, Russia, and Mexico; global food firms account for more than 75% of the functional food market and include key players such as Coca Cola Co., Dean Foods, General Mills, Inc., Kellogg Co., Kraft Foods, Nestle S.A., and Pepsi Co. Key growth functional food categories include health snacks and breakfast products such as cereals and grains (Industry Arc, 2014). With sales of US\$ 43.9 billion in 2012 representing a 6.9% increase from 2011, the United States dominates the functional food market; it is expected to record a compound annual growth rate (CAGR) of 8.7% globally by 2020 (Euromonitor, 2014; Nutrition Business Journal, 2013). Moreover, Europe accounts for an additional 20.2% of the global market value, with European regional sales dominated by the United Kingdom, France, and Germany (Nunes, 2014). The Chinese functional food market was valued at US\$ 24.6 billion in 2012 and is expected to continue growing, but by a reduced percentage by 2017. The Brazilian market would also be an important market for functional foods and health supplements, valued at US\$ 8.7 billion in 2014 and expected to increase by 12% by 2019 (Euromonitor, 2015; Martins, 2015).

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1.4 Vision

“All people in Cambodia will have better health and wellbeing by effectively using local supplements to strengthen economic competency, quality of life, society, and environment in a sustainable manner.”

Prime Minister Samdech Hun Sen gave specific recommendations for dealing with malnutrition including the need for us to “Continue to invest in strengthening food diversification... by focusing on the diversification of climate-resilient, nutrition-rich crops and seeds, increase in the production of vegetables, fruit, poultry, and fish, food systems, and changes of attitude in order to promote food security and nutrition and food safety and to improve food quality for the people.

A vision statement spells out the role by which the Ministry of Health steers the development of the strategic intent to serve the Cambodian people, with emphasis and concentration on sector priorities and influence on the sector’s resources potential and core competencies to achieve the defined health development goals. The Ministry of Health reaffirms its strong commitment to “managing and leading the entire health sector effectively to ensure that quality health services are geographically and financially

accessible and socio-culturally acceptable to all people in Cambodia.” To achieve this long-term vision for the development of high-quality health, Cambodia is envisaged to achieve 3 goals.

1.5 Goals and objectives

Goal 1. Utilizing extraction, encapsulation, and spray dryer method to develop the supplement products in order to enhance the quality of life, wellness, and environment

- Objective 1.1: Support and promote encapsulation technology R&D and its applications in improving the quality of life, health, and medicine.
- Objective 1.2: Build collaborative networks among the public sector, private sector, and human resource sector in order to raise awareness; apply, integrate, and disseminate the knowledge acquired from nanotechnology development within the organization’s strategies
- Objective 1.3: Create a bio-health data information center

Goal 2. Improving the formulation technology and manufacturing industry to meet the market demand through extraction and encapsulation

- Objective 2.1: Value creation of local products
- Objective 2.2: Mechanisms for developing the ability for technology absorption and technology transfer from overseas; possessing legal-enabling factors; adoption of incentive creation measures to attract investment; and improvement of financial regulations

Goal 3. Ranking among ASEAN’s Top 3 in nanotechnology research and education

- Objective 3.1: Rank among the top three food supplement and agriculture exporters in ASEAN by increasing resource management efficiency in the production process and by reducing production costs
- Objective 3.2: Awareness of the negative and ethical impacts resulting from the application of nanotechnology

- Objective 3.3: Correct knowledge and understanding of the application of nanotechnology in consumer goods
- Objective 3.4: Acquisition of new patterns of learning development and social education

1.6 Scope and limitation (10 years): 2020-2030

The scope of TRM bio health is to provide clear focus and long-term direction for supporting the Fast Track Roadmap for Improving Nutrition and development partners, which will enable effective and coordinated response to improve maternal and young child undernutrition. Covering the wider supplement product market and value chains, it takes a focused approach to analyzing the main hurdles, enablers, and required actions toward realizing Nanotechnology's potential for Cambodia and linking the current strategy with multisectoral policies and documents and identifying areas requiring stronger partnerships and collaboration.

It has been decided that the analyses will focus on the arrival of novel nano-based food materials, innovative dietary supplement, food packaging, intelligent delivery mechanisms of nutrients and bioactive materials, implementation of green nanotechnologies for crop production, and nano-biosensors to provide safer foods, with a strategy that outlines a well-defined, evidence-based "roadmap" for Nanoencapsulation technology and extraction technology for nutrients including the overall goal, objectives, strategies, and activity plan from 2020-2030 by focusing specifically on what the ministry can do to address maternal and young child undernutrition. Therefore, it is limited in scope to address the full spectrum of causes of undernutrition, which requires a broad coalition of multisectoral interventions.

This review aims to present the scope and boundary of discussion on the recent innovations of nano-based Nano supplement technologies and Nanoencapsulation for drug delivery as patented knowledge across a number of applications, future trends, and impact in the food industry for 2020-2030.

Part 2. TRM Development

2.1. Future prospect

The Bio Health Roadmap (2020-2030) clearly defines the goals, strategies, measures, and directions for the country, with expectations on building Cambodia's capabilities to pursue advancements in technology and to cope with changes that may occur in the future. From the development of this TRM, the following are expected to benefit the country:

ECONOMIC SECTOR ON PEOPLE AND SOCIETY

- Acquisition of new patterns of learning development and social education
- Increase in abilities to control, prevent, and reduce the rate of problematic diseases
- Correct knowledge and understanding of the application of nanotechnology in consumer goods
- Awareness of the negative and ethical impacts resulting from the application of nanotechnology

CAMBODIA'S SCIENCE AND TECHNOLOGY CAPABILITIES

- R&D expenditure on nanotechnology increase
- Ratio of nanotechnology R&D personnel to population is 2.5 to 1,000
- Percentage of R&D personnel in business sector is 50%
- Ability to register no fewer than 100 patents and have no fewer than 500 papers published in international academic journals
- Mechanisms to create the ability for technology absorption and technology transfer from overseas; possessing legal-enabling factors; adoption of incentive creation
- Measures to attract investment; improvement of financial regulations

2.2. Environmental analysis

2.2.1. Trends/Drivers

STEEP	Driver
Social	Lifestyle change Age distribution of population, Longest life expectancy Impact/Influence on the quality of life (safety/security) Influence on the creation of job opportunities
Technological	New Product, New Technology, and New development in technology transfer
Economic	Money supply, gross product trends, unemployment levels, wage/price controls Impact on economic growth and productivity Influence on the creation and growth potential of SMEs Economic sustainability National competitiveness Economic transformation
Environmental	Sustainable development pressure, raw material sources, reuse and recycling industries, and related processes and regulations, biotechnology (from agriculture to drugs) Impact on the reduction of pollution, protection from pollution, and evaluation of impact from pollution Effect on Climate Change (adaptation and mitigation)
Political	Special income, Tax laws, Environmental protection laws, inter-ministerial harmonization Support in the administration/state policy and regulation (soft infrastructure)

2.2.2. Opportunities/Threats

Driver	Opportunity/Threat
A. Lifestyle change	Progress itself is up-to-date, civilization trend (O)
B. Age distribution of population	Pyramid 18-35 40% (O)
C. New developments in technology transfer	Obsolete Technology development is limited (T)
D. Licensing protection	No specific accountable institutions (T)
E. Gross domestic product trends	Affects local vendors (T)
F. Disposable income	Adequate and stable income (O)
G. Reuse and recycling	Increase of waste (T)
H. Biotechnology (from agriculture to drugs)	Limited resources and lack of domestic supply (T)
I. Special incentive	- Strengthening of one-stop service (T) - No tax for high technology transfer (O)
J. Inter-ministerial harmonization	Collaboration strengthening (T)
K. Input of raw materials	High agricultural production for agro-processing inputs of raw materials (O) Khmer Traditional medicinal plants are abundant (O)
L. Demand & Supply	Food safety consumption is increased (O) Local and international markets are increased (O) Food safety particularly processed food safety is limited (T) Supply chain infrastructure is limited (T)

2.3. Strategic products and services

2.3.1. Product feature concepts

In general, supplements that provide a mixture of micronutrients have been more successful in preventing stunting and are simpler to take and distribute. Multiple micronutrients together with energy and macronutrients are also needed for the management of moderate malnutrition. Multiple micronutrients may be delivered in the form of medicine-like supplements but may also be combined with food -- for instance, in milk drinks, fortified dried cereal mixes used to supplement complementary foods, or lipid

nutrition supplements.

Linear growth retardation or stunting may occur with or without low weight-for-age; in both cases, however, stunted or moderately malnourished children are deficient in micronutrients. Pregnancy and the first 2 years are critical periods. Dietary deficiency of zinc, iron, calcium, and vitamin A are especially common, often occurring together. Zinc is essential for adequate growth, and supplements have been shown to increase intrauterine femur length and to prevent stunting. The latter also provides essential fats necessary for growth. Micronutrient powders for home fortification are effective in preventing anemia, but the present combinations do not prevent stunting. Improving the diets of infant and young children is also possible, and increased intake of animal source foods can improve growth.

Therefore, the product concept of multiple micro-vitamins is the main point of development to help support pregnant women and children under the age of 5 through multiple micro-nutrition supplement. To transform this concept product and delivery into the customer target, some objectives and strategic need to be identified to develop the product market as follows:

- Objective 1: Promote health benefits among pregnant women and children under the age of 5 through multiple micro-nutrition supplement.
 - Strategic 1.1: Nutrition assessment especially focuses on nutrition that causes stunting, wasting, and underweight among children and mortality of mothers and children.
 - Strategic 1.2: Analysis of data and interpretation of information
 - Strategic 1.3: Product design: In many forms (Tablet, Powder, Liquid, Jelly...) in order to make the people enjoy taking them.
 - Strategic 1.4: Product cost: affordable price to enable the low-income family to buy the products as well.

- Objective 2: Guarantee that the micronutrients supplied do not exceed the FDA standard. Recommendation and basic daily nutrition requirements for pregnant women and children.
 - Strategic 2.1: Recognized by the national health ministry.
 - Strategic 2.2: Labeling (side effect, nutrition fact...) on packaging.
- Objective 3: Establish an R&D center with related technology and ability to come up with new innovative ideas in creating a product of the highest quality and which is trustworthy.
 - Strategic 3.1: Applied technology (Biotechnology, Nanotechnology, Extraction technology...)
 - Strategic 3.2: Continuous research and development of new multiple micronutrition supplements.
- Objective 4: All data and information are collected well and kept for further research plan and to follow up on existing activities, to check the deviation step, and to act based on the agreed-upon plan.
 - Strategic 4.1: Create a case control study before products are fully applied.
- Objective 5: Rank among ASEAN's top 3 in technology research and study on nutrition supplements and reduce malnutrition in women and children under 5 years to the targeted 15% by 2030.
 - Strategic 5.1: Educated target people and their families on the general knowledge of multiple micronutrition deficiency affecting health.
 - Strategic 5.2: Use media and social media to promote products.
 - Strategic 5.3: Collaborate with industries for the private sector to invest in new technology.
 - Strategic 5.3: Collaborate with related organization.

2.3.2. Grouping

There are 8 key product feature concepts as shown in Figure 1:

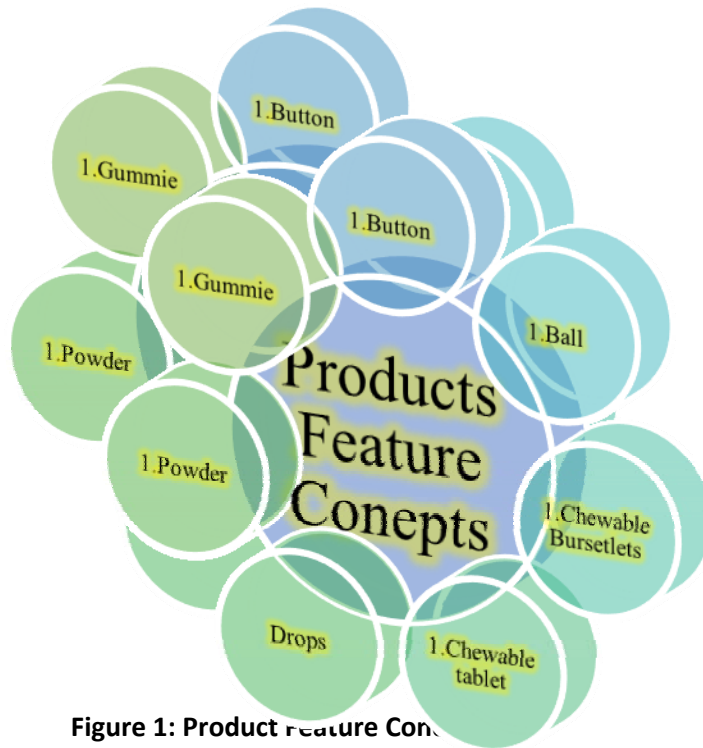


Figure 1: Product Feature Concepts

including dried products, frozen products, ready-to-eat products, bottled products, vacuum products, and packaged products.

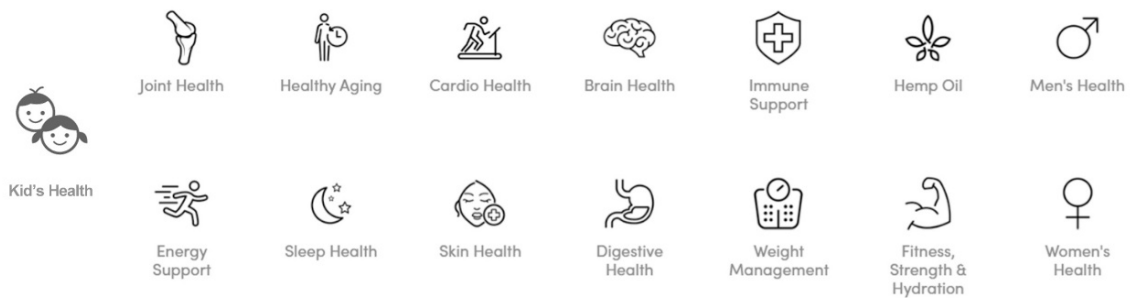


Figure 2: Group Product Supplements

Table 10: Identification and Prioritization of Key Products

OPPORTUNITIES & THREATS	OPPORTUNITIES			THREATS	
	Supplement Type	Target	Low	Middle	High
DIETARY SUPPLEMENT (MULTIPLE MICRONUTRIENTS)	Ball Buton Chewable Bursetlets Chewable tablet Drops Gummy Powder Liquid	Kids	<u>Multi-Vitamin, Vitamin B, Vitamin C, Vitamin D, Vitamin E, Vitamin A, and Vitamin K.</u> Multivitamin + High Dha Fish Oil	Micronutrient powder milk drinks lipid nutrition supplements. Adult Vita Gummies Daily Probiotic Vitamins Iron Supplement with Vitamin A, B, and C	Multiple micronutrient Multivitamins Dietary Supplements
		Women	Iron deficiency Vitafusion Calcium, Gummy Vitamins	Anti-Stress Night Multivitamin, Vitamin K12 Relieve joint pains Superfoods Beauty Greens	Dietary Supplements Vitamin B Complex Multivitamins for Women over 50 Nano Products
		Man	Men's Multi-vitamin Vitagummies Fish Oil, Calcium, Vitamin D, Fiber, Coenzyme Q10, Magnesium	Adult Vita Gummies Daily Fiber Folic Acid (vitamin B-9) Selenium, Whey Protein, Creatine	Dietary Supplements Coenzyme Q10 Omega-3 Fatty Acids Lycopene

2.3.3. Prioritization

The key products described above are classified as low, middle, and high priority. Among all key products, only Multiple micronutrient is classified as high priority.

SUGGESTED RAW MATERIAL AND FUTURE SMART MULTIPLE VITAMINS & SUPPLEMENTS

1. Asiatic centella (*Centella asiatica* (L.) Urb.) commonly known as Centella, Asiatic pennywort, or Gotu kola, is a herbaceous, frost-tender perennial plant from the flowering plant family Apiaceae. It is native to the wetlands in Asia and is used as a culinary vegetable and as a medicinal herb.



- Food benefits:
Leaves are bittersweet and rich in vitamins B1, B2, B3, B6, and K as well as minerals such as calcium, sodium, magnesium, and zinc. It enhances and strengthens the immune system and helps the brain improve memory.

2. Drumstick Tree or Moringa (*Moringa oleifera* Lam.) is a fast-growing, drought-resistant tree of family Moringaceae and is native to the tropical and subtropical regions of South Asia. Common names include moringa, drumstick tree (from the long, slender,

triangular seed pods), horseradish tree (from the taste of the roots, which resembles horseradish), and ben oil tree or benzolive tree (from the oil derived from the seeds). It is widely cultivated for its young seed pods and leaves used as vegetables and for traditional herbal medicine. It is also used for water purification. Oleifera is considered to be an aggressive invasive species. Moringa tree – perfect nutritional superfood for everyone Meet the moringa tree, the most important tree on this planet with the most amazing properties that ancient medicine claims prevent 300 diseases. New evidence and preliminary studies on Moringa oleifera suggest that Moringa leaves could be a miracle cure for malnutrition as well as a host of many illnesses (cancer, diabetes, liver and kidney problems, ulcers, anemia, high blood pressure, stress and much more). Moringa tree contains the perfect combination of nutrients, vitamins, minerals, and proteins (at least 90 nutrients, 20 amino acids, 46 anti-oxidants (boost memory, cancer-preventive) and 36 anti-inflammatory agents for everyone, especially for those who need extra energy boost and who are leading busy lives.



- Food benefits of Moringa oleifera:
 - Enhanced vigor: boost your body with naturally occurring nutrients to make your energy last longer
 - Mental, emotional, and mood balance: balance the chemicals that govern your thoughts and emotions. Experience new clarity focus and peace of mind.
 - Rapid recovery: Moringa oleifera's complete amino profile, in addition to a powerhouse of vitamins and minerals, is the perfect supplement after grueling workouts and physical activities.

- Nutrient-dense mother's milk: Iron, potassium, vitamins A, B, C, E, and many other important nutrients are all easily absorbed from *Moringa oleifera*, migrating from mother to the growing newborn through mother's milk. In Africa, it is called "Mother's Best Friend."
 - Healthy blood sugar levels: Antioxidants and other regulating compounds support healthy blood sugar levels and keep the blood free of toxic substances.
 - Beauty: Moringa tree oil has been used in hundreds of cosmetics and beauty products. In Africa, it is called "Never dying tree" because of its amazing resilience and beauty and healing properties. Moringa is rich in zeatin, a plant hormone with anti-aging effect on skin cells.
 - Jet lag: Because Moringa has high antioxidant content, it is perfect for combating jet lag
 - Moringa also helps with: increased joint flexibility, improved sensory perception (eyesight especially since it contains lots of vitamin A), better sleep, and enhanced memory.
 - Aphrodisiac: Moringa has traditionally been used to boost sexual stamina and desire and to treat male sexual disorders. A study (published in the International Journal of Pharmacy and Pharmaceutical Sciences) found that it enhanced sexual behavior in rats. Moringa contains saponins, chemical compounds that have been shown in other studies to support libido and levels of the sex hormone testosterone.
3. Kangkong (*Ipomoea aquatic* Forssk), commonly known as water spinach, is a crop that has been farmed in Asian cultures for generations. Due to its relatively simple farming process and growing demand, it is the crop of choice for many farmers. Kangkong is in high demand in Thailand, Indonesia, and Malaysia.



- Food benefits of Kangkong:

- High in Vitamins & Minerals

Kangkong is known for being a great provider of vitamins and minerals.

Kangkong is a great source of vitamin A when about 100 grams of kangkong is eaten. That's a whopping 210% of your daily vitamin A requirement. Vitamin A is essential for the health of skin, hair, and vision.

Another major vitamin found in kangkong is Vitamin C, which is great for protecting the body from various diseases. Vitamin C is also known to be great for skin and hair repair.

Apart from vitamins, kangkong is a leading source of vital minerals such as iron, magnesium, and calcium. Magnesium and calcium are extremely important for the health of bones and teeth. Kangkong contains 21% of your daily calcium and 18% of your daily magnesium needs per 100 grams..

These are just a few of the vitamins and minerals found in kangkong. Aside from these vitamins and minerals, kangkong is a great source of Folates, Niacin, Pantothenic acid, Pyridoxine, Thiamin, Riboflavin, Copper, Manganese, Phosphorus, Selenium, Zinc.

- Skin Health

The abundant vitamin C found in kangkong leaves does wonders for skin and eye health. Vitamin C has great anti-aging properties. Kangkong can reduce wrinkles significantly and can even speed up the healing process and formation of scar tissue. Vitamin C is also known to reduce the duration of skin redness.

4. Wild Vigna (*Vigna umbellata*)

Origin and distribution: Annual form, mostly grow in upland areas and is drought-tolerant.



Part of plant used: Seed used for soup, glutinous rice cake, and dessert.

Nutrition and health benefit: High protein content, high Sulfur amino acids and consequently high chemical score. Moreover, the level of resistant starch was also high (64–75%). Trypsin inhibitors suggest that it helps prevent cancer.

5. Sweet Potato (*Ipomoea batatas*)

Distribution and availability: Perennial vine and drought-tolerant, mostly planted in sandy soils in lowland areas.

Part of plant used: The roots are boiled, fried, or made into dried chips, and the leaves are used for cooking and animal feeding.



Nutrition and health benefit: High vitamin A, energy and carbohydrates, vitamin B6, vitamins C and D, and magnesium, a relaxation and anti-stress mineral.

6. Sleek Bash or Ivy Gourd (*Coccinia grandis*)

Distribution and availability: Found among the wild plants near the village along the fences.

Part of plant used: Leaf used for soup. Boiled leaves can be used for salad and eaten with fermented fish.



Nutrition and health benefit: Rich in vitamins and minerals including iron, Vitamin B2, Vitamin B1, dietary fiber, and calcium. Reportedly lowers blood sugar level, prevents obesity, prevents kidney stones, lowers heart disease and risk of stroke.

7. Romduol (Latin botanical name: *Sphaerocoryne affinis*, *Mitrella mesnyi*, or *Popowia aberrans*) is a Southeast Asian flower that grows up to 3-15 m high. The bark is smooth, flat, or flattened, splitting the leaflets, but opposite, finely fused; The lower part of the leaves are light green, in the lower branches and the arches are yellow, 2-lobed, with similar thickness, thick and hard. The dorsal flowers are shaped like 6 cucumbers, each 2-lobed and 3-lobed; the inner lobes are smaller and more curved than the apex, with the flowers blooming from December to March. 1-2 capsules are ripe, black, sour, and sweet. Romduol is a great herbal remedy for the following: fever, dizziness, heart disease, blood thinner, and blood pressure. In ancient times, Romduol flower was used as a female lip balm because of its fragrant aroma. Meanwhile, the flowers are used for producing herbal medicines, and the bark can be used as a home remedy.

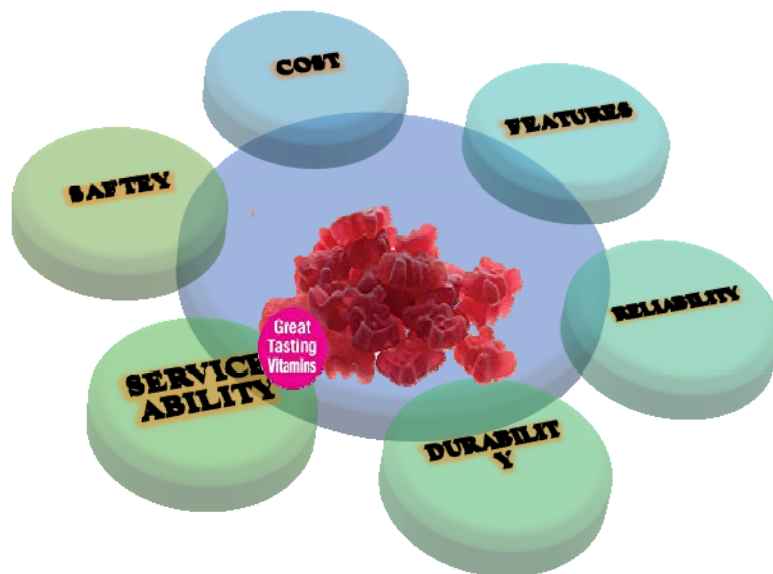


Cambodia has selected Romduol flowers to represent the national identity as per the Royal Decree dated 21 March 2005.

2.3.4. Performance dimensions

Quality is multidimensional. Product and service quality consist of a number of dimensions that determine how customer requirements are achieved. Therefore, it is essential that all dimensions that may be important to the customers be considered.

Figure 3: Products' Dimensions

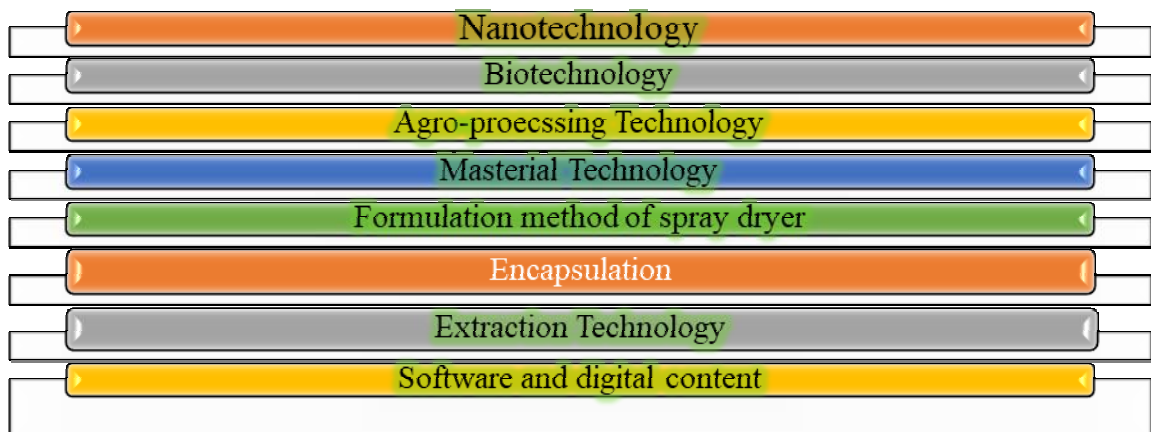


- Cost – Reasonable price and competitiveness compared to neighboring countries
- Purity – Has a strong market competitive
- Assurance - Assurance refers to the level of certainty a customer has regarding the quality of the service provided.
- Features – The product possesses all of the features specified or required for its intended purpose, product design, and performance specifications to enable familiarity with its intended uses and maintain close relationships with the end-users.

- Safety: no side effects on people
- Tangibles - Tangibles refer to a service's look or feel.
 - Design: Tablets, capsules, liquid, powder, jelly.... etc.
 - Good look, colorful
- Reliability refers to the dependability of the service providers and their ability to keep their promises.
- Durability is closely related to warranty and product, i.e., work longer and work well under special conditions.

The quality of products and services can be measured by their dimensions. Evaluating all dimensions of a product or a service helps determine how well the service holds up against meeting customer requirements.

Technology alternatives

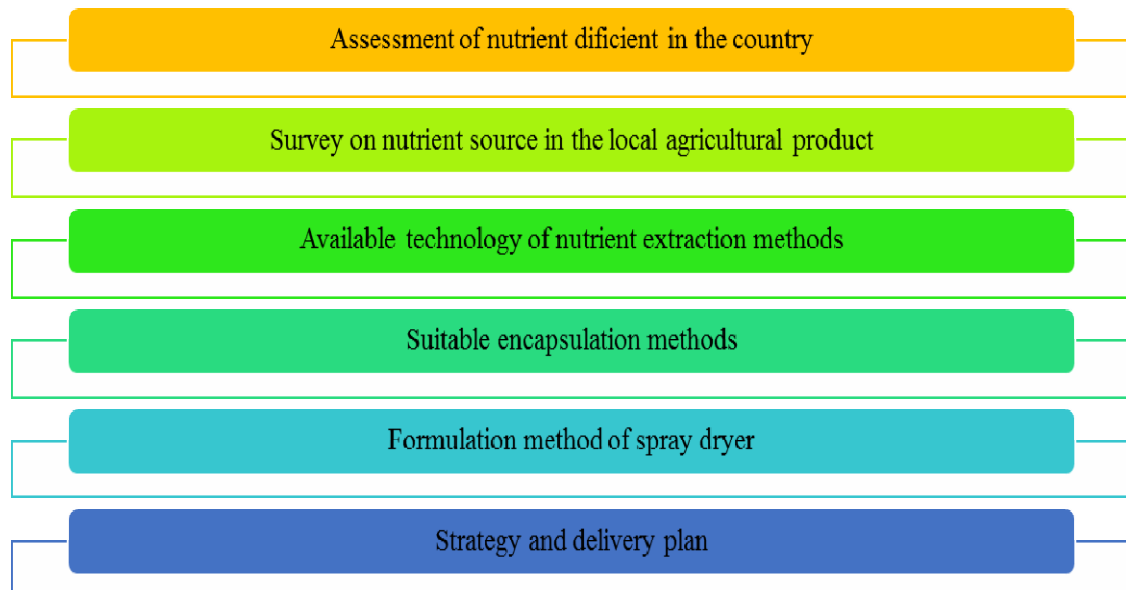


2.4. Key technologies

2.4.1 Technology solutions

Technology step is key to achieving and sustaining the objective of the bio-health project. The engagement from different related stakeholders in the private and public sectors pose another challenge in improving the long-term planning of product and technology. Moreover, the research institution and database management play an important role in the development of technology to be implemented and create a new idea of product delivery. The following diagram shows the doable step of technology from the local product to the supplement product involved in the step technology, such as nutrients extraction method, encapsulation, and spray dryer.

2.4.2 Grouping



2.4.3 Priorities

To define the high-priority technologies for developing the product and increase national competitiveness we need some technology to identify the development elements as follows:

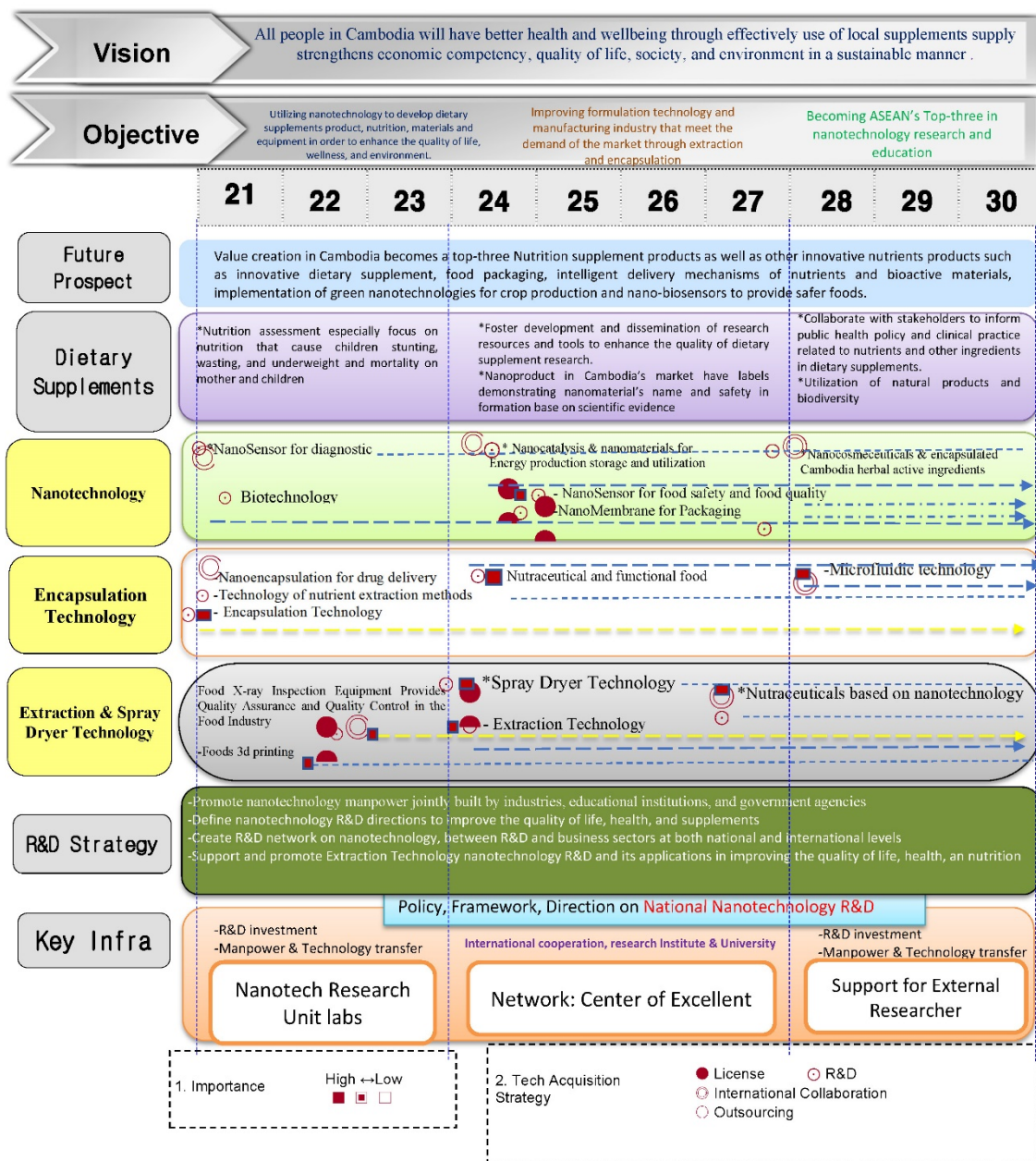
1. Defining the socioeconomic development of the two target clusters

- Agriculture and Food Sector;
- Multiple micro vitamin, Healthcare, and dietary Supplement products;

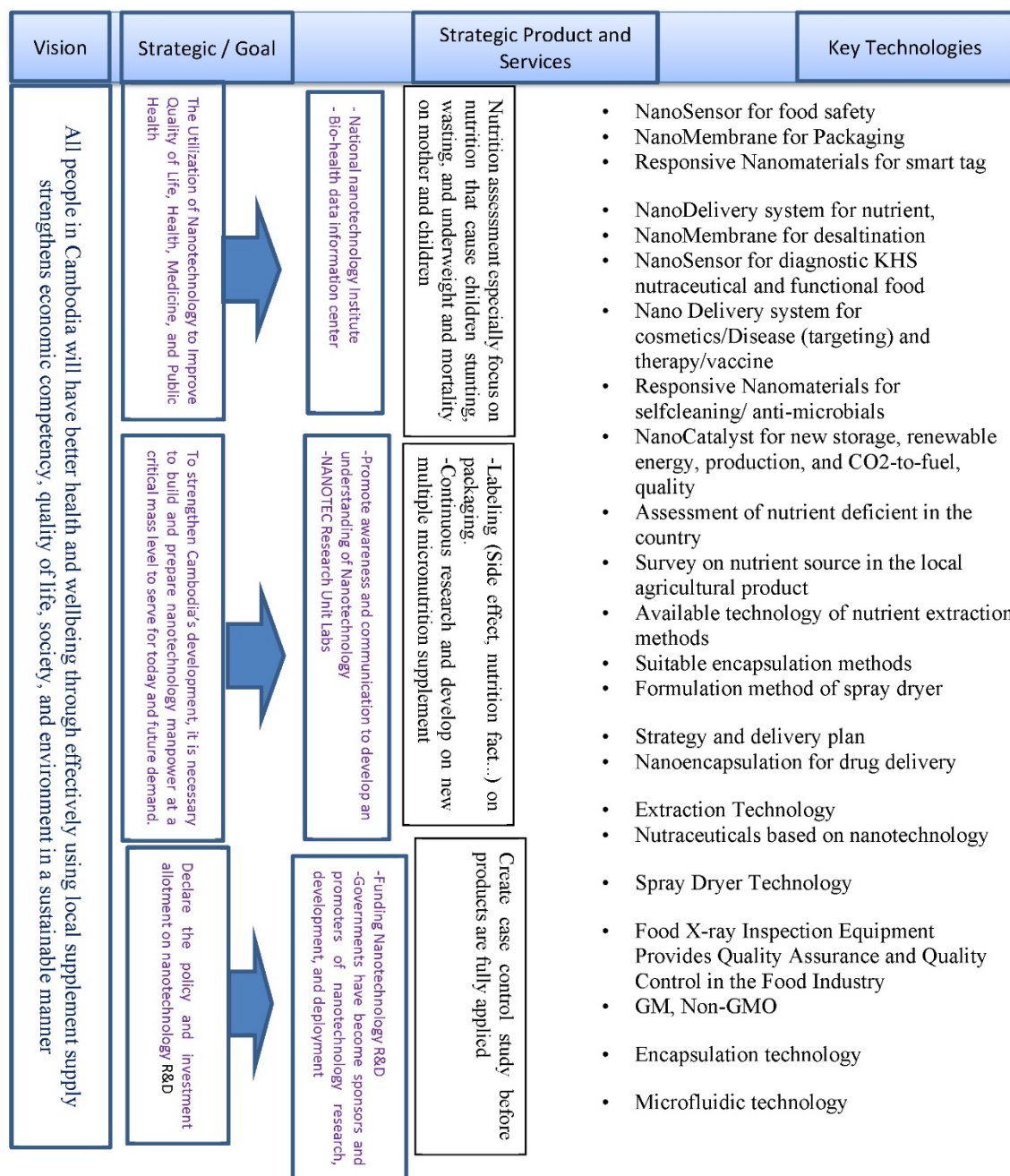
2. Defining the fundamental academic disciplines in nanotechnology and classifying core technologies into three areas:

- Encapsulation technology, technology of nutrient extraction methods, and spray dryer.
- Nanomaterials, Nanosensors for dietary supplement products, and Bionanotechnology.
- The technology platform is classified into three areas:
 - Nano Coating Technology,
 - Nanoencapsulation for drug delivery, and
 - Functional Nanostructure.

2.5 Macro TRMs for strategic products



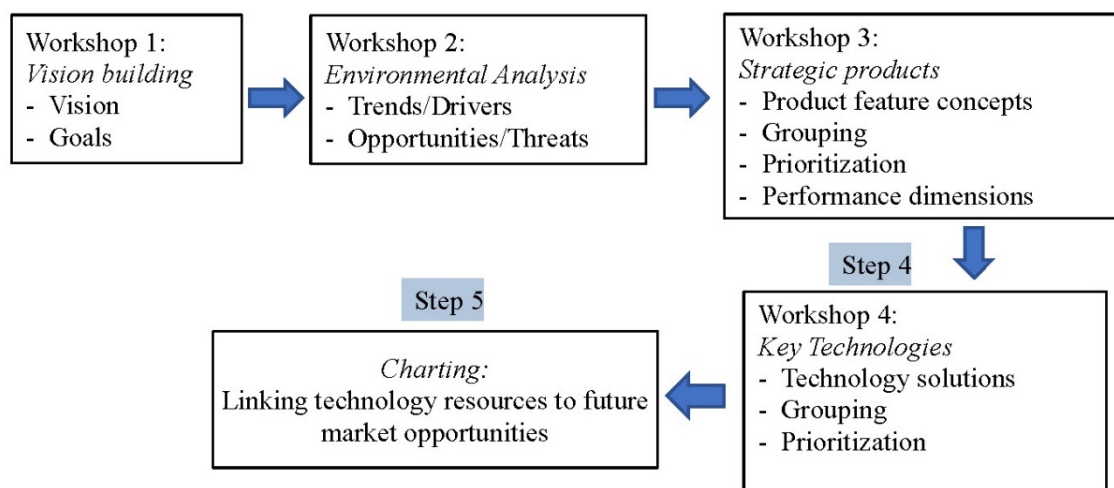
2.6. Micro TRMs for key technologies



Part 3. Recommendations

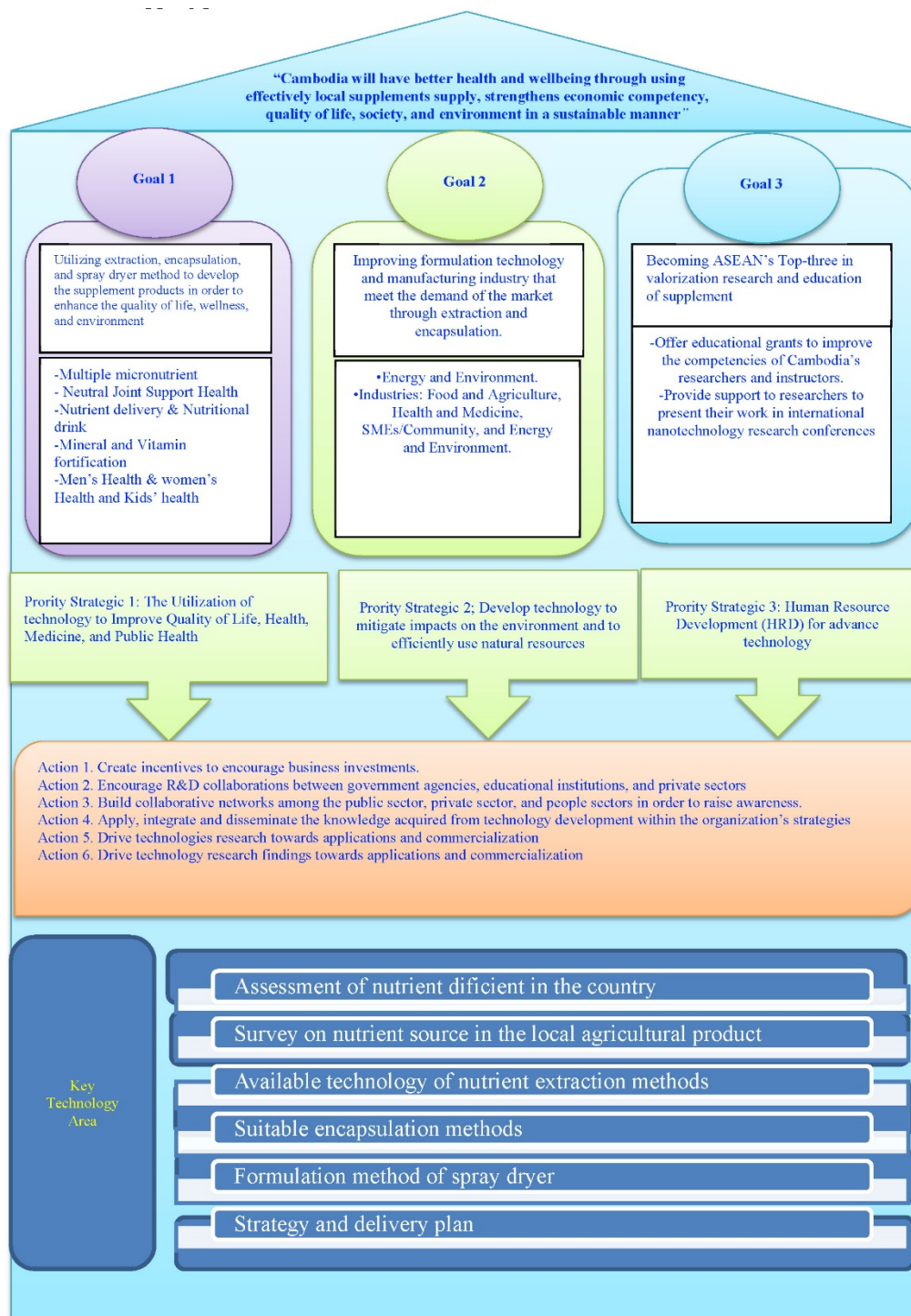
3.1. Plan to implement recommendations

Recommendation	Responsible Body	Schedule (in year)									
		1	2	3	4	5	6	7	8	9	10
Establishing Research Institute for Nanotech	NSTC, ITC, Ministry of Health, NIPH										
Curriculum Design	Ministry of Education, ITC										
Improving Intellectual Property Rights	NSTC, Ministry of commercial, Ministry of Handicraft , MoME										
Aligning encapsulation technology R&D Strategies with National Plans	NSTC, ITC, MoH, Ministry of Education, NIPH,										
Establishment of Research and Development Centers	New R&D Institute for ITC										
Establishment bio data centre	NSTC, ITC, MoEY, NIPH										
Capacity Building	NSTC, Ministry of Education, ITC										
Infrastructure Development	RICT, MINISTRY OF ICT										
Transferring Technological Knowledge	NSTC, MINISTRY OF ICT,										
National Nanotechnology Policy, Nanotechnology Master plan	NSTC, MoH, ITC, NIPH										
Nanotechnology Literacy and Awareness Campaigning	NSTC, ITC										
Promotion of Private Nanotechnology Industry, SMEs	ITC, Ministry of Labor, MINISTRY OF ICT, Ministry of Health, MoC										



Part 4. Appendices

4.1. Roadmapping process



4.2. Participants

Name	Position	Responsibility	Organization	e-mail
Mr. SOK PHAL	Department Director	Vision, Goal, Objective, Scope	GS-NSTC	sokphal7778@gmail.com
Dr. Kuok Fidero	Dean of the Faculty of Chemical and Food Engineering	Current Industry, Market Trend, Environment Analysis	ITC, Techno	fidero@itc.edu.kh
Dr. Tourt Sodanea	Nutritionist	Strategy Product, Macro TRMs	Private Sector	sodaneatuot@gmail.com
Dr. Ung Porsry	Researcher of Food Science	Background, Key Technology, Micro TRM		ungporsry@yahoo.com
Mr. SEK SOKNA	Deputy Chief of Bureau	Reference	Ministry of Health	sokna168@gmail.com

Technical Note:

Food X-ray Inspection Equipment Provides Quality Assurance *and* Quality Control in the Food Industry

The ability of x-ray inspection systems to provide unsurpassed detection of a wide range of **physical contaminants**, including **metal, glass, mineral stone, calcified bone**, and **high-density plastics**, means that they can be used to keep contaminants out of the production process as well as catch contaminants before the products leave the factory. It simply depends on where they are located in the production process, i.e., critical control points (CCPs).



Nanosensors are any biological, chemical, or surgical sensory points used to convey information about nanoparticles to the macroscopic world. Their use mainly includes various medicinal purposes and as gateways to building other nanoproducts, such as computer chips that work at the nanoscale and nanorobots. Presently, there are several ways proposed to make nanosensors, including top-down lithography, bottom-up assembly, and molecular self-assembly.

The nanosensors used in food analyses combine knowledge of biology, chemistry, and nanotechnology. Nanosensors are placed in food packaging to control the internal and external conditions of food from a microbiological point of view; the main objective is to

reduce the nanosensors' pathogen detection time from days to hours or even minutes.

A smart tag is the selection of special content that can be used for hyperlinking or adding to various utilities in Microsoft Outlook or other software programs. By identifying smart tags, users can save time by turning various words and phrases into proper nouns that can function as data objects in a Microsoft interface.

Nanomembrane or filtration (NF) is a relatively recent membrane filtration process used most often with low total dissolved solids water such as surface water and fresh groundwater, for the purpose of softening (polyvalent cation removal) and removal of disinfection by-product precursors such as natural organic matter and synthetic organic matter. Nanofiltration is also becoming more widely used in food processing applications such as dairy for simultaneous concentration and partial (monovalent ion) demineralization.

Nanodelivery system for nutrient provides a means of controlling stability, solubility, and bioavailability and also supports the controlled release of food bioactives. There are two main types of nanodelivery systems: liquid and solid. Liquid nanodelivery systems include nanoemulsions, nanoliposomes, and nanopolymersomes. Solid nanodelivery systems include nanocrystals, lipid nanoparticles, and polymeric nanoparticles.

GM stands for "genetically modified." An organism, such as plant, animal, or bacterium, is considered genetically modified if its genetic material has been altered through any method, including conventional breeding. A "GMO" is a genetically modified organism.

non-GMO applies to any foods or ingredients that have not been genetically modified. There are many GMO products on the supermarket shelves that you might not be aware of unless you read the ingredient labels. Note, however, that many GMO-free foods now make it a point to label their products as non-GMO as there is intense debate on the overall health risks associated with GMO vs. non-GMO foods.

Supplements Production Line



Outline of Technology Roadmap for Agro-processing

4.2.3 Agro-Processing TRM (draft)



ព្រះរាជាណាចក្រកម្ពុជា
ជាតិ សាសនា ព្រះមហាក្សត្រ

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Science and Technology Orientation for Growth and
Product Development Marketing

រៀបចំដោយ ៖ **ការិយាល័យគ្រប់គ្រងការកែច្នៃ**

អង្គភាព ៖ **នាយកដ្ឋានកសិ-ឧស្សាហកម្ម**

ក្រសួង-ស្ថាប័ន ៖ **កសិកម្ម រុក្ខាប្រមាញ់ និងនេសាទ**

រាជធានីភ្នំពេញថ្ងៃទី ០១ ខែមិថុនា ឆ្នាំ២០១៩

1st draft

Executive Summary

Part 1: Introduction

1.1 Background: technology, socioeconomic, environment, existing policy

In a recently released Cambodia Policy Note titled [*“Benefitting from the Digital Economy.”*](#) the World Bank assesses four interconnected policy areas in Cambodia: Entrepreneurship and Innovation; Digital Economy; Microfinance and Household Welfare; and Social Assistance.

Science, technology, and innovation (STI) are important drivers of innovative capacity and global competitiveness and critical inputs for structural economic transformation. There is a close relationship between economic dynamism and technological innovation performance. This is particularly evident in the stellar economic performance and technical prowess of China, Hong Kong, Singapore, South Korea, and Taiwan, all of which invest heavily in STI through efforts in promoting higher education and technical training to create a highly qualified, skilled, and disciplined workforce at all levels to meet the demand of inclusive, sustainable, rapid industrialization, which led them to achieve exceptional long-term growth.

A country’s capacity to harness the potential of STI for industrialization, growth, and development depends on an enabling innovation ecosystem including policy and regulatory framework, physical infrastructure, human capital development, and research and development (R&D) investment.

Acknowledging that Cambodia has seen great success in poverty reduction and shared prosperity, the report notes that the focus on tourism, agriculture, construction, and garment sectors as growth drivers must be diversified to ensure sustained economic growth for the next 20 years. The report highlighted 4 key areas that need to be addressed for Cambodia to reap greater benefits from the digital economy:

- Low fixed-broadband subscriptions: High mobile penetration is not matched by the widespread use of high-speed fixed-broadband services, making Cambodia lag behind in regional and global averages.
- Low digital adoption in the corporate sector: Less than one-fourth of businesses in Cambodia had a website in 2017 compared to the world media of 46 percent. This is compounded by the lack of service data servers, low IT skills, and incomplete legal framework.
- Lack of leadership in digital government development: Despite the recent establishment of a Government Data Center, Cambodia does not have a guiding framework and lacks security standards for websites and cloud services.
- Investment needed in infrastructure, regulations, and skills: Focus should be on the following: mandating greater passive infrastructure sharing among Cambodia's telecom operators; creating a Digital Skills Readiness Strategy; adopting laws in e-commerce, cyber security, and data protection and privacy; and better support for the Digital Government Strategy.

The biggest takeaway from the report is that there are plenty of policy options and untapped potential that Cambodia can leverage, including from the entrepreneur sector.

“While the country’s economic outlook remains positive, Cambodia could begin to explore new growth drivers that will create jobs and boost prosperity over the next 20 years. The next economic transformation of Cambodia will depend on its ability to empower domestic entrepreneurs and citizens with capabilities and tools for the modern economy of tomorrow.”

Nonetheless, the responsibilities of Science, Technology and Innovation are crucially relevant interaction with cross-cutting issue from various organizations and institutions. In 2014, the Royal Government of Cambodia established through Royal Decree NS/RKT/0916/1136 dated 08 September 2014 the National Science and Technology Council (NSTC), which consists of 22 ministries as members including the Supreme National Economic Council, Cambodia Chamber of Commerce, and research

representative. Its mission is to enable Cambodia's technological capability to catch up with that of other countries in the region and the world. Cambodia is moving in the right direction. As announced by the Ministry of Posts and Telecommunications in early 2018, the government aims to transform the country into a predominantly digital economy by 2023. The country is taking steps to develop its S&T sector including ICT infrastructure. The Ministry of Education, Youth, and Sports is promoting STEM (science, technology, engineering, mathematics) education from primary school to post-secondary levels. This is in keeping with IDP 2015–2025, which emphasizes the need to embrace modern S&T and invest in R&D and highlights the linkages between S&T and other priority sectors. Cambodia's new TVET Policy 2017–2025 sets four goals: to improve TVET quality to meet national and international market demand; to increase equitable access to TVET for employment generation; to promote public-private partnerships and aggregate resources from stakeholders to support the sustainable development of the TVET system; and to improve governance of the TVET system. All in all, extra effort and policy coordination and visionary, inspiring, and determined leadership at all levels are still needed for industrialization. The Cambodia National Science and Technology Master Plan 2014-2020 was launched in October 2014 by the Ministry of Planning with support from the Korea International Cooperation Agency (KOICA). Industrial innovation is a key focus, **particularly in the areas of agriculture**, primary industries, and information and communication technology (ICT).

At the same time, the Ministry of Agriculture, Forestry, and Fisheries (MAFF) has the Department of Agro-Industry (DAI) as the secretariat unit responsible for the agro-industry development sector; in particular, agro-processing promotion and development are being pursued to contribute to the IDP 2015-2025, which mentioned clearly archiving 12 percent of the agricultural processed product export by 2025. Recently, DAI has been formulating the **AI4.0 Strategic Plan** (Strategic Plan for Cambodian Agro-Industrial Sector Development 2019-2030) under the support of the international technical assistant from the Agriculture Service Program for Innovation, Resilience, and Extension. **AI4.0** aims to improve and modernize the agro-processing small and medium enterprise during the interaction period of the scientific and innovative diversification and transformation era in the three stages.

Technology roadmapping is a needs-driven technology planning process to help identify, select, and develop technology alternatives to satisfy a set of product needs. A technology roadmap is the document generated by the technology roadmapping process. There are 2 types of technology roadmap (TRM): TRM from market-driven planning and TRM from technology-driving planning. In particular, TRM from market-driven planning should be considered for Cambodia's context. First, industry trends analysis should be the focus, followed by technology roadmap development.

Therefore, Cambodia's economy will record double growth derived from science, technology, and innovation as the main driver of industrial modernization and economic catch-up. Mostly importantly, the small and medium enterprise will develop considerably and gain the most, with the latest digital transformation and technologies (artificial intelligence, big data, Internet of Things, block chain technology, virtual and augmented reality) propelling manufacturing and services toward Industrial Revolution 4.0.

1.2 The current industry: products, customers, suppliers, and manufacturing processes

In Cambodia, there have been few food industries that create innovative solutions for the poultry processing industry, meat processing, advanced food processing, and fish processing industry, driving excellence in performance, food safety, and sustainability. STI ensures that machinery engineering will last longer, perform better, and enhance business productivity.

Cambodia spent about \$200 million on food and beverage imports to supply the domestic market last year, recording an increase of more than 10 per cent from the previous year according to official data from the Ministry of Commerce in 2013.

The top 10 companies in the field of agro-processing and F&B in terms of modern technology and innovation are as follows:

Table 1: Top 10 Companies in the Field of Agro-processing

No.	Name of company	No.	Name of company
1	C.P. Cambodia Co., Ltd.	6	Cambodia Brewery Limited
2	Ly Ly Food Industry Co., Ltd.	7	La plantation organic processing industry
3	Khmer Foods Group Co., Ltd.	8	Cambodia Beverages Company, Ltd.
4	Khmer & Korean Trading Co., Ltd.	9	Golden Rice Cambodia
5	Hyundai Agro (Cambodia) Co., Ltd	10	Kirirom Food Production

**Figure 1: Rice Miller Machine at Golden Daun Keo Rice****- C.P. Cambodia Co., Ltd.**

C.P. CAMBODIA CO., Ltd. was established in 1996 through investment by Charoen Pokphand Foods Public Company Limited. The company in Cambodia started running a business dealing with animal feed and poultry. After that, the company expanded to the swine business and the integrated animal business in 2008. The company built the food processing plants to produce food products for the Cambodian people. They consume the products that they have produced by themselves. Currently, the company has total assets valued at more than 150 million dollars, over 1,500 Cambodian employees, and farmers

and over 1,000 families who join the animal farm business with the company all over Cambodia.

Type of Business of C.P. CAMBODIA CO., Ltd.:

The integrated agro-industrial business of agriculture and animal feed is categorized into 3 sections:

1. Animal Feed Business: the company is the pioneer and leader in feed production and sale in Cambodia. The main products of the company are swine and chicken feed. The company has feed-mill plants located in many provinces throughout Cambodia. The feed-mill plants of the company are standard plants. We have professional engineers and high technology to control our production. The Company sells to small independent farmers and large farms, both directly and via appointed agents or distributors across Cambodia.

2. Farm Business: Separated into 2 parts, i.e., poultry and swine business. The company is the leader in the research and development of natural genetic selection, with the aim of obtaining quality breeds that are healthy, sturdy, and suitable for farming conditions in Cambodia. The company produces the parent stock of the animals as well as piglets, broiler chicks, layer chicks, and layers by using the company's feed for feeding. The company has built a standard farm system with a good prophylactic system before entering into the farm, and it also has veterinarians who are able to provide animal healthcare and consult regularly in every contracted farm along with controlling animal breeding efficiency. The farmers can get advice from the veterinarians at all times.

3. Food business: The company adds value to its basic processed meat through additional processing in order to produce cooked, semi-cooked, and ready-to-eat food products such as chicken, pork, other parts of animals, or processed food products such as sausage, meatball, pork pate, and Jayor chicken. The products are then packed, chilled or frozen, and distributed either under the CP brand via convenience stores, supermarkets, fast food stores, and retailers and wholesalers.

CPF and Sustainability

The company is dedicated to the research and development of innovations that add values and benefits to enhance quality and food safety and is able to respond to trend changes. This is to ensure that innovations will enhance the worth and value of the business, society, and environment. Meanwhile, the company also focuses on promoting knowledge, capabilities, career, and life among all employees and business partners as well as the community, including the conservation of natural resources and the environment, which incur significant food production costs. The company believes that innovation and efficiency will enable it to enhance quality and food safety and create a healthy, self-reliant society regardless of the limited natural resources. This way, the company can reduce adverse environmental impact as well as reinforce social and economic sustainability based on our vision of **“The kitchen of the world.”**

The Company produces livestock feed in the form of concentrate, powder, and pellets for broilers, layers, swine, and duck. Agricultural products such as corn, soybean meal, and broken rice are used in the production of livestock feed under fully computerized formulation and production processes to ensure consistent quality as required. Quality control at different stages of production helps raise confidence in the quality of the Company's products. As the leader in the business, the Company's products are well-known among customers. Therefore, an important marketing strategy is to maintain high product quality as well as provide excellent customer service continually.

Ly Ly Food Industry Co., Ltd.

Ly Ly Food Industry Co., Ltd. has developed many rice cracker products including milk cheese rice crackers, ginger rice crackers, original rice crackers, spicy rice crackers, snow rice crackers, durian rice crackers, fried short tube snack (tomato flavor, BBQ and seafood flavor, squid flavor, spicy flavor, onion flavor, crab flavor, burger flavor, roasted duck flavor, potato flavor, chive chicken flavor, fish flavor, sweet corn flavor, and shrimp flavor), short stick snack (chocolate flavor, jackfruit flavor, and durian flavor), long tube snack (chocolate flavor, milk flavor), and dried fruits-vegetables in different packaging. Ly Ly Food Industry

Co., Ltd. exports its products to the following international markets: USA, Myanmar (Burma), Vietnam, Canada, Korea, Australia, Malaysia, Italy, China, Thailand, Indonesia, and Mauritius.

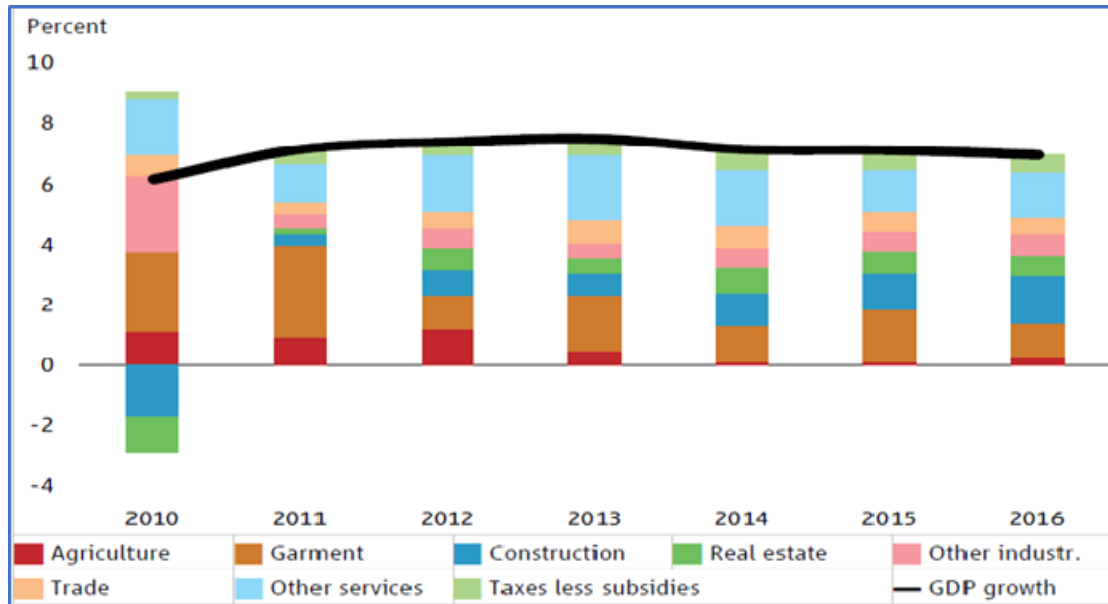
As a joint venture between local company Ly Ly Food Industry Co., Ltd. and Japanese firm Kameda Seika Co., Ltd., Ly Ly Kameda Co., Ltd., was officially launched on Saturday. The \$16 million factory, which can produce more than 3,000 tons of crackers annually for export, is 49 per cent owned by Ly Ly Food Industry and 51 per cent owned by Kameda Seika. It employs about 250 workers. Ly Ly Food Industry's first factory also has a production capacity of more than 3,000 tons a year, and about 65 per cent of the factory's products are sold locally with the remaining 35 per cent exported.



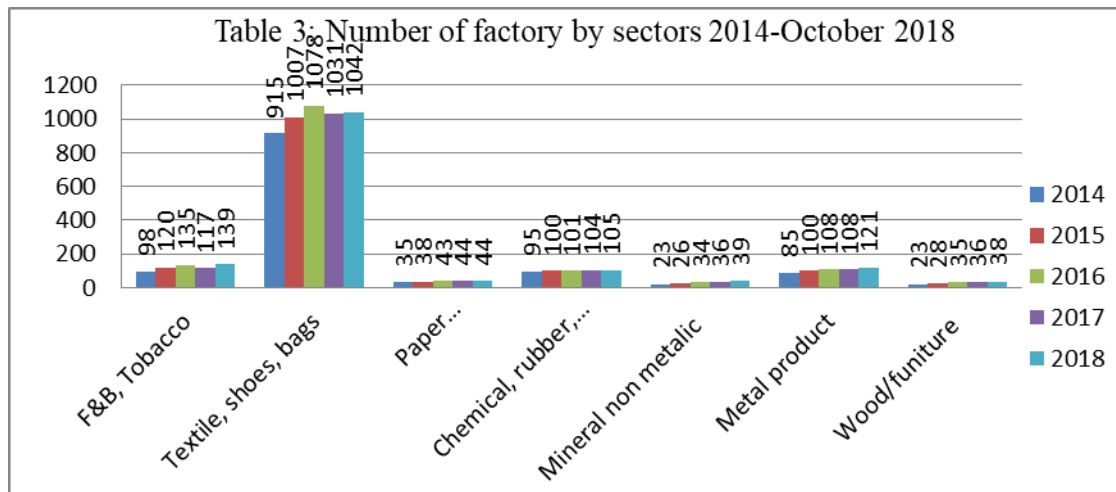
Figure 2: Semi-automatic Cutting Machine of Cassava Chips

- Khmer Foods Group Co., Ltd.

KFG is now one of the largest rice suppliers and exporters from Cambodia. KFG supplies rice to a diversified clientele, from local consumers to supermarkets and to producers of rice overseas. There are 2 types of rice including fragrant rice (Pkar Romdoul and Sen Kra Oub) and non-fragrant rice (long grain white rice).

Table 2: Share of Manufacturing Sector in GDP at the Current Prices (%)


Source: Ministry of Industry and Handicraft

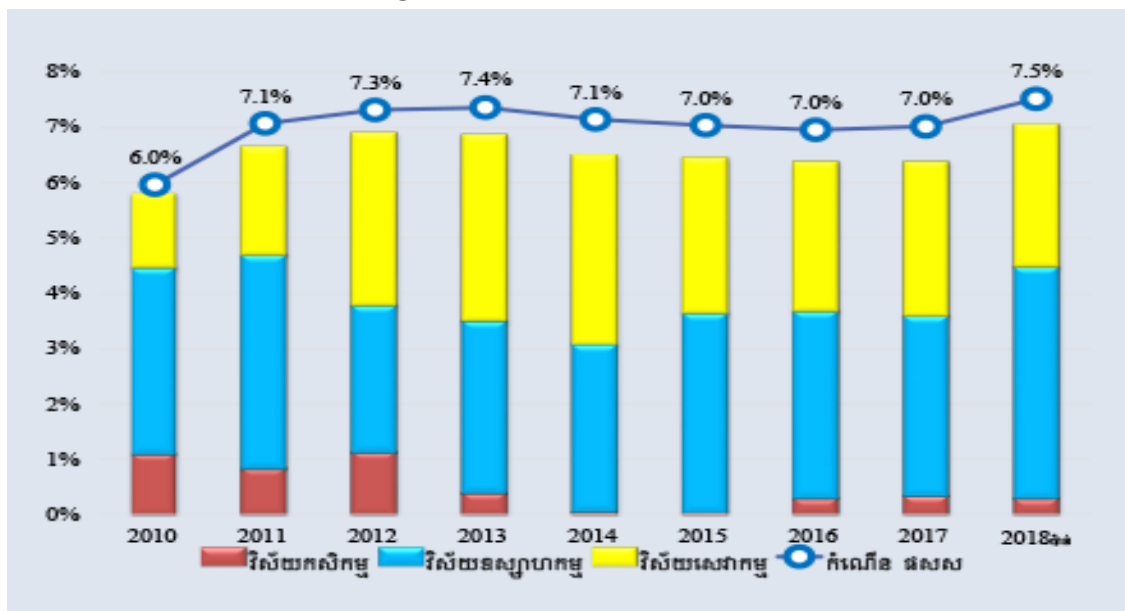


Source: Ministry of Industry and Handicraft, 2018

a. Market trends and projections (ASEAN and global)

In Cambodia, the economy has been growing by 7.2% in the last two decades. This growth is forecast to continue stably for the next decade when peace and political security and stability are maintained.

Figure 3: GDP Growth 2010-2018



Source: Ministry of Economic and Finance

b. Vision

Agro-Foods Processing will be equipped with modernized facilities, automation, professional skills demanded at all levels, high productivity, quality, standard, and safety, inclusive and challenging in the free markets.

c. Goals and objectives

The science, technology, and innovation roadmap 2019-2030 will be the long term national strategic action plan to contribute to socioeconomic growth and share in

achieving the goals below.

- Contribute to poverty reduction and improve the living standard of the Cambodian people.
- Contribute to Cambodia's macroeconomic development and achieve the long-term goal and vision for 2030 and 2050.
- Contribute to increasing and diversifying processed products and commodities in order to enhance food nutrition and promote the quality, safety, and export of the manufactured product (non-garment).

The objectives of this roadmap are:

- Product development, career creation, and commercialization.
- Coordinate the supplier system with high quality of raw materials and inputs, stability, and competitive price and sustainability for agro-processing production and increase the value added.
- Develop the roadmap in the field of science, technology, and innovation by aligning the needs of the food industry from food engineering and advanced food engineering and/or nanotechnology in food processing.
- Promote Digital, Artificial Intelligence and Block Chain Technology (AI).
- Promote R&D in robotics and automation in food science and technology.

d. Scope and limitation

The scope of this roadmap covers three priority sub-agricultural sectors:

- **Agricultural subsector:** Crop and industrial crop production, horticulture (fruits and non-fruits) and vegetable production
- **Animal and husbandry subsector:** Animal husbandry and poultry production

- **Fishery subsector:** Fisheries production involving both seafood and freshwater food processing production.

The roadmap for agro-processing will cover the pre-harvesting and post-harvesting areas, the primary processing stages, and the secondary processing stages. Those stages include: harvesting in every step semi-manually and automatically; post-harvesting steps such as grading, cleaning, smoking, dehydration, fermentation, packaging, etc. The primary and secondary steps include all processes such as shelling, peeling, filling and seaming, pre-cooling, chilling, pre-packing, cleaning and disinfection, pasteurization, sterilization, packaging, preservation, loading, palletizing, etc. until the delivery stages of the finished products.

This roadmap does not cover various agro-processing production areas such as:

- Salt production
- Water treatment production
- Medical and cosmetic production
- Tobacco production

Part 2: TRM Development

2.1 Future prospect

In the Cambodian context, the main keys of success in science, technology, and innovation are strongly committed and supported as follows:

- A technology-driven growth strategy led by strong government agencies that are able to formulate, implement, and coordinate STI strategies and policies.
- An ecosystem conducive for creating a strong national innovation system for industrialization that engages all stakeholders in technology-driven growth.
- Adequate financial resources to support STI institutions.
- In addition, essential physical infrastructure at the national level should have been established such as the following:

Table 4: Future Expectation Regarding Physical Infrastructure Development by 2050

Name in Khmer language	Name in English
រាជបណ្ឌិតសភាវិទ្យាសាស្ត្រ និងបច្ចេកវិទ្យាកម្ពុជា	Royal Academy of Science and Technology of Cambodia
មជ្ឈមណ្ឌលវិទ្យាសាស្ត្រ បច្ចេកវិទ្យា និងស្រាវជ្រាវ នវានុវត្តន៍កម្ពុជា	Cambodia Center for Science, Technology, and Research Innovation (CSTAR)

2.2 Environmental analysis

2.2.1 Trends/Drivers

Table 5: STEEP Analysis of Trends & Drivers

STEER	Drivers
Social	Labor force, Human resources
Technological	Technology, Qualification & Standards, R&D on STI
Economic	GDP (annual income), Demand & Supply, Input of raw materials
Environmental	Global warming
Political	Peace and stability, Policy support

2.2.2 Opportunities/Threats

Table 6: Analysis of Drivers

<i>Drivers</i>	<i>Opportunity (ies)/Threat (s)</i>
1. Peace and stability	- Development of SME in the agriculture sector; Agro-Processing subsector (O) - Instability of the markets (T)
2. Policy support	- RGC has set a clear target to archive around 12% of agricultural processed exports by 2025 (O)
3. GDP (annual income)	- The employment rate is high, and income is increasing each year (O)
4. Labor force	- Middle-age persons account for approximately 80% of the total population (O) - Technical and professional skills are limited (T) - Immigrant (T)
5. Technology	- Technology transfer from developed countries (O) - SMEs in Agro-Processing face limitations in terms of technology, equipment, and facilities (T) - Agro-Processing machinery and engineering company are limited (T) - No Digital, Artificial Intelligence and Block Chain Technology (T)
6. Demand & Supply	- Food safety consumption is increased (O) - Local and international markets are increased (O) - Food safety and Processed food safety are limited (T) - Supply chain infrastructure is limited (T) - Price is higher (T)
7. Qualification & Standards	- SMEs in Agro-Processing have low qualification and standards such as GHP, GMP, HACCP, and ISO standards (T)
8. R&D on STI	- R&D investment from the government and private sector is still lower than that of ASEAN Member States and developed countries* (T)
9. Input of raw materials	- High agricultural production for agro-processing inputs of raw materials (O) - Sell the inputs of raw materials to neighboring countries at a low price (T)
10. Global warming	- Low productivity leads to low yield and lack of production supply (T) - Lack of water supply to the plantation areas (T)
11. Human resources	- Food science and technology supervision is limited (T) - Lack of physical infrastructure (T)

***Note:** Cambodia's spending on R&D is low at 0.12 percent of GDP in 2015 compared to Malaysia (1.30 percent), Singapore (2.18 percent in 2014), and Thailand (0.62 percent).

(Source: Cambodia Outlook 2018)

2.3 Strategic products and services

2.3.1 Product feature concepts

Table 7: Concepts of Key Products' Features

Concepts of Key Products' Development Features
Dried products
Frozen products
Ready-to-eat products
Canned products
Bottled products
Vacuum products (aseptic and modified atmosphere)
Packaged products

Table 8: Key Technologies

Linkage of Key Technologies	
Semi-Finished/Processed Products	Finished/Processed Products
Cleaning	Cleaning
Grading	Grading
Peeling	Peeling
Shelling	Grinding
Vacuum packing	Extraction/Fermentation/Drying/Smoking
Labeling	Mixing
Coding	Pasteurization/Sterilization
Registration	Cooling/Freezing
Storage and preservation	Canning/Bottling
Logistics: Transportation and delivery (Cool chain supply and cooling plant warehouse)	Labeling
	Registration
	Packing/Drop packer/Wrap-around
	Palletizing
	Logistics: Transportation and delivery (cool chain supply and cooling plant warehouse)

2.3.2 Grouping

Table 9: Development of Key Products

Key Product Development	
Semi-Finished/Processed Products	Finished/Processed Products
Crops (industrial and horticulture) Cassava: Dried cassava chips and starch Black pepper: Dried white and black pepper Maize: Dried maize Cashew: Dried cashew Soybean: Dried soybean Banana: Fresh banana Mango: Fresh mango Others: Meat Production Frozen meat, smoked products, ready-to-eat sausage Fisheries Production Frozen seafood, smoked products	Crops (industrial and horticulture) Mango: Juice, Jam, Wine, Pickles, etc. Cassava: Chips, starch, ethanol, etc. Longan: Bottle or can-processed longan, wine Black pepper: dried, white and black pepper, pickles, etc. Dragon fruit: Wine, fresh fruit, juice, etc. Maize: Feed, juice, crackers, snacks, etc. Cashew: Dried Cashew apple: Wine, juice, vinegar, low-alcohol soft drink, etc. Soybean: Juice, sauce, NATTO, MISO, oil, fermented soybean, etc. Sugarcane: Wine, ethanol, sugar, etc. Pineapple: Juice, jam, wine, etc. Banana: Chips, candy, wine, crackers, etc. Others: Rice, cassava Vegetables Cucumber: Pickles (canned or bottles) Mustard green: Pickles Pumpkin: Noodles, powder, wine, cake, etc. Others: Meat Production Sausage, meat boil, hotdog, dried meat, pate, ham and others Fisheries product Fish sauce, fermented fish (Pro hok and Pro ork), smoked fish, etc.

2.3.3 Prioritization

Table 10: Identification and Prioritization of Key Products

Key Product Development Identification & Prioritization				
Opportunities & Threats	Opportunities		Threats	
	The Royal Government of Cambodia has already set in Industrial Development Policy 2015-2025 the export of about 12% of processed agricultural products by 2025 out of the total export. The government has committed to upgrading the Cambodian economy for Cambodia to become an upper middle-income country by 2030 and a high-income country by 2050. Cambodia has amended the investment law to attract investors under the Council Development of Cambodia (CDC).		Lack of capital investments from both the national budget law and the private sector. Investors and agro-processing companies are limited in the country. The Agro-Processing Small and Medium Enterprises (SMEs) are not yet clustered and mobilized like a special economic zone (SEZ) and/or an agricultural processing zone (APZ).	
Products/Services	Subsector/	Low	Middle	High
	Agriculture	Mungbean	Dragon fruit Longan Cashew apple Black pepper Pineapple Vegetables	Cashew Mango Cassava Soybean Banana Sugarcane Maize Rice
	Animal		Oat Goat/Sheep	Cow Chicken Pig Duck
	Fishery		Frozen seafood Processed crocodile	Freshwater fishes Sea products

2.3.4 Performance dimensions

Table 11: Identification of Performance Dimension

Performance Dimension	Performance Target
Productivity	Higher than 50% improvement
Capacity	More than 15000 rounds per minute (RPM)
Quality	More than 95% total productive quality
Cost saving	Reduction of total input through 50% cost saving
Efficiency	More than 90% production with loss of less than 10%
Engine type	Gas, diesel, and petroleum optional

Table 12: Technology Alternatives

Technology Alternatives
Washing/Cleaning (cleaning tank, automatic washing machine, disinfection tank and unit, etc.)
Dehydration (drum dryer, HTU (high-temperature unit), UV (ultraviolet), solar dehydrator, etc.)
Filtration unit (membrane filter, sand filter, etc.)
Heating unit (pasteurization and sterilization)
Fermentation unit (fermenter tank, cooling tank, storage tank, sedimentation tank, etc.)
Packaging unit (canning, bottling, labeling, coding, packing, weighting, palletizing, etc.)
Transportation unit (cool chain supply, cool chain warehouse and storage)
Delivery and marketing (bar code scanner and registration unit)

2.4 Key technologies

2.4.1 Technology solutions

Table 13: Technology Solutions

Technologies
Cleaning/Washing
Grading
Peeling
Cutting
Grinding
Blending
Extraction/Fermentation/Drying/Smoking
Mixing
Pasteurization/Sterilization
Cooling
Canning/Bottling
Labeling
Registration
Packing
Palletizing
Logistics: Transportation and Delivery (cool chain supply and cooling plant warehouse)

2.4.2 Grouping

Table 14: Technology Tree

Products	Tech 1	Tech 2	Tech 3	Tech 4	Tech 5
Pickle	Grading	Fermentation	Preservation	Packaging	Labeling
Juice	Extraction	Preservation	Pasteurization	Packaging	Labeling
Wine	Fermentation technology	Aging technology	Filtration technology	Packaging technology	Labeling
Dried food products	Peeling	Cutting	Drying	Packaging	Labeling
Smoked products	Washing	Grading	Smoking	Packaging	Labeling
Jam	Blending	Jam making	Preservation	Sterilization	Packaging and labeling
Fermented soybean	Sorting	Fermentation	Preservation	Filtration	Packaging
Fermented fish	Washing	Grading	Fermentation	Packaging	Labeling
Sausage	Blending	Sausage making	Preservation	Packaging	Labeling

2.4.3 Prioritization

Table 15: Prioritization of Key Technologies

Products	Critical needs	Strategic needs	Less important
Pickle	- Fermentation - Preservation	- Packaging - Labeling	Grading
Juice	- Extraction - Preservation - Pasteurization	- Packaging - Labeling	
Wine	- Fermentation - Aging - Filtration	- Packaging - Labeling	
Dried food products	- Drying	- Packaging - Labeling	- Peeling - Cutting
Smoked products	- Smoking	- Packaging - Labeling	- Washing - Grading
Jam	- Jam making - Preservation - Sterilization	- Packaging - Labeling	Blending
Fermented soybean	- Fermentation - Preservation - Filtration	Packaging	Sorting
Fermented fish	- Fermentation	- Packaging - Labeling	- Washing - Grading
Sausage	- Sausage making - Preservation	- Packaging - Labeling	Blending

2.5 Macro TRMs for strategic products

TRM type: Bar

Vision	Agro-Foods Processing will be equipped with modernized facilities, automation, and professional skills demanded at all levels, high productivity, quality, standard, and safety, inclusive and challenging in the free markets.									
Objectives	Product development, career creation, and commercialization. Coordinate the supplier system with high quality of raw materials and inputs, stability and competitive price, and sustainability for agro-processing production and increase the value added. Develop the roadmap in the field of science, technology, and innovation by aligning the needs of the food industry from food engineering and advanced food engineering and/or nanotechnology in food processing. Promote Digital, Artificial Intelligence and Block Chain Technology (AI). Promote R&D in robotics and automation in food science and technology.									
Year	20-21	22	23	24	25	26	27	28	29	30
Future Prospects	Peace and stability, policy support, GDP (annual income), labor force, technology, demand & supply, qualification & standards, R&D on STI, input of raw materials, global warming, and human resources									
Pickle	Fermentation			Preservation			Packaging	Labeling	Grading	
Juice	Extraction			Preservation			Pasteurization		Packaging	Labeling
Wine	Fermentation			Aging			Filtration		Packaging	Labeling
Dried food products	Drying				Packaging		Labeling	Peeling		Cutting
Smoked products	Smoking				Packaging			Labeling	Washing	
Jam	Jam making			Preservation		Sterilization		Packaging	Labeling	Blending
Fermented soybean	Fermentation				Preservation		Filtration	Packaging	Sorting	
Fermented fish	Fermentation				Packaging		Labeling	Washing	Grading	
Sausage	Sausage making				Preservation				Labeling	Blending

Part 3: Conclusion

3.1 Recommendations

This TRM for the development of the Agro-Processing subsector is only part of the agriculture sector, which is mandatory for the Ministry of Agriculture, Forestry, and Fisheries (MAFF) of the Royal Government of Cambodia (RGC). The technical departments of the Department of Agro-Industry (DAI) serve as the secretary of MAFF, and they will be responsible for the implementation, monitoring, and evaluation of all programs, projects, strategic action plan, and other policies to contribute to the development of MAFF and RGC. Regarding its being mandatory, the Development Partners (DPs) should have taken into account working continuously with the technical department under the coordination of the National Science and Technology Council (NSTC). Therefore, it presents a great opportunity to develop Cambodia.

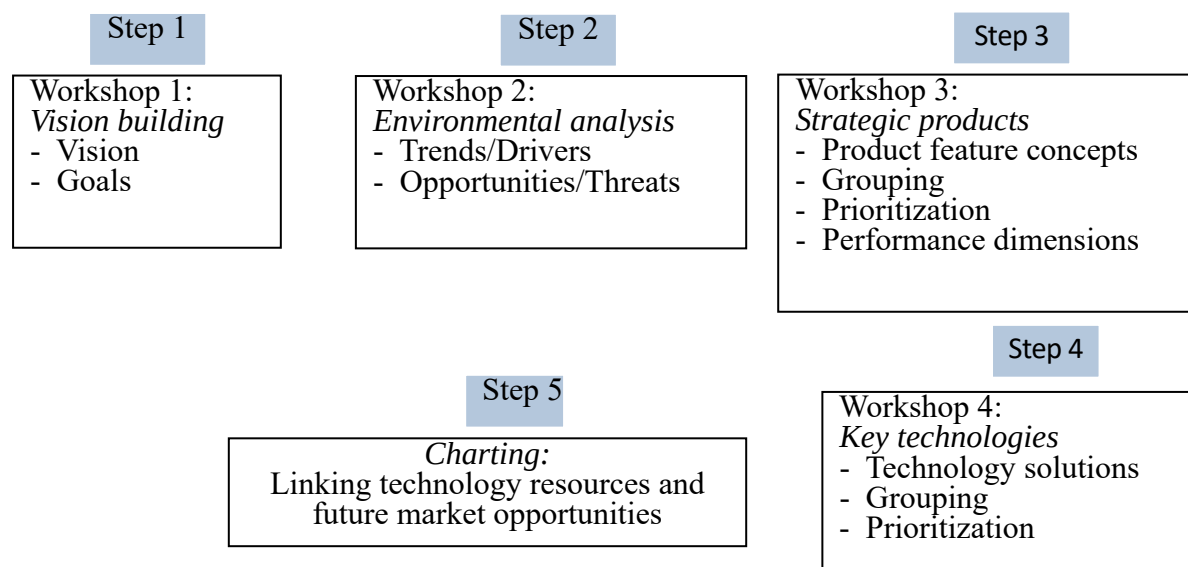
3.2 Implementation

Table 17: Plan of Action Implementation by 2030

Strategic Plan/Program/Activities	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	
Strategy 1 (S1): Strengthen institutional capacity, human resources development, and capacity building												
Program 1:												
Program 2:												
Program 3:												
Strategy 2 (S2): Develop the science, technology, and innovation and marketing orientation												
Program 1:												
Program 2:												
Program 3:												
Program 4:												
Strategy 3 (S3): Integrate and enhance private investment to establish the scientific industry and artificial intelligence												
Program 1:												
Program 2:												
Program 3:												
Program 4:												
Strategy 4 (S4): Promote and invest in research, development, and extension												
Program 1:												
Program 2:												
Program 3:												
Table 18: Financial Support												
												0.000 USD
Strategic Plan/Program/Activities	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	Total Lump Sum
Strategy 1 (S1): Strengthen institutional capacity, human resources development, and capacity building												16,100.00
Program 1:	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	5,500.00
Program 2:			✓	✓	✓	✓	✓	✓	✓	✓	✓	7,100.00
Program 3:			✓	✓	✓	✓	✓	✓	✓	✓	✓	3,500.00
Strategy 2 (S2): Develop the science, technology, and innovation and marketing orientation												6,215.00
Program 1:		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	1,860.00
Program 2:		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	935.00
Program 3:		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	560.00
Program 4:		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	2,860.00
Strategy 3 (S3): Integrate and enhance private investment to establish the scientific industry and artificial intelligence												4,000.00
Program 1:	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	450.00
Program 2:		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	900.00
Program 3:		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	1,200.00
Program 4:	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	1,450.00
Strategy 4 (S4): Promote and invest in research, development, and extension												32,805.00
Program 1:		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	2,500.00
Program 2:		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	1,890.00
Program 3:		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	2,100.00
Total Lump Sum S1 + S2 + S3 + S4												59,120.00

Part 4: Appendices

4.1 Roadmapping process



4.2 Participants

Table 19: List of Participants

No.	Name	Gender	Position	Organization	Ministry/Institution
1	PANN VISAL	M	Deputy-Director	Dept. of National R&D and Infrastructure	NSTC Ministry of Planning
2	PHON RENO	M	Bureau Chief	Dept. of Agro-Industry	Ministry of Agriculture, Forestry, and Fisheries
3	TAN RAKSMEY	F	Head of the Food Technology and Nutrition Research Unit	Research and Innovation Center	Ministry of Education, Youth, and Sports/Institute of Technology of Cambodia
4	TUY VITOU	M	Officer	Dept. of Agro-Industry	Ministry of Agriculture, Forestry, and Fisheries



CHAPTER 4

Conclusion

Policy Consultation on National STI Policy
Development and STI System Management for
Cambodia's Policymakers

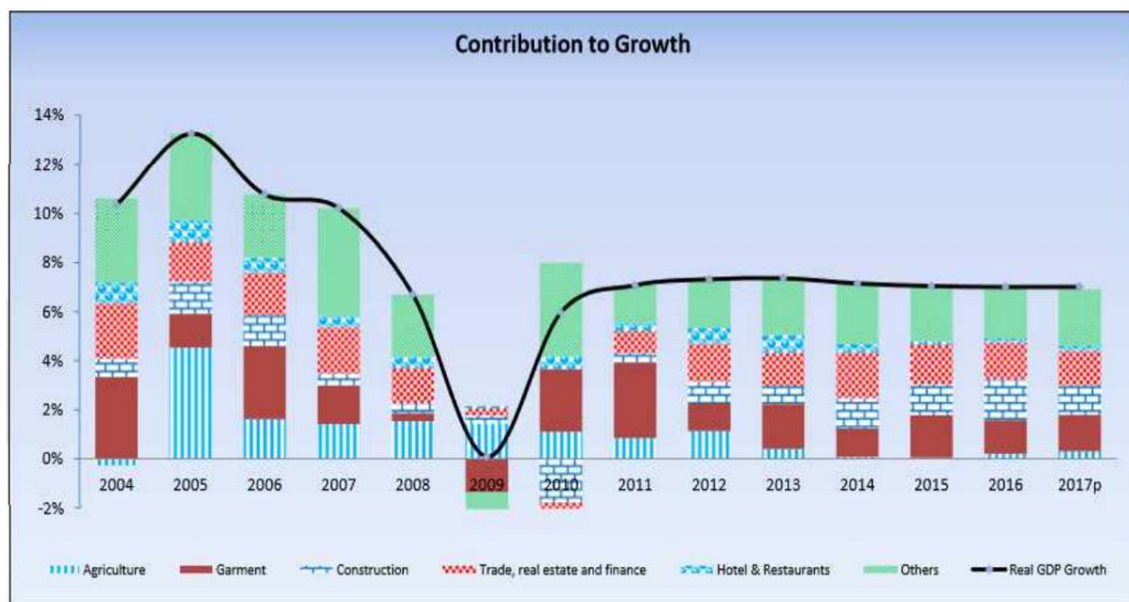


Chapter 4. Conclusion

1. STI Policy

Cambodia has maintained stable economic growth of around 7% in recent years, and it is pushing for a national strategy to develop into a high middle-income country by 2030 and a high-income country by 2050. Note, however, that tourism and real estate development are a major part of the economy, with the industrial sector focusing heavily on textiles and clothing. To realize the national vision, it is necessary for Cambodia to find new growth engines.

[Figure 4-1] Contribution to Growth by Sectors in Cambodia

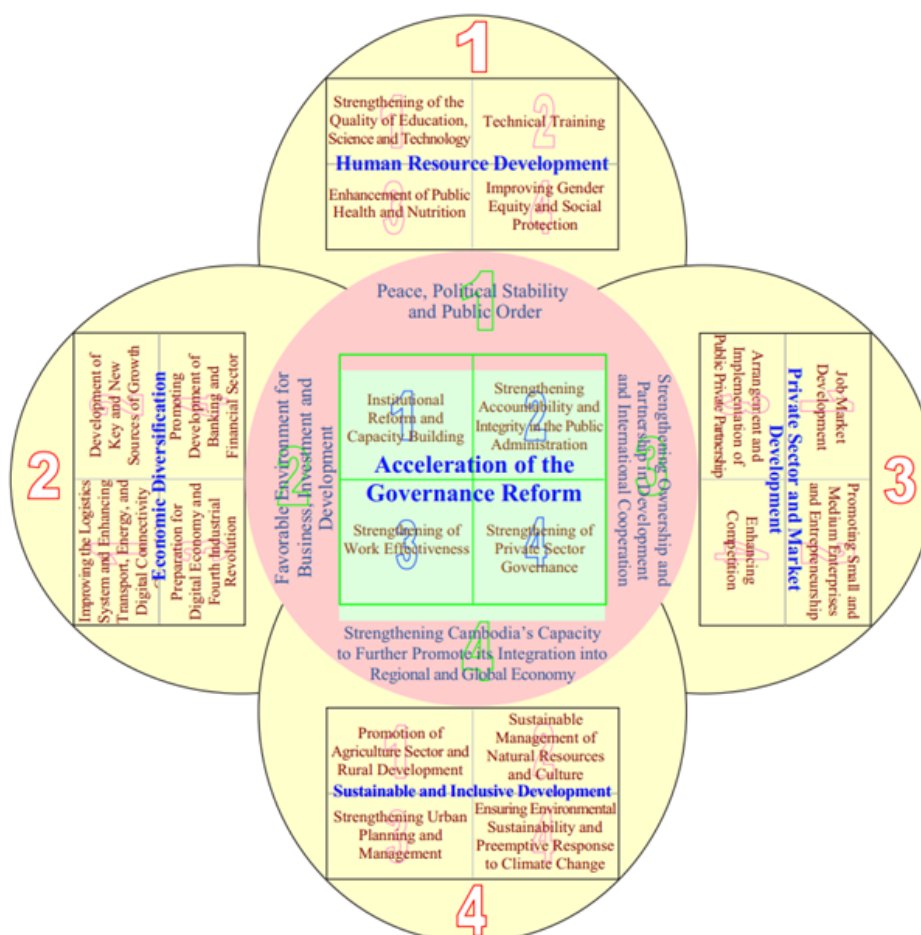


Source: Statistics Office of Cambodia, recited from KOTRA (2019)

New growth engines can mostly be acquired through science and technological innovation. It means that the key for the implementation of Cambodia's national development strategy is science and technological innovation. Cognizant of this background, STEPI has provided policy advice and support in cooperation with GS-NSTC in Cambodia to establish "STI Policy 2019-2025," the first official Cambodian science and technology master plan. The plan aims to secure science and technology personnel, expand science and technology investment, strengthen the legal and institutional foundations, and foster a close relationship between the social system and STI.

A particular area of emphasis in Cambodia's science and technology policy is human resource development (HRD), one of the highest-priority policy areas in the Rectangular Strategy IV as the Cambodian government's national development strategy. In particular, securing qualified scientists, engineers, and technicians in the field of science and technology is one of the most fundamental and critical success factors in the realization of STI policy. That is why STEM education and related vocational training should be emphasized. Looking at the current situation in Cambodia, however, there is a shortage of qualified teachers in science and technology, including lack of facilities and equipment for science and technology education. Considering Korea's experience, it is recommended that vocational education for the technicians and higher education system in science and technology for researchers be developed in parallel, from the mid- and long-term perspectives.

[Figure 4-2] Diagram of Rectangular Strategy Phase IV



Source: Royal Government of Cambodia, Rectangular Strategy Phase IV, 2018. 9

The “STI Policy 2019-2025” plan is currently in the process of obtaining approval from the Cambodian government. Considering the characteristics of science and technology, where environmental changes in technology and markets are very rapid, however, the time period from the establishment of the national STI policy or STI plan to its implementation through the approval of government should be shortened. Cambodia can miss the most effective time for STI implementation. Cambodia's STI system needs to be upgraded by establishing an appropriate strategy that meets the national development goals and implementing it effectively and quickly.

Upgrading Cambodia's STI system requires strengthening the governance of STI policy. Cambodia currently does not have a central administrative body that leads and coordinates STI policies. The National Science and Technology Council (NSTC) and its General Secretariat under the Ministry of Planning is now playing such role. Under this governance structure, the importance of STI in policy priorities across the government cannot be emphasized, and it is difficult to get sufficient support. Chaired by the Prime Minister, NSTC is usually held twice a year, making it difficult to address the key agenda of important STI policies in a timely manner. Therefore, it is recommended that the General Secretariat of NSTC be expanded and reorganized to create a dedicated department dealing with science and technology innovation and allocate appropriate budgets and manpower.

As a key driver for achieving Cambodia's national development strategy goals, STI is known to be an area that requires not only science and technology but also the active participation and policy coordination of the relevant ministries. Therefore, it is important to have an implementation system that enables practical policy discussion and coordination. A higher policy coordination system is also needed, such as the Ministerial Conference on Science and Technology Innovation, which can coordinate the major STI-related policies of ministries. The highest STI policy decision making may be submitted to NSTC.

2. Technology Roadmap

The pilot project of Cambodia's Technology Roadmap is a process of learning how to establish a national technology strategy for the implementation of the STI Policy prepared last year. Through this process, the overall concept of technology roadmap is more understandable for the participants and the Cambodian S&T community, but practical training is not yet sufficient for each step of the roadmap process.

In particular, the technology roadmap is a learning process that utilizes the collective intelligence of participating experts. Through this, it is important to communicate with the relevant community and form consensus. In that sense, there are still some activities that are not well-organized.

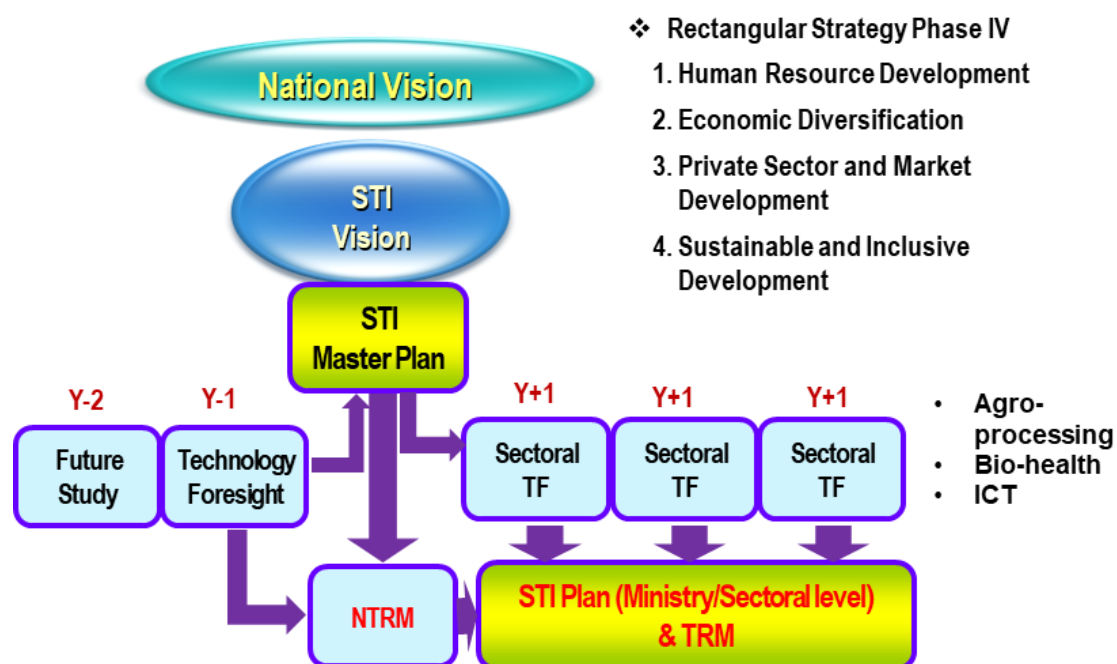
Still, a significant achievement was that the pilot project was made to recognize the importance of the technology roadmap and to understand the basic concepts and procedures by science and technology policy organizations such as GS-NSTC. It is also important to be able to cultivate experts who can play the role of leader or champion in each sector by carrying out the national technology roadmap project in the second phase.

The role of the government is important for the successful development of the national technology roadmap. Governments should not directly interfere with the roadmap contents but facilitate and coordinate instead the roadmap process itself. And because the technology roadmap reflects strategic priorities at the national level, there is a high likelihood of conflicting interests between technologies. For this reason, governments should always manage the process so that it is fair and transparent during the roadmap process. Another role of the government is to support sufficient resources, such as supporting staff and budget, for the technology roadmap. The government should not pressure the roadmap team to finish the report too soon but should allow time for sufficient discussion and consensus. Finally, the end-user of the technology roadmap is the company, so it is always necessary to encourage the experts from the private sector to participate actively in the roadmap.

On the other hand, an organization or an institution in charge of the technology roadmap should ensure that the roadmap process is planned systematically at each stage and select the most appropriate experts in each field. The most critical success factor is to select experts with technical expertise and strong understanding of the industry and who are willing to participate. In addition, the opinions of a few specific experts should not be unilaterally reflected during the roadmap process. Instead, agreed-upon opinions should be drawn through the collective intelligence of the team. Finally, the technology roadmap is an iterative learning process; it is important to finish and submit a report, but regularly updating and implementing it in accordance with economic, social, and technological changes are more important and effective.

The STI plan and policy, such as last year's "STI Policy 2019-2025" and this year's technology roadmap, should be implemented in line with the national development goals especially when an independent STI ministry is established. For example, the government set the national future strategies based on future study. Technology foresight follows to show the direction of the national STI master plan. The STI plan includes the vision, goals, and strategies of STI development as well as the key technologies needed to achieve the national development goals. For the next step, it needs the national technology roadmap for these key technologies. Finally, each ministry or institution in charge of each sector establishes and executes a strategy that includes a technology roadmap for each sector.

[Figure 4-3] Linking the National Vision and STI Plan



Source: Lee, Jungwon, National Strategic Planning for STI Development, 2019

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